Indian Crops: Diseases, Treatments, Soil and Nutrition Guide

A field reference for 160 Indian crops: best season, soil, farming details (duration, seed rate, water, yield), fertilizer, nutrient deficiencies, common diseases and pests, chemical doses and biological options. Hindi and Telugu names are given in Roman letters here; the Excel and JSON files hold them in native script.

How to use this guide

- All spray doses are **per litre of water** unless stated. For a **15 L knapsack pump, multiply by 15** (for example 2.5 g/L = 37.5 g per pump). For a 20 L pump multiply by 20.
- Doses and fertilizer figures are commonly recommended ranges from ICAR and state agricultural university packages of practices. **They vary by state, soil and variety, so do a soil test and follow the product label.** Where a row says "Follow label and local advice", ask your Krishi Vigyan Kendra (KVK) or agriculture officer.
- Fertilizer is shown as kg per hectare of N : P2O5 : K2O (1 hectare is about 2.5 acres), or per tree where stated.
- "Bio / organic option" means biopesticides, bio-control agents and botanicals. "Antibiotic" means products like Streptocycline, used only against bacterial diseases.
- Virus diseases cannot be cured by spraying. The sprays shown for them control the insect that spreads the virus.
- In a PDF reader, use the bookmarks panel to jump straight to any crop.

Crops covered

Cereals, Millets And Fodder Crops: 1. Rice; 2. Wheat; 3. Maize; 4. Sorghum; 5. Pearl millet; 6. Finger millet; 7. Barley; 8. Minor millets; 9. Quinoa; 10. Buckwheat; 11. Amaranth grain; 12. Oats; 13. Berseem; 14. Lucerne; 15. Napier grass

Cash, Fibre And Green Manure Crops: 16. Cotton; 17. Sugarcane; 18. Jute; 19. Tobacco; 20. Sugar beet; 21. Sunn hemp; 22. Dhaincha

Vegetables And Tuber Crops: 23. Tomato; 24. Chilli; 25. Potato; 26. Onion; 27. Brinjal; 28. Okra; 29. Cabbage and Cauliflower; 30. Cucurbits; 31. Capsicum; 32. French bean and Dolichos; 33. Garden pea; 34. Carrot; 35. Radish; 36. Beetroot; 37. Leafy vegetables; 38. Drumstick; 39. Sweet potato; 40. Cassava; 41. Garlic; 42. Colocasia; 43. Cluster bean; 44. Sweet corn and Baby corn; 45. Elephant foot yam; 46. Pointed gourd; 47. Bottle gourd; 48. Bitter gourd; 49. Cucumber; 50. Broccoli; 51. Ivy gourd; 52. Ridge gourd; 53. Snake gourd; 54. Ash gourd; 55. Pumpkin; 56. Knol-khol; 57. Turnip; 58. Lettuce; 59. Yardlong bean; 60. Curry leaf

Fruits: 61. Mango; 62. Banana; 63. Grapes; 64. Pomegranate; 65. Citrus; 66. Papaya; 67. Guava; 68. Apple; 69. Pineapple; 70. Watermelon and Muskmelon; 71. Sapota; 72. Custard apple; 73. Litchi; 74. Jackfruit; 75. Strawberry; 76. Peach and Plum; 77. Ber; 78. Amla; 79. Dragon fruit; 80. Fig; 81. Pear; 82. Date palm; 83. Lemon; 84. Sweet orange; 85. Jamun; 86. Tamarind; 87. Avocado; 88. Kiwi; 89. Bael; 90. Apricot; 91. Walnut; 92. Cherry; 93. Passion fruit

Pulses (Seed Crops): 94. Chickpea; 95. Pigeon pea; 96. Green gram and Black gram; 97. Lentil; 98. Field pea; 99. Horse gram; 100. Cowpea; 101. Rajmash; 102. Moth bean

Oilseeds: 103. Groundnut; 104. Soybean; 105. Mustard; 106. Sunflower; 107. Sesame; 108. Safflower; 109. Castor; 110. Linseed; 111. Niger; 112. Toria and Taramira

Spices And Plantation Crops: 113. Turmeric; 114. Ginger; 115. Black pepper; 116. Cardamom; 117. Coriander; 118. Cumin; 119. Fennel; 120. Fenugreek; 121. Ajwain; 122. Clove; 123. Nutmeg; 124. Cinnamon; 125. Vanilla; 126. Saffron; 127. Tea; 128. Coffee; 129. Rubber; 130. Coconut; 131. Arecanut; 132. Cashew; 133. Oil palm; 134. Cocoa

Flowers: 135. Rose; 136. Marigold; 137. Jasmine; 138. Chrysanthemum; 139. Tuberose; 140. Gerbera; 141. Carnation; 142. Gladiolus; 143. Anthurium; 144. Orchids; 145. Crossandra

Medicinal And Aromatic Crops: 146. Ashwagandha; 147. Aloe vera; 148. Stevia; 149. Betel vine; 150. Mushroom; 151. Mulberry; 152. Lemongrass; 153. Mentha; 154. Tulsi; 155. Palmarosa; 156. Patchouli; 157. Citronella; 158. Safed musli; 159. Giloy; 160. Brahmi

Which crop suits which soil

Soil type	Features	Best crops	Cautions
Alluvial (river plains)	Deep, fertile, good water holding	Rice, wheat, sugarcane, maize, jute, mustard, potato, pulses, most vegetables	Check drainage in low spots
Black cotton (regur)	Clay-rich, holds moisture, cracks when dry	Cotton, soybean, sorghum, chickpea, pigeon pea, sunflower, safflower, onion, citrus, grapes, pomegranate	Waterlogs when wet and is hard to work; needs drainage and timely tillage
Red soil	Light, well-drained, low in nitrogen and organic matter	Groundnut, ragi, millets, chilli, tomato, tobacco, castor, horse gram, mango, sapota, vegetables with irrigation	Add FYM and zinc; mulch to save moisture
Laterite	Acidic, well-drained, low fertility, high rainfall regions	Cashew, coconut, rubber, tea, coffee, arecanut, pineapple, cassava, jackfruit, black pepper	Lime and organic matter for acidity; mulch
Sandy / desert	Very light, drains fast, low nutrients	Bajra, moth bean, cluster bean, cumin, sesame, watermelon, carrot, ber, date palm, aloe vera, groundnut	Drip irrigation and organic manure essential
Saline and alkaline	High salts or high pH, poor structure	Barley, sugar beet, cotton, date palm, ber, guava, amla, tamarind; rice after gypsum	Gypsum and drainage; avoid sensitive crops like beans and chilli
Hill and forest (acidic)	Humus-rich, cool, slopes	Tea, coffee, cardamom, apple, pear, peach, kiwi, strawberry, potato, peas, rajmash	Terraces to stop erosion; lime if too acidic
Clay and heavy soils	Sticky when wet, slow drainage	Rice, sugarcane, wheat, cabbage, banana (with drainage)	Not good for carrot, radish, groundnut or most root crops
Loam (balanced soil)	Ideal mix of sand, silt and clay	Almost all vegetables, fruits, pulses and cereals	Maintain organic matter

Which crop suits which season

Season	Crops that do best
Kharif (monsoon: sow Jun-Jul, harvest Sep-Nov)	Rice, maize, sorghum, bajra, ragi, cotton, jute, sugarcane (planted earlier), groundnut, soybean, pigeon pea, green and black gram, sesame, castor, cowpea, okra, brinjal, chilli, turmeric, ginger
Rabi (winter: sow Oct-Dec, harvest Mar-May)	Wheat, barley, chickpea, lentil, field pea, mustard, linseed, safflower, potato, onion, garlic, cabbage, cauliflower, carrot, radish, coriander, cumin, fennel, fenugreek, strawberry
Zaid / summer (sow Feb-Mar, harvest Apr-Jun)	Watermelon, muskmelon, cucumber, bottle gourd, bitter gourd, okra, cluster bean, green gram, sunflower, sesame, fodder crops, mentha
Year-round with irrigation or polyhouse	Tomato, chilli, brinjal, leafy vegetables, banana, papaya, flowers, mushroom, gerbera, carnation, capsicum
Perennial and plantation	Mango, citrus, guava, pomegranate, coconut, arecanut, cashew, tea, coffee, rubber, black pepper, cardamom, oil palm, mulberry, drumstick

Rule of thumb: pick the crop that suits your soil and rainfall first, then fit the season. A soil test (pH, organic carbon, N, P, K, zinc) from your KVK or soil health card is the best starting point.

Best crop choices for common farm situations

Situation	Good crop choices	Why
Very low rainfall or dryland (under 500 mm)	Bajra, sorghum, moth bean, cluster bean, horse gram, castor, cumin, taramira, ber, custard apple, tamarind, bael, aloe vera	Deep roots or short duration; need little irrigation
Heavy rain or waterlogged fields	Rice, jute, colocasia (arbi), dhaincha, brahmi, pointed gourd, sugarcane, banana (with drainage)	These tolerate standing water better than most crops
Small land, quick cash (30-90 days)	Leafy vegetables, radish, coriander leaf, okra, cucumber, marigold, green gram, oyster mushroom	Short crop cycle; frequent harvest and sales
Small land, steady income	Banana, papaya, drumstick, curry leaf, jasmine, crossandra, betel vine, Napier or berseem fodder with a cow or buffalo	Regular harvest or dairy-linked demand
High-value horticulture (needs capital and skill)	Capsicum and cucumber in polyhouse, gerbera, anthurium, orchids, strawberry, dragon fruit, avocado, kiwi, vanilla, saffron	High price but higher cost, risk and technical need
Salty or alkaline soil	Barley, sugar beet, cotton, date palm, ber, guava, amla, dhaincha (to improve soil)	Better salt tolerance than most crops
Hills and cool climate	Apple, pear, peach, apricot, walnut, cherry, potato, peas, rajmash, buckwheat, tea, cardamom, strawberry	Need winter chill or cool summers
Livestock and fodder	Napier grass, berseem, lucerne, oats, fodder sorghum and maize, cowpea	High green fodder per hectare

Situation	Good crop choices	Why
Improve the soil (green manure and rotation)	Dhaincha, sunn hemp, cowpea, green gram, berseem; rotate legumes with cereals	Legumes add nitrogen and organic matter
Grow under trees (intercrop)	Cocoa, black pepper, ginger, turmeric, pineapple, banana, vanilla under coconut or arecanut	Uses shade and space for extra income

Nutrient deficiency guide

Nutrient	Symptoms	Common in	Quick fix
Nitrogen (N)	Pale yellow older leaves, stunted growth	All cereals, leafy vegetables, sugarcane	Urea topdressing in splits; foliar urea 20 g/L (2%) for quick fix
Phosphorus (P)	Dark green or purplish leaves, poor roots, late maturity	Pulses, wheat, maize on acid or very alkaline soils	SSP or DAP at sowing (basal); foliar DAP 20 g/L (2%), filter before spraying
Potassium (K)	Scorched edges on older leaves, weak stems, poor fruit quality	Potato, banana, sugarcane, tomato, coconut	MOP or SOP basal; foliar potassium nitrate 10 g/L (1%)
Calcium (Ca)	Blossom end rot, tip burn, empty pods (groundnut pops), cracked fruit	Tomato, chilli, cabbage, groundnut, pomegranate	Gypsum 400-500 kg/ha for groundnut at flowering; foliar calcium nitrate 5 g/L
Magnesium (Mg)	Yellow between veins of older leaves; reddening in cotton	Cotton, citrus, banana, sandy soils	Magnesium sulphate 10 g/L (1%) foliar, two sprays; or 25-50 kg/ha soil
Sulphur (S)	Pale young leaves, low oil in oilseeds, poor nodulation	Mustard, groundnut, soybean, onion, pulses	Gypsum or elemental sulphur 20-40 kg S/ha basal
Zinc (Zn)	Khaira in rice, white bud in maize, little leaf, short internodes	Rice, maize, wheat, citrus, mango, guava	Zinc sulphate 25 kg/ha soil; foliar 5 g/L (0.5%) with 2.5 g/L lime
Iron (Fe)	Yellow young leaves with green veins (iron chlorosis)	Groundnut, sugarcane, citrus, sorghum, high-pH soils	Ferrous sulphate 5 g/L (0.5%) foliar plus citric acid 1 g/L; repeat after 10 days
Boron (B)	Hollow stem, brown curd, fruit cracking, poor fruit set	Cauliflower, tomato, sunflower, litchi, coconut	Borax 10 kg/ha soil in deficient fields, or foliar 1-2 g/L
Manganese (Mn)	Yellow between veins of young leaves, grey specks in cereals	Wheat, sugarcane, light alkaline soils	Manganese sulphate 2.5 g/L (0.25%) foliar
Molybdenum (Mo)	Whiptail leaves in cauliflower, poor nodules in legumes	Cauliflower, soybean, pulses on acid soils	Seed treatment ammonium molybdate 1 g/kg or foliar 0.5 g/L

Foliar sprays correct a deficiency fast but only for a short time. Confirm with a soil or leaf test, avoid spraying in hot sun, and do not tank-mix with pesticides unless the labels allow it. Too much micronutrient can burn leaves.

Safety basics (read before spraying)

- **Toxicity colour triangle on the label:** Red = extremely toxic, Yellow = highly toxic, Blue = moderately toxic, Green = slightly toxic. Prefer Green or Blue products and bio options where they work.
- **Waiting period (PHI):** minimum days between last spray and harvest. Never harvest earlier. Bio products usually have 0 days. Be extra strict for leafy vegetables, herbs, betel leaf, tea, stevia and mulberry (silkworm feed).
- Wear gloves, mask, full sleeves and boots. Do not eat, drink or smoke while spraying. Wash well afterwards.
- Spray in the **early morning or evening**, not in strong sun, wind or rain. Do not spray during full flowering when bees are active.
- Do not mix products unless the label allows it. Do not mix bio-agents (Trichoderma, Pseudomonas, Bt) with fungicides or copper in the same tank.
- **Rotate chemical groups** to avoid resistance. Do not repeat the same product every spray.
- **2,4-D is a weedicide**, not a crop medicine. Drift or spray from the same pump badly damages tomato, chilli, cotton, grapes and many other crops. Use a separate, clearly marked pump for herbicides.
- Keep pesticides in original containers, away from children and food. Triple-rinse and puncture empty containers.

Bio-control and organic products at a glance

Product	What it controls	Typical dose
Trichoderma viride / harzianum	Root rot, wilt, damping off, collar rot	Seed 4-10 g/kg; soil 2.5 kg with 100 kg FYM per ha; drench 5 g/L
Pseudomonas fluorescens	Blast, sheath blight, wilts, leaf spots	Seed 10 g/kg; foliar 5-10 g/L; soil 2.5 kg/ha with FYM
Bacillus subtilis	Leaf spots, blights, powdery mildew, root diseases	Foliar 2-5 g/L; seed 10 g/kg
Ampelomyces quisqualis (Ampelo-Biocure type)	Powdery mildew	About 5 ml/L as per label (75 ml per 15 L pump)
Bacillus thuringiensis (Bt)	Caterpillars: fruit borer, armyworm, diamondback moth	1-2 g/L (or 2 ml/L liquid)
Beauveria bassiana	Aphids, thrips, whitefly, hoppers, borers	5 g/L (or 5 ml/L liquid)
Metarhizium anisopliae	Planthoppers, white grub, termites, beetles	5 g/L; soil 2.5 kg/ha with FYM
Lecanicillium (Verticillium) lecanii	Whitefly, aphids, scales, thrips, mealybug	5 g/L
NPV (HaNPV, SlNPV)	*Helicoverpa* (fruit and pod borer), *Spodoptera*	250 larval equivalents (LE) per ha, evening spray
Trichogramma egg cards	Stem borers, fruit borers, bollworm eggs	1-1.5 lakh eggs per ha, 4-6 weekly releases
Neem oil (1500 ppm type)	Sucking pests, mites, early caterpillars	3-5 ml/L with 1 ml/L liquid soap
Neem seed kernel extract (NSKE) 5%	Aphids, leaf miner, young caterpillars	50 g kernel powder per litre, soaked overnight, strained
Pheromone and sticky traps	Monitoring and mass trapping	5 pheromone traps per acre for borers; 10-12 yellow sticky traps per acre; methyl eugenol / cue-lure traps 8-10 per acre for fruit flies
Neem cake	Soil pests, nematodes, soil-borne diseases	150-250 kg per ha in soil, mixed before sowing or planting
Bordeaux mixture (1%)	Downy mildew, canker, rots (a traditional copper mix)	10 g copper sulphate plus 10 g lime per litre of water, freshly prepared

Bio products work best as prevention or at the first signs. They act slower than chemicals and are sensitive to sun and heat, so spray in the evening and store as the label says.

CEREALS, MILLETS AND FODDER CROPS

1. Rice (Paddy)

Hindi / Telugu name	dhān / cāval / vari
Best season	Kharif (Jun-Jul sowing); Rabi in south and east
Best soil	Clay loam to clay that holds water; alluvial
Soil pH	5.5-7.0
Main growing states	West Bengal, UP, Punjab, AP, Telangana, Odisha, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	100-120 : 50-60 : 40-50 kg/ha
Nutrient tip	Zinc (khaira disease) and iron deficiency are common; apply zinc sulphate 25 kg/ha
Crop duration	100-150 days (variety)
Seed rate / spacing	Seed 40-50 kg/ha (nursery); spacing 20 x 15 cm
Water need	High (1200-1600 mm)
Typical yield	4-6 t/ha grain

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Blast	Spindle-shaped grey spots on leaves, neck rot, white empty panicles	Tricyclazole 75 WP: 0.6 g/L; Isoprothiolane 1.5 ml/L	*Pseudomonas fluorescens* seed treatment 10 g/kg and foliar 5-10 g/L; avoid excess nitrogen
Sheath blight	Oval greenish-grey lesions on sheath near water level	Hexaconazole 5 EC: 2 ml/L; Validamycin 3 EC: 2.5 ml/L	*P. fluorescens* foliar 5-10 g/L; *Trichoderma* soil application
Bacterial leaf blight	Yellow-white stripes from leaf tips, wilting leaves	Streptocycline 0.1-0.15 g/L plus copper oxychloride 2.5 g/L	Resistant varieties; avoid excess nitrogen; drain the field
Brown spot	Brown oval spots with grey centre, mostly in poor soils	Mancozeb 2.5 g/L	*Trichoderma* seed treatment 4 g/kg; balanced nutrition
Stem borer	Dead hearts in young plants, white ears at flowering	Cartap hydrochloride 50 SP: 2 g/L; Chlorantraniliprole 18.5 SC: 0.3 ml/L	*Trichogramma japonicum* cards, 5-6 releases; pheromone traps
Brown planthopper	Circular dry patches (hopper burn)	Pymetrozine 50 WG: 0.6 g/L; Imidacloprid 17.8 SL: 0.3 ml/L	*Metarhizium* or *Beauveria* 5 g/L; drain water; avoid excess nitrogen
Leaf folder	Leaves folded lengthwise, whitish scraped streaks	Chlorantraniliprole 0.3 ml/L; Cartap 2 g/L	*Trichogramma chilonis*; neem oil 5 ml/L
Tungro virus	Orange-yellow stunted plants, poor tillers	No cure. Control green leafhopper vector with Imidacloprid 0.3 ml/L	Resistant varieties; synchronous planting; remove infected plants
Gall midge	Silver shoots (onion shoots), no panicle	Cartap hydrochloride 2 g/L or Chlorantraniliprole 0.3 ml/L	Early planting; resistant varieties; parasitoids protected
Sheath rot and false smut	Rotted sheath at flag leaf; orange-green balls on grains	Propiconazole 1 ml/L at boot and heading; Copper oxychloride 3 g/L	Avoid excess nitrogen; seed treatment with *P. fluorescens*

2. Wheat

Hindi / Telugu name	gehūṃ / godhuma
Best season	Rabi (Oct-Dec sowing)
Best soil	Well-drained loam to clay loam; alluvial, black
Soil pH	6.0-7.5
Main growing states	UP, Punjab, Haryana, MP, Rajasthan, Bihar
Fertilizer (N : P2O5 : K2O)	120 : 60 : 40 kg/ha
Nutrient tip	Zinc and manganese deficiency in sandy or high-pH soils
Crop duration	110-140 days
Seed rate / spacing	Seed 100-125 kg/ha; rows 20-22.5 cm
Water need	Moderate (4-6 irrigations)
Typical yield	4-5 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Yellow (stripe) rust	Yellow powder in stripes along leaves	Propiconazole 25 EC: 1 ml/L; Tebuconazole 25.9 EC: 1 ml/L	Resistant varieties; early sowing

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Brown (leaf) rust	Orange-brown round pustules on leaves	Propiconazole 1 ml/L; Mancozeb 2.5 g/L	Resistant varieties; avoid late sowing
Powdery mildew	White powdery growth on leaves	Wettable sulphur 2.5 g/L; Propiconazole 1 ml/L	*Bacillus subtilis* 2-5 g/L
Loose smut	Black powdery ear heads at flowering	Seed treatment: Carboxin 2.5 g/kg or Tebuconazole 1.5 g/kg	*Trichoderma* seed treatment 4 g/kg; hot-water seed treatment
Karnal bunt	Black powder and fishy smell in some grains	Propiconazole 1 ml/L at ear emergence	Certified seed; avoid irrigation at flowering
Aphids	Sap-sucking clusters on ears and leaves	Imidacloprid 17.8 SL: 0.3 ml/L	Neem oil 5 ml/L; *Beauveria* 5 g/L; ladybird beetles help
Termites	Wilting and cut plants in dry fields	Seed treatment: Imidacloprid 70 WS 1 g/kg (label dose)	Well-rotted FYM; neem cake 250 kg/ha; *Metarhizium*
Foot rot and root rot	Yellowing, rotted crown and roots	Seed treatment: Carboxin plus Thiram 2.5 g/kg	*Trichoderma* 4 g/kg seed; crop rotation

3. Maize

Hindi / Telugu name	makkā / mokkajonna
Best season	Kharif and Rabi (south); Zaid in some areas
Best soil	Well-drained fertile loam; red and alluvial; avoids waterlogging
Soil pH	5.5-7.5
Main growing states	Karnataka, MP, Maharashtra, AP, Telangana, Bihar
Fertilizer (N : P2O5 : K2O)	120-150 : 60 : 40 kg/ha
Nutrient tip	Zinc deficiency causes white bud; apply zinc sulphate 25 kg/ha
Crop duration	90-120 days
Seed rate / spacing	Seed 20-25 kg/ha; 60 x 20 cm
Water need	Moderate (500-800 mm)
Typical yield	5-8 t/ha (hybrids)

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fall armyworm	Ragged holes in whorl leaves, sawdust-like frass	Emamectin benzoate 5 SG: 0.4 g/L; Spinetoram 11.7 SC: 0.5 ml/L; Chlorantraniliprole 0.3 ml/L	NSKE 5% or neem oil 5 ml/L early; *Bt* 2 g/L; *Metarhizium rileyi*; pheromone traps
Turcicum leaf blight	Long cigar-shaped grey-brown lesions	Mancozeb 2.5 g/L; Propiconazole 1 ml/L	*P. fluorescens* 10 g/L; resistant hybrids
Downy mildew	Pale stripes, white fuzz under leaves, stunting	Seed treatment: Metalaxyl 2.4 g/kg	Early sowing; remove infected plants
Stem borer	Dead heart, shot holes in leaves	Chlorantraniliprole 0.3 ml/L	*Trichogramma chilonis* 1 lakh/ha, 2-3 releases
Banded leaf and sheath blight	Banded lesions with sclerotia on sheath and cob	Hexaconazole 2 ml/L or Validamycin 2.5 ml/L	Remove lower leaves; *P. fluorescens*
Stalk rot	Lodging, rotted lower stalk before maturity	Avoid excess nitrogen; drench Carbendazim 1 g/L if early	Balanced potash; plant at proper density

4. Sorghum (Jowar)

Hindi / Telugu name	jvār / jonna
Best season	Kharif and Rabi
Best soil	Medium to deep black soils; also red loams; tolerates drought
Soil pH	6.0-8.5
Main growing states	Maharashtra, Karnataka, Rajasthan, MP, Telangana
Fertilizer (N : P2O5 : K2O)	80 : 40 : 40 (rainfed 40 : 20 : 20) kg/ha
Nutrient tip	Iron chlorosis in calcareous soils
Crop duration	100-120 days
Seed rate / spacing	Seed 10-12 kg/ha; 45 x 15 cm
Water need	Low

Typical yield	2-3.5 t/ha grain

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Shoot fly	Dead hearts in seedlings within 3-4 weeks	Seed treatment: Imidacloprid 70 WS 5 g/kg	Timely early sowing; higher seed rate; resistant varieties
Grain mold	Pink-black mouldy grains in humid weather at maturity	Mancozeb 2.5 g/L at flowering	Early-maturing varieties; harvest on time
Anthracnose and leaf blight	Red-brown spots and stripes on leaves	Mancozeb 2.5 g/L; Carbendazim 1 g/L	*P. fluorescens* 10 g/L; crop rotation; clean residue
Stem borer and fall armyworm	Dead hearts, shot holes, whorl feeding	Chlorantraniliprole 0.3 ml/L; Emamectin benzoate 0.4 g/L (armyworm)	*Trichogramma chilonis*; *Bt* 2 g/L; neem oil 5 ml/L
Downy mildew	White downy growth, stripes, shredded leaves	Seed treatment: Metalaxyl 6 g/kg	Resistant hybrids; remove infected plants

5. Pearl millet (Bajra)

Hindi / Telugu name	bājarā / sajja
Best season	Kharif (Jun-Jul)
Best soil	Light sandy and sandy loam; poor soils; drought tolerant
Soil pH	6.5-8.0
Main growing states	Rajasthan, UP, Haryana, Gujarat, Maharashtra
Fertilizer (N : P2O5 : K2O)	60-80 : 30-40 : 30 kg/ha
Nutrient tip	Responds well to nitrogen; zinc in sandy soils
Crop duration	75-90 days
Seed rate / spacing	Seed 4-5 kg/ha; 45 x 12 cm
Water need	Very low (250-400 mm)
Typical yield	2-3 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Downy mildew (green ear)	Yellow-green stripes, ears turn into leafy structures	Seed treatment: Metalaxyl (Apron 35 SD) 6 g/kg	Resistant hybrids; rogue out infected plants early
Ergot	Honey-like drops on ear, dark hard bodies in grains	Mancozeb 2.5 g/L at flowering	Float seed in 10% salt water to remove ergot bodies; avoid late sowing
Rust	Orange-brown pustules on leaves	Mancozeb 2.5 g/L	Resistant varieties
Earhead caterpillar	Larvae feeding on ear heads	Emamectin benzoate 0.4 g/L	neem oil 5 ml/L; *Bt* 2 g/L

6. Finger millet (Ragi)

Hindi / Telugu name	rāgī / maṃḍuā / rāgi
Best season	Kharif (Jun-Jul); Rabi in south
Best soil	Red loam and sandy loam; light well-drained soils
Soil pH	5.0-8.2
Main growing states	Karnataka, Tamil Nadu, AP, Uttarakhand
Fertilizer (N : P2O5 : K2O)	60 : 30 : 30 (rainfed 40 : 20 : 20) kg/ha
Nutrient tip	Few problems; add FYM 5-10 t/ha
Crop duration	100-120 days
Seed rate / spacing	Seed 8-10 kg/ha (line sowing); 22.5 x 10 cm
Water need	Low
Typical yield	2-3 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Blast	Spindle spots on leaves, neck and finger blast	Tricyclazole 0.6 g/L; Carbendazim 1 g/L	*P. fluorescens* seed treatment 10 g/kg; resistant varieties
Foot rot	Rotting at base, plants collapse	Drench Carbendazim 1 g/L	*Trichoderma* seed treatment 4 g/kg; avoid waterlogging
Mosaic and streak virus	Mottled, streaked leaves, stunting	No cure. Control leafhopper vector: Imidacloprid 0.3 ml/L	Rogue infected plants; resistant varieties
Stem borer and leaf folder	Dead hearts, folded leaves	Chlorantraniliprole 0.3 ml/L	*Trichogramma* cards; neem oil 5 ml/L

7. Barley

Hindi / Telugu name	jau / bārlī
Best season	Rabi (Oct-Nov)
Best soil	Light to medium loam; tolerates saline and alkaline soils better than wheat
Soil pH	6.5-8.5
Main growing states	Rajasthan, UP, MP, Haryana, Punjab
Fertilizer (N : P2O5 : K2O)	60-80 : 30-40 : 20 kg/ha
Nutrient tip	Tolerates moderate salinity
Crop duration	110-130 days
Seed rate / spacing	Seed 75-100 kg/ha; rows 22.5 cm
Water need	Low to moderate
Typical yield	3-4 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Yellow rust	Yellow stripes of powder on leaves	Propiconazole 1 ml/L	Resistant varieties; early sowing
Loose smut	Black powdery ear heads	Seed treatment: Carboxin 2.5 g/kg or Tebuconazole 1.5 g/kg	*Trichoderma* seed treatment 4 g/kg; hot-water seed treatment
Net blotch and spot blotch	Brown net-like or oval spots on leaves	Mancozeb 2.5 g/L	Certified seed; crop rotation
Aphids	Clusters on leaves and ears	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

8. Minor millets (Foxtail, Kodo, Little millet)

Hindi / Telugu name	kaṃganī / kodo / kuṭakī / korra / arikelu / sāmalu
Best season	Kharif
Best soil	Poor, shallow, light soils; red and lateritic
Soil pH	5.5-8.0
Main growing states	Karnataka, MP, Chhattisgarh, Tamil Nadu, AP
Fertilizer (N : P2O5 : K2O)	40 : 20 : 20 kg/ha
Nutrient tip	Grows on poor soils; add FYM
Crop duration	70-100 days
Seed rate / spacing	Seed 8-10 kg/ha; 25 x 10 cm
Water need	Very low
Typical yield	1-2 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Blast	Spots on leaves, neck rot	Carbendazim 1 g/L; Tricyclazole 0.6 g/L	*P. fluorescens* seed treatment 10 g/kg
Smut	Black powder replacing grains	Seed treatment: Carbendazim 2 g/kg	Hot-water seed treatment; certified seed
Shoot fly and stem borer	Dead hearts in young plants	Seed treatment: Imidacloprid 70 WS 5 g/kg; Chlorantraniliprole 0.3 ml/L	Early sowing; *Trichogramma* cards

9. Quinoa

Hindi / Telugu name	kvinoā / kvinovā
Best season	Rabi (Oct-Nov)
Best soil	Light well-drained sandy loam; tolerates poor and moderately saline soils
Soil pH	6.0-8.5
Main growing states	Rajasthan, UP, Uttarakhand, Karnataka
Fertilizer (N : P2O5 : K2O)	60 : 40 : 20 kg/ha
Nutrient tip	Avoid waterlogging; harvest before rain to prevent sprouting
Crop duration	90-120 days
Seed rate / spacing	Seed 8-10 kg/ha; 45 x 10 cm
Water need	Low
Typical yield	1.5-2.5 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Downy mildew	Yellow patches, grey growth under leaves	Metalaxyl plus Mancozeb 2.5 g/L	*P. fluorescens* 10 g/L; wide spacing; avoid late irrigation
Aphids	Clusters on tender shoots and heads	Imidacloprid 0.3 ml/L (observe waiting period)	neem oil 5 ml/L; *Beauveria* 5 g/L
Leaf miner and caterpillars	Mined leaves, chewed foliage	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L

10. Buckwheat (Kuttu)

Hindi / Telugu name	kuṭṭū / bakvīṭ
Best season	Kharif and Rabi in hills (Mar-Jun or Aug)
Best soil	Light well-drained acidic to neutral soils; grows on poor soils
Soil pH	5.0-7.0
Main growing states	Uttarakhand, Himachal, J&K;, Sikkim, Arunachal
Fertilizer (N : P2O5 : K2O)	40 : 20 : 20 kg/ha
Nutrient tip	Hardy; low-input crop for hills
Crop duration	80-100 days
Seed rate / spacing	Seed 40-60 kg/ha; rows 30 cm
Water need	Low
Typical yield	0.8-1.2 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot and blight	Brown spots on leaves	Mancozeb 2.5 g/L	Clean seed; crop rotation
Aphids	Colonies on tender shoots	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L
Hairy caterpillar	Defoliation	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; hand-pick

11. Amaranth grain (Rajgira)

Hindi / Telugu name	rājagirā / rājagirā
Best season	Kharif and Rabi
Best soil	Well-drained loam; tolerates poor soils
Soil pH	6.0-7.5
Main growing states	Uttarakhand, Himachal, MP, Maharashtra, Karnataka
Fertilizer (N : P2O5 : K2O)	40-60 : 30 : 20 kg/ha
Nutrient tip	Drought tolerant; thin to correct spacing
Crop duration	90-120 days
Seed rate / spacing	Seed 2-3 kg/ha; 45 x 15 cm
Water need	Low
Typical yield	1-2 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Damping off	Seedling collapse in nursery or field	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed; well-drained beds
Leaf spot and rust	Brown spots, orange pustules	Mancozeb 2.5 g/L	Remove infected leaves
Leaf webber and caterpillars	Rolled, webbed and chewed leaves	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L

12. Oats (fodder)

Hindi / Telugu name	jaī / oṭs
Best season	Rabi (Oct-Nov)
Best soil	Well-drained loam to clay loam
Soil pH	5.5-7.5
Main growing states	Punjab, Haryana, UP, Rajasthan
Fertilizer (N : P2O5 : K2O)	80-100 : 40 : 20 (cut-based nitrogen in splits) kg/ha
Nutrient tip	Good winter green fodder
Crop duration	60-90 days to cutting
Seed rate / spacing	Seed 75-100 kg/ha; rows 25 cm
Water need	Moderate
Typical yield	35-50 t/ha green fodder

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Rust and leaf blight	Orange pustules or brown spots on leaves	Mancozeb 2.5 g/L	Resistant varieties; early sowing
Smut	Black powdery grain heads	Seed treatment: Carboxin 2.5 g/kg	Clean seed; hot-water treatment
Aphids	Colonies on leaves	Imidacloprid 0.3 ml/L only if label allows for fodder	Neem oil 5 ml/L; avoid sprays close to feeding

13. Berseem (Egyptian clover)

Hindi / Telugu name	barasīm / barsīm
Best season	Rabi (Oct-Nov); 5-6 cuts
Best soil	Heavy loam and clay loam with good drainage; tolerates mild salinity
Soil pH	6.0-8.0
Main growing states	Punjab, Haryana, UP, Rajasthan, Gujarat
Fertilizer (N : P2O5 : K2O)	20 : 80 : 40 (phosphorus basal) kg/ha
Nutrient tip	Rhizobium culture; good winter green fodder
Crop duration	150-180 days (5-6 cuts)
Seed rate / spacing	Seed 25-30 kg/ha; broadcast
Water need	Moderate; irrigate after each cut
Typical yield	80-100 t/ha green fodder

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Stem rot (*Sclerotinia*)	White cottony rot at stem base	Drench Carbendazim 1 g/L	Crop rotation; *Trichoderma* seed treatment
Root rot	Wilting, rotted roots	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed; drainage
Aphids and weevils	Curled leaves, notched leaf edges	Use only if label allows for fodder	neem oil 5 ml/L; timely cutting

14. Lucerne (Alfalfa)

Hindi / Telugu name	rijakā / lūsarn / lūsarn
Best season	Sow Oct-Nov; perennial for 3-4 years
Best soil	Deep well-drained loam; avoids waterlogging and acid soils
Soil pH	6.5-8.0
Main growing states	Gujarat, Rajasthan, Punjab, Maharashtra, Karnataka
Fertilizer (N : P2O5 : K2O)	20 : 60 : 40 (phosphorus basal) kg/ha
Nutrient tip	Rhizobium culture; deep roots survive dry spells
Crop duration	3-4 years; cut every 25-30 days
Seed rate / spacing	Seed 15-20 kg/ha; rows 25-30 cm
Water need	Moderate
Typical yield	80-100 t/ha green fodder per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot and downy mildew	Brown spots, yellow patches	Mancozeb 2.5 g/L	Cut on time; clean harvest
Aphids (blue-green)	Curled tips, sticky leaves	Imidacloprid 0.3 ml/L if label allows for fodder	neem oil 5 ml/L; *Beauveria* 5 g/L; ladybirds help
Dodder (parasitic weed)	Yellow thread-like vine over plants	Use clean seed; remove infested patches	Clean seed; burn infested patches; shallow cut

15. Napier grass (Hybrid Napier)

Hindi / Telugu name	nepiyara ghās / nepiyar gaḍḍi
Best season	Plant Jun-Aug or Feb-Mar; perennial
Best soil	Fertile loam; wide soil range; no waterlogging
Soil pH	5.5-7.5
Main growing states	AP, Tamil Nadu, Karnataka, Maharashtra, Kerala, Punjab
Fertilizer (N : P2O5 : K2O)	150-200 : 50 : 40 per year in splits after cuts
Nutrient tip	Heavy nitrogen feeder; cut every 40-50 days
Crop duration	First cut at 60-70 days; later cuts every 40-50 days
Seed rate / spacing	About 40,000 two-node stem cuttings/ha; 90 x 60 cm
Water need	Moderate
Typical yield	150-250 t/ha green fodder per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf blight	Brown streaks and leaf drying	Mancozeb 2.5 g/L	Clean cutting tools; remove infected leaves
Stem borer	Dead hearts in shoots	Chlorantraniliprole 0.3 ml/L only if label allows	neem oil 5 ml/L; cut and destroy affected shoots
Termites	Wilting clumps, damaged roots	Follow label	Neem cake in soil; *Beauveria* 5 g/L

CASH, FIBRE AND GREEN MANURE CROPS

16. Cotton

Hindi / Telugu name	kapās / patti
Best season	Kharif (May-Jul sowing)
Best soil	Deep black cotton (regur) soils; also alluvial and red loams with drainage
Soil pH	6.0-8.0
Main growing states	Gujarat, Maharashtra, Telangana, AP, Karnataka, Punjab
Fertilizer (N : P2O5 : K2O)	120-150 : 60 : 60 kg/ha
Nutrient tip	Magnesium (red leaves in autumn) and boron (boll shedding); spray magnesium sulphate 10 g/L
Crop duration	150-180 days
Seed rate / spacing	Hybrid seed 1-2 kg/ha; 90-120 x 60 cm
Water need	Moderate (700-1000 mm)
Typical yield	2-3 t/ha seed cotton

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Whitefly	White flies under leaves, sticky honeydew, sooty mould	Diafenthiuron 50 WP: 1.2 g/L; Flonicamid 50 WG: 0.3 g/L	Neem oil 5 ml/L; *Lecanicillium* 5 g/L; yellow sticky traps
Aphids and jassids	Curled, reddish, burned leaf edges	Imidacloprid 17.8 SL: 0.3 ml/L; Thiamethoxam 25 WG: 0.2 g/L	NSKE 5%; *Beauveria* 5 g/L
Pink bollworm	Rosette flowers, damaged seeds in bolls	Emamectin benzoate 0.4 g/L; Spinosad 0.3 ml/L	Pheromone traps 5 per acre; timely sowing; clear stubble
American bollworm	Holes in squares and bolls	Chlorantraniliprole 0.3 ml/L; Emamectin 0.4 g/L	HaNPV 250 LE/ha; *Bt*; *Trichogramma*
Bacterial blight	Angular water-soaked spots, black arm on stem	Streptocycline 0.1 g/L plus copper oxychloride 3 g/L	Acid-delinted treated seed; resistant varieties
Root rot / wilt	Sudden drying of plants	Drench Carbendazim 1 g/L at plant base	*Trichoderma* 4 g/kg seed plus 2.5 kg/ha in FYM
Cotton leaf curl virus	Upward curled thick leaves, vein thickening	No cure. Control whitefly with Diafenthiuron 1.2 g/L	Resistant varieties; remove weeds; yellow sticky traps
Mealybug (*Phenacoccus*)	White cottony masses on stems and bolls	Profenofos or Buprofezin as per local advice	Release *Aenasius arizonensis* parasitoid; neem oil
Alternaria leaf spot	Brown spots with rings, leaf drop	Mancozeb 2.5 g/L	Balanced potash; remove infected leaves

17. Sugarcane

Hindi / Telugu name	gannā / ceraku
Best season	Feb-Mar spring planting; Jul-Sep autumn; Oct-Nov preseasonal
Best soil	Deep fertile loam and clay loam with good drainage and irrigation
Soil pH	6.5-7.5
Main growing states	UP, Maharashtra, Karnataka, Tamil Nadu, Gujarat
Fertilizer (N : P2O5 : K2O)	250 : 100 : 120 kg/ha
Nutrient tip	Iron chlorosis (yellow-white leaves); spray ferrous sulphate 5 g/L
Crop duration	10-14 months
Seed rate / spacing	About 35,000-40,000 three-bud setts/ha; rows 90-120 cm
Water need	High (1500-2500 mm)
Typical yield	80-120 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Red rot	Red patches with white cross bands inside stalk, drying top leaves	Dip setts in Carbendazim 1 g/L for 10 minutes; no cure once field is infected	Resistant varieties; healthy seed; *Trichoderma* in soil; burn infected clumps
Smut	Black whip-like shoot from the top	Dip setts in Propiconazole 1 ml/L	Hot-water treatment of setts (50 C, 2 hours); rogue out clumps

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Early shoot borer	Dead hearts in young shoots	Chlorantraniliprole 18.5 SC: 0.375 ml/L (drench)	*Trichogramma chilonis* 50,000/ha, 6-8 releases; earthing up
Internode borer	Bore holes and tunnels in stalks	Chlorantraniliprole as per label	*Trichogramma* releases; trash mulching
Woolly aphid	White woolly patches under leaves	Only if severe, as per label and local advice	Protect predators (*Dipha aphidivora*, *Micromus*); avoid broad-spectrum sprays
Grassy shoot disease	Many thin grass-like shoots, white leaves	No cure	Disease-free setts; hot-water treatment; rogue clumps
Pokkah boeng	Twisted, torn top leaves, rotted top	Carbendazim 1 g/L plus Copper oxychloride 3 g/L	Resistant varieties; avoid excess nitrogen
Whitefly (*Aleurolobus*) and pyrilla	Sooty mould, yellowing	Follow label	*Epiricania melanoleuca* parasitoid for pyrilla; neem oil

18. Jute

Hindi / Telugu name	jūṭ / paṭasan / janapanāra
Best season	Mar-May sowing (Kharif)
Best soil	Fertile alluvial loam near rivers; well-drained
Soil pH	6.0-7.5
Main growing states	West Bengal, Bihar, Assam, Odisha
Fertilizer (N : P2O5 : K2O)	60 : 30 : 30 kg/ha
Nutrient tip	Needs enough moisture at retting
Crop duration	120-150 days
Seed rate / spacing	Seed 5-6 kg/ha (line sowing)
Water need	High rainfall
Typical yield	2-3 t/ha fibre

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Stem rot	Dark lesions on stem near soil, wilting	Carbendazim 1 g/L drench	*Trichoderma* seed treatment 4 g/kg; crop rotation; drainage
Anthracnose	Sunken dark patches on stem and fibre damage	Mancozeb 2.5 g/L; Carbendazim 1 g/L	Clean seed; remove infected plants
Hairy caterpillar and semilooper	Leaf defoliation	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L; hand-pick egg masses
Yellow mite	Downward curling, bronzing of leaves	wettable sulphur 2.5 g/L	neem oil 5 ml/L

19. Tobacco

Hindi / Telugu name	taṃbākū / pogāku
Best season	Rabi (Oct-Nov planting) for FCV and country tobacco
Best soil	Light sandy loam; red and black soils depending on type; low chloride
Soil pH	5.5-7.5
Main growing states	AP, Karnataka, Gujarat, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	60-100 N, 40-60 P2O5, 60-100 K2O (type-dependent) kg/ha
Nutrient tip	Avoid chloride fertilizers (use sulphate of potash)
Crop duration	120-150 days after transplanting
Seed rate / spacing	About 20,000 plants/ha; 100 x 50 cm
Water need	Moderate
Typical yield	2-3 t/ha cured leaf

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Damping off (nursery)	Seedlings collapse at ground level	Drench Metalaxyl plus Mancozeb 2 g/L	*Trichoderma* 4 g/kg seed; raised nursery beds
Frogeye leaf spot	Round spots with light centre on leaves	Mancozeb 2.5 g/L	*P. fluorescens*; avoid dense planting

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Tobacco caterpillar (*Spodoptera*)	Large irregular holes in leaves	Emamectin benzoate 0.4 g/L	SlNPV 250 LE/ha; hand-pick egg masses; pheromone traps
Tobacco mosaic virus	Mottled, distorted leaves	No cure. Control aphids with Imidacloprid 0.3 ml/L	Remove infected plants; wash hands before handling
Black shank	Black stem base, sudden wilt	Drench Metalaxyl plus Mancozeb 2.5 g/L	Crop rotation; resistant varieties

20. Sugar beet

Hindi / Telugu name	cukaṃdara (śugara bīṭ) / ṣugar bīṭ
Best season	Rabi (Oct-Nov)
Best soil	Deep, fertile, well-drained loam; tolerates some salinity
Soil pH	6.5-8.0
Main growing states	Maharashtra, Punjab, Haryana, UP
Fertilizer (N : P2O5 : K2O)	120 : 60 : 60 kg/ha
Nutrient tip	Boron deficiency causes heart rot; borax 10 kg/ha
Crop duration	150-180 days
Seed rate / spacing	Seed 4-5 kg/ha; 45 x 20 cm
Water need	Moderate
Typical yield	40-60 t/ha roots

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Cercospora leaf spot	Small round spots with brown border	Mancozeb 2.5 g/L; Carbendazim 1 g/L	*P. fluorescens* 10 g/L; crop rotation
Root rot	Wilting, rotted roots and crown	Drench Carbendazim 1 g/L	*Trichoderma* 4 g/kg seed; good drainage
Leaf-eating caterpillar	Defoliation	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L
Aphids	Curled leaves, virus spread	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

21. Sunn hemp (Sanai)

Hindi / Telugu name	sanaī / san / janumu
Best season	Kharif (Apr-Jun) as green manure
Best soil	Almost any soil; tolerates poor and dry soils
Soil pH	6.0-8.0
Main growing states	All states
Fertilizer (N : P2O5 : K2O)	Not needed; fixes nitrogen
Nutrient tip	Plough in at 45-60 days before flowering to add nitrogen
Crop duration	45-60 days (green manure); 100-120 days (fibre)
Seed rate / spacing	Seed 25-30 kg/ha green manure; rows 25-30 cm
Water need	Low
Typical yield	8-10 t/ha green biomass

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Hairy caterpillar	Defoliation	Emamectin benzoate 0.4 g/L only if needed	*Bt* 2 g/L; hand-pick
Wilt and root rot	Yellowing and wilting	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed
Stem borer	Bored stems	Follow label	neem oil 5 ml/L

22. Dhaincha (Sesbania)

Hindi / Telugu name	ḍhaiṃcā / jīlugu
Best season	Kharif (Apr-Jul) as green manure
Best soil	Wide range; tolerates waterlogging, saline and alkaline soils
Soil pH	5.0-9.0
Main growing states	All states (rice-growing regions especially)
Fertilizer (N : P2O5 : K2O)	Not needed; fixes nitrogen
Nutrient tip	Excellent before rice; also helps reclaim alkaline soils
Crop duration	45-55 days
Seed rate / spacing	Seed 40-50 kg/ha; broadcast or rows 25 cm
Water need	Low to high (tolerates standing water)
Typical yield	10-15 t/ha green biomass

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Hairy caterpillar	Defoliation	Emamectin benzoate 0.4 g/L only if needed	*Bt* 2 g/L; hand-pick
Stem borer	Bored stems	Follow label	neem oil 5 ml/L
Root rot	Wilting in waterlogged patches	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed

VEGETABLES AND TUBER CROPS

23. Tomato

Hindi / Telugu name	ṭamāṭar / ṭamāṭā
Best season	Year-round with irrigation; Kharif, Rabi and summer plantings
Best soil	Well-drained sandy loam to red loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	AP, Karnataka, MP, Maharashtra, West Bengal
Fertilizer (N : P2O5 : K2O)	120-150 : 60-75 : 60-75 kg/ha
Nutrient tip	Calcium deficiency causes blossom end rot; spray calcium nitrate 5 g/L; boron for fruit set
Crop duration	60-90 days of harvest after transplanting (seedlings 25-30 days)
Seed rate / spacing	Seed 300-400 g/ha (hybrids 100-150 g); 60-90 x 45 cm
Water need	Moderate; drip is best
Typical yield	25-60 t/ha (hybrids more)

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Early blight	Concentric dark rings on older leaves	Mancozeb 2.5 g/L; Chlorothalonil 2 g/L; Azoxystrobin 1 ml/L	*Trichoderma* and *P. fluorescens* 5-10 g/L; *Bacillus subtilis*
Late blight	Large dark watery patches, white fuzz in humid weather	Metalaxyl plus Mancozeb 2.5 g/L; Cymoxanil plus Mancozeb 3 g/L	Prevention is key; copper oxychloride 3 g/L before disease; avoid overhead water
Leaf curl virus	Upward curling, yellow, stunted leaves	No cure for virus. Control whitefly: Imidacloprid 0.3 ml/L or Diafenthiuron 1.2 g/L	Yellow sticky traps 10-12 per acre; neem oil 5 ml/L; virus-free seedlings; remove infected plants
Fruit borer (*Helicoverpa*)	Round holes in fruit, caterpillar inside	Emamectin benzoate 0.4 g/L; Chlorantraniliprole 0.3 ml/L	HaNPV 250 LE/ha; *Bt* 2 g/L; pheromone traps 5 per acre; marigold border
Bacterial wilt	Sudden wilting of green plants, brown inside stem	No effective spray. Drench copper oxychloride 3 g/L as support	*P. fluorescens* soil application; rotate with cereals; resistant rootstock
Fusarium wilt	Yellowing from lower leaves, one-sided wilt	Drench Carbendazim 1 g/L	*Trichoderma* 2.5 kg/ha in FYM; raised beds
Tomato leaf miner (*Tuta absoluta*)	Mines in leaves, bored fruit	Chlorantraniliprole 0.3 ml/L; Spinosad 0.3 ml/L	Pheromone traps; neem oil 5 ml/L; remove infested leaves
Tomato spotted wilt virus	Bronzed leaves, ring spots on fruit	No cure. Control thrips: Fipronil 1.5 ml/L	Blue sticky traps; remove infected plants
Root knot nematode	Galls on roots, stunting, yellowing	Follow label	Neem cake 250 kg/ha; marigold border; *Paecilomyces lilacinus*
Septoria leaf spot	Many small circular spots with dark border	Chlorothalonil 2 g/L; Mancozeb 2.5 g/L	Crop rotation; remove lower leaves

24. Chilli

Hindi / Telugu name	mirc / mirapa
Best season	Kharif (Jun-Jul) and Rabi (Oct-Nov)
Best soil	Well-drained red loam and black soils; avoids waterlogging
Soil pH	6.0-7.5
Main growing states	AP, Telangana, Karnataka, Maharashtra, Odisha
Fertilizer (N : P2O5 : K2O)	120 : 60 : 60 kg/ha
Nutrient tip	Calcium and boron deficiency cause flower and fruit drop; zinc for leaf curling look
Crop duration	150-180 days
Seed rate / spacing	Seed 1-1.5 kg/ha; 60 x 45 cm
Water need	Moderate
Typical yield	Green 10-15 t/ha; dry 2-3 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Anthracnose / fruit rot / die-back	Sunken dark spots on fruit, twigs drying from tip	Difenoconazole 25 EC: 0.5 ml/L (7.5 ml per 15 L pump); Azoxystrobin 1 ml/L; copper oxychloride 3 g/L	*P. fluorescens* and *Bacillus subtilis*; clean seed; remove infected fruit

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew (*Leveillula taurica*)	White or yellow patches on leaf, powder underneath	Wettable sulphur 2.5 g/L; Hexaconazole 1 ml/L	*Ampelomyces quisqualis* product about 5 ml/L (for example 75 ml per 15 L pump)
Leaf curl / murda (thrips and mites)	Curling, shrunken leaves, stunted growth	Spiromesifen 240 SC: 1 ml/L; Fipronil 5 SC: 1.5-2 ml/L	Blue sticky traps; neem oil 5 ml/L; *Beauveria* 5 g/L; rotate chemicals
Yellow mites	Downward-curled leaves, yellowish shiny underside	Propargite 57 EC: 2 ml/L; Fenpyroximate 5 EC: 1 ml/L	Neem oil 5 ml/L; wettable sulphur 2 g/L carefully
Fruit borer	Holes in fruit, caterpillar inside	Emamectin benzoate 0.4 g/L	HaNPV; *Bt* 2 g/L; pheromone traps
Damping off (nursery)	Seedlings collapse at ground level	Drench Metalaxyl plus Mancozeb 2 g/L, or copper oxychloride 3 g/L	*Trichoderma* seed treatment 4-10 g/kg; solarized nursery beds
Bacterial leaf spot	Small water-soaked spots turning brown	Streptocycline 0.1 g/L plus Copper oxychloride 3 g/L	Clean seed; crop rotation
Cercospora leaf spot	Round spots with grey centre	Mancozeb 2.5 g/L	*P. fluorescens* 10 g/L
Wilt (Fusarium / Phytophthora)	Sudden wilting and root rot	Drench Carbendazim 1 g/L or Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma* 4 g/kg seed; raised beds; rotation

25. Potato

Hindi / Telugu name	ālū / baṃgāḷāduṃpa
Best season	Rabi (Oct-Nov) in plains; summer crop in hills
Best soil	Loose, well-drained sandy loam and loam; avoids heavy clay
Soil pH	5.0-6.5
Main growing states	UP, West Bengal, Bihar, Punjab, Gujarat
Fertilizer (N : P2O5 : K2O)	150-180 : 60-80 : 100 kg/ha
Nutrient tip	Needs high potash; magnesium and zinc in sandy soils
Crop duration	90-120 days
Seed rate / spacing	Seed tubers 2-3 t/ha; 60 x 20 cm
Water need	Moderate (500-600 mm)
Typical yield	20-30 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Late blight	Dark watery patches, white fuzz below leaf, fast field collapse	Preventive: Mancozeb 2.5 g/L; Cymoxanil plus Mancozeb 3 g/L; Metalaxyl-M plus Mancozeb 2.5 g/L	Copper oxychloride 3 g/L preventive; weather-based timing; resistant varieties
Early blight	Concentric ring spots on older leaves	Mancozeb 2.5 g/L; Chlorothalonil 2 g/L	*Trichoderma* and *Bacillus subtilis*
Black scurf / common scab	Black crust or rough patches on tubers	Seed tuber dip: boric acid 3% for 20 minutes	*Trichoderma* 10 g/kg tuber or soil use; crop rotation
Aphids (virus carriers)	Clusters on leaves, mosaic and leaf roll	Imidacloprid 0.3 ml/L	Neem oil 5 ml/L; certified seed; yellow sticky traps
Tuber moth	Tunnels in stored tubers	Follow label	*Bt* spray; keep tubers well earthed up
Bacterial wilt (brown rot)	Wilting, bacterial ooze from eyes	No effective spray	Healthy seed; rotation; avoid infested fields
Cutworm and white grub	Cut seedlings; holes in tubers	Follow label for soil application	*Metarhizium* in soil; deep ploughing
Common scab	Corky rough patches on tuber	Seed treatment with boric acid 3%	Keep soil moist at tuber initiation; rotation

26. Onion

Hindi / Telugu name	pyāj / ullipāya
Best season	Kharif, late Kharif and Rabi
Best soil	Well-drained loam and sandy loam rich in organic matter
Soil pH	6.0-7.5
Main growing states	Maharashtra, Karnataka, MP, Gujarat, Rajasthan
Fertilizer (N : P2O5 : K2O)	100 : 50 : 50 (plus sulphur 30 kg/ha)
Nutrient tip	Sulphur improves pungency and yield; avoid excess nitrogen late

Crop duration	100-150 days
Seed rate / spacing	Seed 8-10 kg/ha (nursery); 15 x 10 cm
Water need	Moderate
Typical yield	25-40 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Purple blotch	Purple-centred sunken spots on leaves	Mancozeb 2.5 g/L; Azoxystrobin 1 ml/L; Propiconazole 1 ml/L	*P. fluorescens* 10 g/L; avoid dense planting
Downy mildew	Pale patches with grey-purple fuzz in cool humid weather	Metalaxyl plus Mancozeb 2.5 g/L	Remove infected leaves; proper spacing
Basal rot / Fusarium	Yellowing, bulb rots at base	Drench Carbendazim 1 g/L	*Trichoderma* 5 g/L drench; crop rotation
Thrips	Silvery streaks, curled leaf tips	Fipronil 1.5 ml/L; Spinosad 0.3 ml/L	Blue sticky traps; neem oil 5 ml/L; *Beauveria* 5 g/L
Stemphylium blight	Small yellow-orange flecks turning to dark patches	Mancozeb 2.5 g/L; Propiconazole 1 ml/L	Avoid dense planting; good drainage
Soft rot (bacterial)	Foul-smelling rot in bulbs	Copper oxychloride 3 g/L as preventive	Proper curing; avoid injury and excess irrigation near harvest

27. Brinjal (Eggplant)

Hindi / Telugu name	baiṃgan / vaṃkāya
Best season	Year-round with irrigation
Best soil	Fertile well-drained loam to clay loam
Soil pH	5.5-6.5
Main growing states	West Bengal, Odisha, Gujarat, AP, Karnataka
Fertilizer (N : P2O5 : K2O)	100-150 : 50 : 50 kg/ha
Nutrient tip	Calcium and boron for fruit quality; avoid waterlogging
Crop duration	150-180 days
Seed rate / spacing	Seed 400-500 g/ha; 75 x 60 cm
Water need	Moderate
Typical yield	30-50 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Shoot and fruit borer	Wilting shoot tips, bore holes in fruit with frass	Emamectin benzoate 0.4 g/L; Chlorantraniliprole 0.3 ml/L	Cut and destroy affected shoots; pheromone traps 5-8 per acre; *Bt* 2 g/L; neem oil 5 ml/L
Little leaf (phytoplasma)	Tiny narrow leaves, bushy stunted plants, no fruit	No cure. Control leafhopper vector: Imidacloprid 0.3 ml/L	Uproot and destroy infected plants; yellow sticky traps; resistant varieties
Bacterial wilt	Sudden wilting of green plants	No effective spray. Drench copper oxychloride 3 g/L as support	*P. fluorescens* soil use; rotate with cereals; resistant rootstock
Phomopsis blight / fruit rot	Brown leaf spots, sunken rotting fruit	Mancozeb 2.5 g/L; Carbendazim 1 g/L	*Trichoderma* seed treatment 4-10 g/kg; clean seed; remove infected fruit
Whitefly, jassids, spider mites	Yellowing, curling, webbing, sticky leaves	Diafenthiuron 1.2 g/L; Spiromesifen 1 ml/L (mites)	Neem oil 5 ml/L; yellow sticky traps; *Beauveria* 5 g/L
Root knot nematode	Galls on roots, stunting	Follow label	Neem cake 250 kg/ha; marigold; resistant rootstock
Epilachna beetle	Skeletonised leaves	Follow label	Neem oil 5 ml/L; hand-pick grubs

28. Okra (Bhendi / Lady's finger)

Hindi / Telugu name	bhiṃḍī / beṃḍa
Best season	Kharif (Jun-Jul) and Zaid (Feb-Mar)
Best soil	Well-drained sandy loam to clay loam
Soil pH	6.0-6.8
Main growing states	UP, Bihar, West Bengal, Gujarat, AP
Fertilizer (N : P2O5 : K2O)	100 : 50 : 50 kg/ha
Nutrient tip	Responds to nitrogen in splits; zinc and boron in poor soils
Crop duration	90-120 days
Seed rate / spacing	Seed 8-10 kg/ha; 60 x 30 cm
Water need	Moderate
Typical yield	10-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Yellow vein mosaic virus	Yellow network of veins, small hard yellow fruits	No cure. Control whitefly: Imidacloprid 0.3 ml/L; Diafenthiuron 1.2 g/L	Resistant or tolerant varieties; yellow sticky traps; remove infected plants early
Powdery mildew	White powder on leaves, leaf drying	Wettable sulphur 2.5 g/L; Hexaconazole 1 ml/L	*Bacillus subtilis* 2-5 g/L
Shoot and fruit borer	Drooping shoot tips, bored fruit	Emamectin benzoate 0.4 g/L; Spinosad 0.3 ml/L	Pheromone traps; *Bt* 2 g/L; remove affected shoots and fruit
Jassids and aphids	Curled, yellowing, burnt leaf edges	Imidacloprid 0.3 ml/L	Neem oil 5 ml/L; *Beauveria* 5 g/L

29. Cabbage and Cauliflower

Hindi / Telugu name	pattā gobhī / phūla gobhī / kyābejī / kālīphlavar
Best season	Rabi mostly (Aug-Nov planting); early varieties in monsoon
Best soil	Fertile loam rich in organic matter; good moisture
Soil pH	6.0-7.0
Main growing states	West Bengal, Odisha, Bihar, Haryana, UP
Fertilizer (N : P2O5 : K2O)	120-150 : 60-75 : 60 kg/ha
Nutrient tip	Boron (hollow stem, brown curd) and molybdenum (whiptail); borax 10-15 kg/ha
Crop duration	90-120 days
Seed rate / spacing	Seed 400-500 g/ha; 45 x 45 cm
Water need	Moderate
Typical yield	25-40 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Diamondback moth	Small holes and windowing in leaves	Spinosad 0.3 ml/L; Emamectin benzoate 0.4 g/L; Chlorantraniliprole 0.3 ml/L	*Bt* 2 g/L; mustard trap crop (2 rows per 25 rows); pheromone traps; rotate chemicals
Black rot	V-shaped yellow patches from leaf edge, black veins	Streptocycline 0.1 g/L plus copper oxychloride 3 g/L	Hot-water seed treatment (50 C, 30 min); crop rotation; clean seed
Clubroot	Swollen, club-shaped roots, wilting in heat	No effective spray; soil treatment as per local advice	Liming to raise soil pH; *Trichoderma* in soil; rotate for 3-4 years
Downy mildew	Yellow patches, white fuzz under leaf	Metalaxyl plus Mancozeb 2.5 g/L	Wider spacing; avoid overhead irrigation
Aphids	Clusters on leaves, curling, honeydew	Imidacloprid 0.3 ml/L	Neem oil 5 ml/L; *Beauveria* 5 g/L; yellow sticky traps
Cabbage head borer and semilooper	Holes in leaves and heads	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; pheromone traps
Alternaria leaf spot	Concentric spots on outer leaves	Mancozeb 2.5 g/L	Clean seed; rotation

30. Cucurbits (Cucumber, Bitter / Bottle / Ridge gourd, Pumpkin, Melon)

Hindi / Telugu name	kaddūvargīya sabjiyāṃ / dosajāti kūragāyalu

Best season	Zaid (Feb-Mar), Kharif (Jun-Jul)
Best soil	Well-drained sandy loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	UP, Bihar, West Bengal, Gujarat, Karnataka
Fertilizer (N : P2O5 : K2O)	80-100 : 50 : 50 kg/ha
Nutrient tip	Magnesium and boron in poor soils; mulch to conserve moisture
Crop duration	90-120 days
Seed rate / spacing	Seed 2-5 kg/ha (crop dependent); rows 1.5-2.5 m
Water need	Moderate
Typical yield	15-30 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves, drying	Hexaconazole 1 ml/L; wettable sulphur 2-2.5 g/L (test on a few plants first; sulphur can scorch some cucurbits)	*Bacillus subtilis* 2-5 g/L; *Ampelomyces* product as per label
Downy mildew	Angular yellow patches, grey-purple fuzz underneath	Metalaxyl plus Mancozeb 2.5 g/L; Cymoxanil plus Mancozeb 3 g/L	Wider spacing; avoid overhead water; resistant varieties
Fruit fly	Maggots in fruit, fruit rots and drops	Follow label; avoid late sprays near harvest	Cue-lure traps 8-10 per acre; collect and bury infested fruit; bagging of fruit
Red pumpkin beetle	Holes in young leaves and flowers	Imidacloprid 0.3 ml/L as per label	Neem oil 5 ml/L; hand-pick beetles; early morning dusting of ash
Mosaic virus	Mottled, curled leaves, small fruits	No cure. Control aphids and whitefly: Imidacloprid 0.3 ml/L	Yellow sticky traps; virus-free seed; remove infected plants

31. Capsicum (Bell pepper)

Hindi / Telugu name	śimalā mirc / kyāpsikam
Best season	Rabi and polyhouse year-round; hills in summer
Best soil	Well-drained loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	Karnataka, Himachal, Maharashtra, AP
Fertilizer (N : P2O5 : K2O)	120-150 : 60 : 60 kg/ha
Nutrient tip	Calcium for fruit quality; avoid water stress
Crop duration	120-150 days
Seed rate / spacing	Seed 200-300 g/ha; 60 x 45 cm
Water need	Moderate
Typical yield	20-30 t/ha open field (polyhouse 80+)

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves, drying	Hexaconazole 1 ml/L; wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L; *Ampelomyces* product as per label
Phytophthora blight	Water-soaked stem and fruit rot, wilt	Metalaxyl plus Mancozeb 2.5 g/L drench	*Trichoderma*; raised beds; good drainage
Bacterial spot	Small dark spots on leaves and fruit	Streptocycline 0.1 g/L plus copper oxychloride 3 g/L	Clean seed; crop rotation
Thrips and mites	Silvery leaves, curling, bronze fruit	Spiromesifen 1 ml/L; Fipronil 1.5 ml/L	Blue sticky traps; neem oil 5 ml/L; *Beauveria* 5 g/L
Fruit borer	Holes in fruit with caterpillar	Emamectin benzoate 0.4 g/L	HaNPV; *Bt* 2 g/L; pheromone traps

32. French bean and Dolichos (Field) bean

Hindi / Telugu name	phreṃca bīn / sem / phreṃc bīns / cikkuḍu
Best season	Kharif and Rabi
Best soil	Well-drained loam; avoids waterlogging and salinity
Soil pH	6.0-7.5
Main growing states	Karnataka, Maharashtra, Himachal, UP
Fertilizer (N : P2O5 : K2O)	25 : 60 : 50 kg/ha
Nutrient tip	Starter nitrogen only; Rhizobium helps; molybdenum in acid soils
Crop duration	60-90 days
Seed rate / spacing	Seed 60-80 kg/ha (bush); 30 x 10 cm
Water need	Moderate
Typical yield	8-12 t/ha green pods

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Rust	Reddish-brown pustules under leaves	Hexaconazole 1 ml/L; Mancozeb 2.5 g/L	Resistant varieties; crop rotation
Anthracnose	Dark sunken spots on pods and stems	Carbendazim 1 g/L; Mancozeb 2.5 g/L	Clean seed; *Trichoderma* seed treatment 4 g/kg
Aphids and bean fly	Curling, stunted plants, virus spread	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; yellow sticky traps
Pod borer	Bored pods and seeds	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; HaNPV; pheromone traps

33. Garden pea

Hindi / Telugu name	maṭar / baṭhānī
Best season	Rabi (Oct-Nov)
Best soil	Well-drained loam; avoids acidic and waterlogged soils
Soil pH	6.0-7.5
Main growing states	UP, MP, Bihar, Punjab, Himachal
Fertilizer (N : P2O5 : K2O)	20-25 : 50-60 : 40-50 kg/ha
Nutrient tip	Rhizobium culture; sulphur and molybdenum
Crop duration	60-100 days
Seed rate / spacing	Seed 80-100 kg/ha; 30 x 7 cm
Water need	Low to moderate
Typical yield	8-10 t/ha green pods

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves and pods	wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L; resistant varieties
Rust	Orange-brown pustules	Mancozeb 2.5 g/L	Resistant varieties; crop rotation
Leaf miner	Winding white tunnels in leaves	Thiamethoxam 0.2 g/L	neem oil 5 ml/L; remove mined leaves
Wilt and root rot	Yellowing, wilting, rotted roots	Drench Carbendazim 1 g/L	*Trichoderma* seed treatment 4 g/kg; avoid waterlogging

34. Carrot

Hindi / Telugu name	gājar / kyāreṭ
Best season	Rabi (Aug-Nov); hills in summer
Best soil	Deep loose sandy loam; no stones or compact layers
Soil pH	6.0-7.0
Main growing states	Punjab, Haryana, UP, Karnataka
Fertilizer (N : P2O5 : K2O)	80 : 50 : 50 kg/ha
Nutrient tip	Boron for root quality; avoid fresh FYM before sowing
Crop duration	90-120 days
Seed rate / spacing	Seed 5-6 kg/ha; 30 x 6 cm
Water need	Moderate
Typical yield	20-30 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Alternaria leaf blight	Dark brown spots with yellow edge on leaves	Mancozeb 2.5 g/L	*P. fluorescens*; clean seed; crop rotation
Powdery mildew	White powder on leaves	wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L
Root knot nematode	Galls and forking of roots	Follow label and local advice	Neem cake 250 kg/ha in soil; marigold as trap crop; solarization; *Paecilomyces lilacinus*
Aphids	Curling, sticky leaves	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; yellow sticky traps

35. Radish

Hindi / Telugu name	mūlī / mullaṃgi
Best season	Rabi mainly; can be grown most of the year
Best soil	Loose sandy loam
Soil pH	6.0-7.0
Main growing states	Punjab, UP, Bihar, West Bengal
Fertilizer (N : P2O5 : K2O)	50 : 40 : 40 kg/ha
Nutrient tip	Boron deficiency causes hollow roots
Crop duration	30-50 days
Seed rate / spacing	Seed 8-10 kg/ha; 30 x 7.5 cm
Water need	Moderate
Typical yield	15-25 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
White rust	White blisters under leaves	Metalaxyl plus Mancozeb 2.5 g/L	Crop rotation; remove infected leaves
Aphids	Colonies on leaves and stalks	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L
Flea beetle and mustard sawfly	Tiny holes and defoliation	Follow label and local advice	neem oil 5 ml/L; collect larvae; early sowing

36. Beetroot

Hindi / Telugu name	cukaṃdar / bīṭrūṭ
Best season	Rabi (Sep-Nov)
Best soil	Deep loam to sandy loam
Soil pH	6.0-7.5
Main growing states	Karnataka, Tamil Nadu, Maharashtra
Fertilizer (N : P2O5 : K2O)	80 : 60 : 60 kg/ha
Nutrient tip	Boron deficiency causes heart rot
Crop duration	80-100 days
Seed rate / spacing	Seed 6-8 kg/ha; 30 x 10 cm
Water need	Moderate
Typical yield	20-30 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Cercospora leaf spot	Small round spots with reddish border	Mancozeb 2.5 g/L; Carbendazim 1 g/L	*P. fluorescens* 10 g/L; crop rotation
Root rot	Rot at crown and root	Drench Carbendazim 1 g/L	*Trichoderma* seed treatment; good drainage
Aphids and leaf miner	Curled, mined leaves	Imidacloprid 0.3 ml/L	neem oil 5 ml/L

37. Leafy vegetables (Spinach, Amaranth, Fenugreek)

Hindi / Telugu name	pattedāra sabjiyāṃ (pālaka, caulāī, methī) / ākukūralu
Best season	All seasons with irrigation
Best soil	Fertile loam rich in organic matter
Soil pH	6.0-7.5
Main growing states	All states
Fertilizer (N : P2O5 : K2O)	60-80 : 40 : 40 (nitrogen in splits after each cut) kg/ha
Nutrient tip	Nitrogen drives leaf growth; iron and magnesium for colour
Crop duration	25-45 days to first cut
Seed rate / spacing	Seed 8-30 kg/ha (spinach highest); rows 15-20 cm
Water need	Moderate, frequent light irrigation
Typical yield	10-20 t/ha green

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot (Cercospora)	Round spots with light centres	Mancozeb 2.5 g/L only with the waiting period on the label	*P. fluorescens* 10 g/L; remove infected leaves
Downy mildew	Yellow patches, grey fuzz below leaf	Mancozeb 2.5 g/L only if label allows	Wider spacing; avoid overhead irrigation
Leaf webber (amaranth)	Rolled and webbed leaves with larvae	Follow label and local advice	*Bt* 2 g/L; neem oil 5 ml/L; hand removal
Aphids	Clusters on tender leaves	Avoid chemicals where possible	neem oil 5 ml/L; *Beauveria* 5 g/L; wash leaves before eating

38. Drumstick (Moringa)

Hindi / Telugu name	sahajan / munaga
Best season	Jun-Jul or Feb-Mar planting
Best soil	Light well-drained sandy loam or red soils; drought tolerant
Soil pH	6.0-7.5
Main growing states	Tamil Nadu, AP, Karnataka, Maharashtra
Fertilizer (N : P2O5 : K2O)	50 g N, 50 g P2O5, 50 g K2O per tree plus FYM 10 kg
Nutrient tip	Avoid waterlogging
Crop duration	6-8 months to first harvest
Seed rate / spacing	Spacing 2.5 x 2.5 m
Water need	Low to moderate
Typical yield	20-50 t/ha pods

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Hairy caterpillar	Complete defoliation of branches	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L; burn egg masses and larvae
Fruit fly and pod fly	Maggots in pods, pods drop	Follow label and local advice	Methyl eugenol / cue-lure traps; collect fallen pods
Die-back and canker	Drying of branch tips, cankers on stem	copper oxychloride 3 g/L; pruning of infected parts	Pruning in dry weather; paste cut ends
Bud worm	Damaged flower buds	Spinosad 0.3 ml/L as per label	neem oil 5 ml/L

39. Sweet potato

Hindi / Telugu name	śakarakaṃd / cilagaḍaduṃpa
Best season	Kharif, Rabi (south)
Best soil	Well-drained sandy loam; light soils
Soil pH	5.5-6.5
Main growing states	Odisha, UP, Bihar, West Bengal, Gujarat
Fertilizer (N : P2O5 : K2O)	50 : 25 : 50 kg/ha
Nutrient tip	Potash improves tuber size; avoid excess nitrogen
Crop duration	100-120 days
Seed rate / spacing	About 40,000 vine cuttings/ha; 60 x 20 cm
Water need	Low to moderate
Typical yield	15-25 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Sweet potato weevil	Tunnels and bad taste in tubers	Follow label	Pheromone traps; *Beauveria* 5 g/L soil application; clean vines; earthing up
Stem and tuber rot	Black rot at vine base, tuber rot	Dip vines in Carbendazim 1 g/L for 10 minutes	*Trichoderma*; healthy vines; rotation
Leaf spot	Brown spots on leaves	Mancozeb 2.5 g/L	Remove infected leaves
Virus (feathery mottle)	Mottled, small leaves	No cure	Virus-free vines; control whitefly with neem oil 5 ml/L

40. Cassava (Tapioca)

Hindi / Telugu name	kasāvā / ṭaipiokā / karrapeṃḍalaṃ / sarvara
Best season	Mar-May and Sep-Oct planting
Best soil	Light, loose, well-drained red loam and laterite
Soil pH	5.5-7.0
Main growing states	Tamil Nadu, Kerala, AP
Fertilizer (N : P2O5 : K2O)	100 : 50 : 100 kg/ha
Nutrient tip	Potash is important for starch
Crop duration	8-10 months
Seed rate / spacing	About 12,500 setts/ha; 90 x 90 cm
Water need	Low to moderate
Typical yield	25-40 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Cassava mosaic disease	Yellow-green mosaic, distorted small leaves	No cure. Control whitefly with Imidacloprid 0.3 ml/L	Healthy setts; resistant varieties; rogue infected plants
Root and tuber rot	Rotting of roots, wilting	Drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma* in soil; good drainage
Mealybug	White cottony masses on shoot tips	Follow label	Parasitoid *Acerophagus papayae* via KVK; neem oil 5 ml/L
Spider mite	Yellow speckling, webbing in dry weather	Spiromesifen 1 ml/L	Neem oil 5 ml/L; water sprays

41. Garlic

Hindi / Telugu name	lahasun / vellulli
Best season	Rabi (Oct-Nov)
Best soil	Well-drained loam rich in organic matter
Soil pH	6.0-7.5
Main growing states	MP, Gujarat, Rajasthan, UP
Fertilizer (N : P2O5 : K2O)	100 : 50 : 50 (plus sulphur 30 kg/ha)
Nutrient tip	Sulphur improves pungency
Crop duration	120-150 days
Seed rate / spacing	Cloves 500-700 kg/ha; 15 x 7.5 cm
Water need	Moderate
Typical yield	8-12 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Purple blotch	Purple-centred spots on leaves	Mancozeb 2.5 g/L; Azoxystrobin 1 ml/L	*P. fluorescens* 10 g/L; avoid dense planting
Rust	Orange pustules on leaves	Mancozeb 2.5 g/L; Propiconazole 1 ml/L	Crop rotation; remove infected leaves
Basal rot	Yellowing, rotted bulb base	Drench Carbendazim 1 g/L	*Trichoderma* 5 g/L drench; seed clove treatment
Thrips	Silvery streaks, curled tips	Spinosad 0.3 ml/L; Fipronil 1.5 ml/L	Blue sticky traps; neem oil 5 ml/L

42. Colocasia (Arbi)

Hindi / Telugu name	arabī / cāmaduṃpa
Best season	Feb-Mar and Jun-Jul planting
Best soil	Fertile well-drained loam with good moisture
Soil pH	5.5-7.0
Main growing states	UP, Bihar, West Bengal, AP
Fertilizer (N : P2O5 : K2O)	80 : 40 : 80 kg/ha
Nutrient tip	Needs consistent moisture
Crop duration	150-180 days
Seed rate / spacing	Cormels 1.5-2 t/ha; 45 x 30 cm
Water need	Moderate to high
Typical yield	15-25 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf blight (*Phytophthora*)	Large water-soaked spots, leaf collapse in humid weather	Metalaxyl plus Mancozeb 2.5 g/L	Wider spacing; resistant varieties; remove infected leaves
Tobacco caterpillar (*Spodoptera*)	Holes, skeletonised leaves	Emamectin benzoate 0.4 g/L	SlNPV 250 LE/ha; hand-pick egg masses
Corm rot	Rotting of corms in field and storage	Corm dip Mancozeb 2.5 g/L	*Trichoderma* in soil; well-drained soil; healthy corms

43. Cluster bean (Guar)

Hindi / Telugu name	gvār / gorucikkuḍu
Best season	Kharif and Zaid
Best soil	Sandy loam to light soils; drought tolerant
Soil pH	7.0-8.5
Main growing states	Rajasthan, Haryana, Gujarat
Fertilizer (N : P2O5 : K2O)	20 : 40 : 20 kg/ha
Nutrient tip	Rhizobium culture; low nitrogen need
Crop duration	90-120 days
Seed rate / spacing	Seed 15-20 kg/ha; 45 x 15 cm
Water need	Low
Typical yield	1-1.5 t/ha seed; 8-10 t/ha green pods

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Alternaria leaf blight	Brown spots with rings on leaves	Mancozeb 2.5 g/L	*P. fluorescens*; crop rotation
Bacterial blight	Angular water-soaked spots on leaves and pods	Streptocycline 0.1 g/L plus copper oxychloride 3 g/L	Clean seed; avoid overhead water
Powdery mildew	White powder on leaves	wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L
Aphids and jassids	Curling and yellowing	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

44. Sweet corn and Baby corn

Hindi / Telugu name	svīṭa ka□rn / svīṭ kārn
Best season	Year-round with irrigation
Best soil	Fertile well-drained loam
Soil pH	6.0-7.0
Main growing states	Punjab, Karnataka, Maharashtra, UP
Fertilizer (N : P2O5 : K2O)	120 : 60 : 40 kg/ha
Nutrient tip	Zinc deficiency (white bud)
Crop duration	70-90 days
Seed rate / spacing	Seed 8-10 kg/ha; 60 x 20 cm
Water need	Moderate
Typical yield	8-10 t/ha green cobs (sweet corn)

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fall armyworm	Whorl and cob damage	Emamectin benzoate 0.4 g/L; Spinetoram 0.5 ml/L (observe waiting period)	*Bt* 2 g/L; neem oil 5 ml/L; *Metarhizium rileyi*; pheromone traps
Turcicum leaf blight	Long grey-brown lesions	Mancozeb 2.5 g/L	*P. fluorescens* 10 g/L; resistant hybrids
Downy mildew	Pale stripes, white fuzz	Seed treatment: Metalaxyl 2.4 g/kg	Remove infected plants; early sowing
Stem borer	Dead heart, shot holes	Chlorantraniliprole 0.3 ml/L	*Trichogramma chilonis* 1 lakh/ha

45. Elephant foot yam

Hindi / Telugu name	jimīkaṃd / sūran / kaṃda
Best season	Feb-Apr planting
Best soil	Fertile sandy loam, rich in organic matter
Soil pH	5.5-7.0
Main growing states	AP, Kerala, West Bengal, UP
Fertilizer (N : P2O5 : K2O)	100 : 60 : 100 kg/ha
Nutrient tip	Potash for corm size; mulch
Crop duration	7-8 months
Seed rate / spacing	Corm pieces about 500 g; 75 x 75 cm
Water need	Moderate
Typical yield	30-40 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Collar rot (*Sclerotium*)	White growth and rot at collar, wilting	Drench Carbendazim 1 g/L; corm dip Mancozeb 2.5 g/L	*Trichoderma* 4 g/kg corm plus soil use; healthy corms; rotation
Leaf blight	Water-soaked spots turning brown	Metalaxyl plus Mancozeb 2.5 g/L	Wider spacing; drainage
Mosaic virus	Mottled, distorted leaves	No cure	Virus-free corms; rogue plants; control aphids with neem oil 5 ml/L
Tobacco caterpillar	Defoliation	Emamectin benzoate 0.4 g/L	SlNPV; hand-picking

46. Pointed gourd (Parwal)

Hindi / Telugu name	paraval / parval
Best season	Plant Jun-Jul or Oct-Nov (perennial vine)
Best soil	Sandy loam near rivers; well-drained
Soil pH	6.0-7.0
Main growing states	Bihar, UP, West Bengal, Assam
Fertilizer (N : P2O5 : K2O)	80 : 60 : 60 kg/ha
Nutrient tip	Needs good drainage; mulch
Crop duration	Perennial; harvest from 3-4 months
Seed rate / spacing	About 3,000-3,500 vine cuttings/ha; 2 x 1.5 m
Water need	Moderate
Typical yield	10-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Vine and stem rot	Rotting of vine bases, sudden wilting	Drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma*; well-drained soil
Powdery and downy mildew	White powder or yellow patches on leaves	Hexaconazole 1 ml/L; Metalaxyl plus Mancozeb 2.5 g/L	Wide spacing; remove infected leaves
Fruit fly	Maggots in fruits	Follow label	Cue-lure traps 8-10 per acre; collect infested fruit
Red pumpkin beetle	Holes in leaves and flowers	Follow label and local advice	neem oil 5 ml/L; hand-pick beetles

47. Bottle gourd

Hindi / Telugu name	laukī / ghīyā / sorakāya
Best season	Zaid and Kharif
Best soil	Well-drained sandy loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	UP, Bihar, West Bengal, Gujarat, Karnataka
Fertilizer (N : P2O5 : K2O)	80-100 : 50 : 50 kg/ha
Nutrient tip	Boron and magnesium in poor soils
Crop duration	100-120 days
Seed rate / spacing	Seed 3-4 kg/ha; 2.5 x 1 m on trellis
Water need	Moderate
Typical yield	20-30 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Downy mildew	Angular yellow patches, grey-purple fuzz below	Metalaxyl plus Mancozeb 2.5 g/L	Wide spacing; avoid overhead water
Powdery mildew	White powder on leaves	Hexaconazole 1 ml/L; wettable sulphur 2 g/L (test on a few plants first)	*Bacillus subtilis* 2-5 g/L
Fruit fly	Maggots, fruit rot and drop	Follow label	Cue-lure traps 8-10 per acre; collect and bury fruit
Red pumpkin beetle	Holes in young leaves	Follow label	neem oil 5 ml/L; hand-pick

48. Bitter gourd

Hindi / Telugu name	karelā / kākarakāya
Best season	Zaid and Kharif
Best soil	Well-drained sandy loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	UP, Karnataka, Tamil Nadu, Kerala, West Bengal
Fertilizer (N : P2O5 : K2O)	60-80 : 50 : 50 kg/ha
Nutrient tip	Boron and magnesium for fruit quality
Crop duration	100-120 days
Seed rate / spacing	Seed 4-5 kg/ha; 2 x 0.6 m on trellis
Water need	Moderate
Typical yield	10-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves	Hexaconazole 1 ml/L	*Bacillus subtilis* 2-5 g/L
Downy mildew	Yellow patches, grey fuzz underneath	Metalaxyl plus Mancozeb 2.5 g/L	Wide spacing; trellis cultivation
Fruit fly	Maggots in fruit	Follow label	Cue-lure traps; bagging of fruit; collect infested fruit
Mosaic virus (aphid and whitefly spread)	Mottled, curled leaves	No cure. Control vectors with Imidacloprid 0.3 ml/L	yellow sticky traps; remove infected plants

49. Cucumber

Hindi / Telugu name	khīrā / dosakāya
Best season	Zaid, Kharif and polyhouse year-round
Best soil	Well-drained sandy loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	Karnataka, UP, Punjab, Maharashtra, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	80-100 : 60 : 60 kg/ha
Nutrient tip	Magnesium and boron in poor soils
Crop duration	60-90 days
Seed rate / spacing	Seed 2-3 kg/ha; 1.5 x 0.6 m
Water need	Moderate
Typical yield	15-25 t/ha (open field); polyhouse more

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Downy mildew	Yellow angular patches, grey fuzz below	Metalaxyl plus Mancozeb 2.5 g/L	Resistant hybrids; avoid overhead water
Powdery mildew	White powder on leaves	Hexaconazole 1 ml/L	*Bacillus subtilis* 2-5 g/L
Cucumber mosaic virus	Mottled, small, curled leaves	No cure. Control aphids with Imidacloprid 0.3 ml/L	neem oil 5 ml/L; remove infected plants; resistant varieties
Fruit fly and red beetle	Maggots in fruit; holes in leaves	Follow label	Cue-lure traps; hand-picking; bagging

50. Broccoli

Hindi / Telugu name	brokalī / brokalī
Best season	Rabi (Sep-Nov); hills in summer
Best soil	Fertile well-drained loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	Himachal, Karnataka, Punjab, Maharashtra
Fertilizer (N : P2O5 : K2O)	120 : 60 : 60 kg/ha
Nutrient tip	Boron and molybdenum deficiency
Crop duration	90-120 days
Seed rate / spacing	Seed 400-500 g/ha; 45 x 45 cm
Water need	Moderate
Typical yield	12-20 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Diamondback moth	Holes and windowing in leaves	Spinosad 0.3 ml/L; Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; mustard trap crop; pheromone traps
Black rot	V-shaped yellow patches from edge	Streptocycline 0.1 g/L plus copper oxychloride 3 g/L	Hot-water seed treatment (50 C, 30 min); rotation
Downy mildew	Yellow patches, white fuzz below	Metalaxyl plus Mancozeb 2.5 g/L	Wide spacing; avoid overhead water
Aphids	Clusters in heads and leaves	Imidacloprid 0.3 ml/L (observe waiting period)	neem oil 5 ml/L; *Beauveria* 5 g/L

51. Ivy gourd (Tindora / Dondakaya)

Hindi / Telugu name	kuṃdarū / toṃḍalī / doṃḍakāya
Best season	Plant Jun-Jul or Feb-Mar (perennial vine)
Best soil	Well-drained sandy loam; thrives in warm humid climate
Soil pH	6.0-7.0
Main growing states	Karnataka, AP, Telangana, Maharashtra, West Bengal
Fertilizer (N : P2O5 : K2O)	FYM plus about 60 : 40 : 40
Nutrient tip	Easy perennial crop; give support
Crop duration	Perennial; harvest from 3-4 months
Seed rate / spacing	Vine cuttings 2 x 2 m on trellis
Water need	Moderate
Typical yield	10-20 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fruit fly	Maggots in fruit	Follow label	Cue-lure traps; collect fallen fruit
Leaf spot and powdery mildew	Brown spots, white powder	Mancozeb 2.5 g/L; wettable sulphur 2.5 g/L	Remove infected leaves
Mealybug and aphids	Cottony masses, curled shoots	Follow label	neem oil 5 ml/L; *Lecanicillium* 5 g/L

52. Ridge gourd

Hindi / Telugu name	toraī / bīrakāya
Best season	Zaid and Kharif
Best soil	Sandy loam to loam with good drainage
Soil pH	6.0-7.0
Main growing states	Karnataka, UP, AP, West Bengal, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	60-80 : 50 : 50 kg/ha
Nutrient tip	Boron for fruit set; trellis for straight fruit
Crop duration	100-120 days
Seed rate / spacing	Seed 3-4 kg/ha; 2 x 1 m on trellis
Water need	Moderate
Typical yield	12-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Downy mildew	Yellow angular patches, grey fuzz below	Metalaxyl plus Mancozeb 2.5 g/L	Wide spacing; avoid overhead water
Fruit fly	Maggots in fruit	Follow label	Cue-lure traps 8-10 per acre; collect infested fruit
Red pumpkin beetle	Holes in leaves and flowers	Follow label	neem oil 5 ml/L; hand-pick
Mosaic virus	Mottled, curled leaves	No cure. Control aphids with Imidacloprid 0.3 ml/L	neem oil 5 ml/L; remove infected plants

53. Snake gourd

Hindi / Telugu name	ciciṃḍā / poṭlakāya
Best season	Kharif and Zaid
Best soil	Well-drained sandy loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	Kerala, Tamil Nadu, AP, Karnataka
Fertilizer (N : P2O5 : K2O)	60 : 40 : 40 kg/ha
Nutrient tip	Trellis for straight fruit; mulch
Crop duration	100-120 days
Seed rate / spacing	Seed 2-3 kg/ha; 2 x 1.5 m on trellis
Water need	Moderate
Typical yield	15-20 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fruit fly	Maggots, fruit rot	Follow label	Cue-lure traps; collect infested fruit
Epilachna beetle	Skeletonised leaves	Follow label	neem oil 5 ml/L; hand-pick grubs
Powdery and downy mildew	White powder or yellow patches	Hexaconazole 1 ml/L; Metalaxyl plus Mancozeb 2.5 g/L	Wide spacing; remove infected leaves
Mosaic virus	Mottled, small leaves	No cure. Control aphids with Imidacloprid 0.3 ml/L	neem oil 5 ml/L; remove infected plants

54. Ash gourd (White pumpkin)

Hindi / Telugu name	peṭhā / kumhaṛā / būḍida gummaḍi
Best season	Kharif and Zaid
Best soil	Well-drained loam; tolerates a wide range of soils
Soil pH	6.0-7.0
Main growing states	UP, Maharashtra, Karnataka, Kerala, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	60 : 40 : 40 kg/ha
Nutrient tip	Long storage life; avoid waterlogging
Crop duration	120-150 days
Seed rate / spacing	Seed 1.5-2 kg/ha; 2.5 x 1.5 m
Water need	Low to moderate
Typical yield	25-30 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Downy mildew	Yellow patches, grey fuzz underneath	Metalaxyl plus Mancozeb 2.5 g/L	Wide spacing; avoid overhead water
Fruit fly	Maggots in young fruit	Follow label	Cue-lure traps; bagging; bury infested fruit
Fusarium wilt	Sudden vine wilt	Drench Carbendazim 1 g/L	*Trichoderma* in soil; crop rotation
Red pumpkin beetle	Holes in leaves	Follow label	neem oil 5 ml/L; hand-pick

55. Pumpkin

Hindi / Telugu name	kaddū / gummaḍikāya
Best season	Kharif and Zaid
Best soil	Well-drained sandy loam to loam rich in organic matter
Soil pH	6.0-7.5
Main growing states	UP, Odisha, West Bengal, Gujarat, Karnataka
Fertilizer (N : P2O5 : K2O)	60 : 40 : 40 kg/ha
Nutrient tip	Good for rotation; avoid waterlogging
Crop duration	100-130 days
Seed rate / spacing	Seed 3-4 kg/ha; 2.5 x 1 m
Water need	Low to moderate
Typical yield	20-30 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves	Hexaconazole 1 ml/L; wettable sulphur 2 g/L (test first)	*Bacillus subtilis* 2-5 g/L
Fruit fly	Maggots in fruit	Follow label	Cue-lure traps; collect infested fruit
Mosaic virus	Mottled, crinkled leaves	No cure. Control aphids with Imidacloprid 0.3 ml/L	neem oil 5 ml/L; remove infected plants
Red pumpkin beetle	Holes in leaves and flowers	Follow label	neem oil 5 ml/L; hand-pick

56. Knol-khol (Kohlrabi)

Hindi / Telugu name	navalakol / gāṃṭha gobhī / naval kol
Best season	Rabi (Aug-Nov); hills in summer
Best soil	Fertile loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	West Bengal, Bihar, Odisha, Himachal
Fertilizer (N : P2O5 : K2O)	100 : 60 : 60 kg/ha
Nutrient tip	Harvest knobs when tender to avoid woodiness
Crop duration	55-70 days
Seed rate / spacing	Seed 1-1.5 kg/ha; 30 x 15 cm
Water need	Moderate
Typical yield	20-25 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Black rot	V-shaped yellow patches from leaf edge	Streptocycline 0.1 g/L plus copper oxychloride 3 g/L	Hot-water seed treatment (50 C, 30 min); rotation
Diamondback moth	Holes and windowing in leaves	Spinosad 0.3 ml/L; Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; mustard trap crop
Aphids	Clusters on leaves	Imidacloprid 0.3 ml/L (observe waiting period)	neem oil 5 ml/L; *Beauveria* 5 g/L

57. Turnip

Hindi / Telugu name	śalagam / ṭarnip
Best season	Rabi (Aug-Nov)
Best soil	Loose sandy loam, well-drained
Soil pH	6.0-7.0
Main growing states	Punjab, UP, Haryana, Himachal
Fertilizer (N : P2O5 : K2O)	50 : 40 : 40 kg/ha
Nutrient tip	Boron for root quality
Crop duration	45-70 days
Seed rate / spacing	Seed 3-4 kg/ha; 30 x 10 cm
Water need	Moderate
Typical yield	20-25 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
White rust	White blisters under leaves	Metalaxyl plus Mancozeb 2.5 g/L	Crop rotation; remove infected leaves
Aphids	Colonies on leaves	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L
Mustard sawfly	Holes in seedling leaves	Follow label	neem oil 5 ml/L; collect larvae

58. Lettuce

Hindi / Telugu name	salāda pattā / leṭyūs
Best season	Rabi and hills in summer
Best soil	Fertile well-drained loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	Karnataka, Maharashtra, Himachal, Uttarakhand
Fertilizer (N : P2O5 : K2O)	80 : 50 : 50 kg/ha
Nutrient tip	Calcium prevents tip burn; keep moist
Crop duration	60-90 days
Seed rate / spacing	Seed 400-600 g/ha; 30 x 30 cm
Water need	Moderate, frequent light irrigation
Typical yield	15-25 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Downy mildew	Yellow patches, grey fuzz below	Metalaxyl plus Mancozeb 2.5 g/L only if label allows (leafy crop)	*P. fluorescens*; wide spacing; avoid overhead water
Soft rot	Slimy rot of leaves and heads	Copper oxychloride 3 g/L as preventive	Good drainage; avoid injury
Aphids	Clusters in leaves	Avoid chemicals where possible	neem oil 5 ml/L; *Beauveria* 5 g/L; wash leaves

59. Yardlong bean (Long bean)

Hindi / Telugu name	lobiyā (laṃbī phalī) / bobbarlu (poḍavaina)
Best season	Kharif and Zaid
Best soil	Well-drained loam; tolerates poor soils
Soil pH	6.0-7.5
Main growing states	Kerala, Karnataka, AP, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	20 : 60 : 40 kg/ha
Nutrient tip	Rhizobium culture; trellis for long pods
Crop duration	80-120 days
Seed rate / spacing	Seed 20-25 kg/ha; 60 x 30 cm on trellis
Water need	Low to moderate
Typical yield	10-15 t/ha pods

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Aphids	Colonies on shoots and pods	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L
Pod borer (*Maruca*)	Bored flowers and pods	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L
Rust	Reddish-brown pustules	Mancozeb 2.5 g/L	Resistant varieties; rotation
Mosaic virus	Mottled, curled leaves	No cure. Control aphids	neem oil 5 ml/L; remove infected plants

60. Curry leaf

Hindi / Telugu name	karī pattā / kaṛhī pattā / karivepāku
Best season	Plant Jun-Aug; perennial
Best soil	Well-drained red loam and sandy loam; avoids waterlogging
Soil pH	6.0-7.5
Main growing states	AP, Telangana, Tamil Nadu, Karnataka, Kerala
Fertilizer (N : P2O5 : K2O)	FYM plus modest NPK in splits per plant (state advice)
Nutrient tip	Cut and prune regularly for fresh leaves
Crop duration	First harvest at 8-12 months; full yield from year 3
Seed rate / spacing	Spacing 1.5 x 1.5 m
Water need	Moderate
Typical yield	10-15 t/ha fresh leaf (from year 3)

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot (*Cercospora*)	Brown spots on leaves	Mancozeb 2.5 g/L only with the label waiting period	*P. fluorescens*; remove infected leaves
Psyllids, scales and mealybug	Curled shoots, honeydew, white masses	Avoid chemicals where possible (leaf is eaten)	neem oil 5 ml/L; *Lecanicillium* 5 g/L; wash leaves
Lemon butterfly caterpillar	Chewed young leaves	Avoid chemicals	*Bt* 2 g/L; hand-pick caterpillars

FRUITS

61. Mango

Hindi / Telugu name	ām / māmiḍi
Best season	Planting Jun-Aug; flowering Dec-Feb; fruiting Mar-Jun
Best soil	Deep well-drained loam, red and alluvial; avoids waterlogging and saline soils
Soil pH	5.5-7.5
Main growing states	UP, AP, Karnataka, Bihar, Gujarat, Telangana
Fertilizer (N : P2O5 : K2O)	Adult tree: about 1 kg N, 0.5 kg P2O5, 1 kg K2O per year plus FYM 50 kg
Nutrient tip	Zinc and boron deficiency (little leaf, poor fruit set); adjust to soil test
Crop duration	First fruit 3-5 years (grafted)
Seed rate / spacing	10 x 10 m (high density 5 x 5 m)
Water need	Low after establishment
Typical yield	8-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on flowers and tender fruit, flower drop	Wettable sulphur 2 g/L; Hexaconazole 1 ml/L (at flowering start)	*Bacillus subtilis*; avoid sprays at peak bloom when bees are active
Anthracnose	Black spots on flowers, leaves, fruit	Carbendazim 1 g/L; copper oxychloride 3 g/L	Prune dead twigs; *Trichoderma*; careful harvest handling
Mango hopper	Hoppers on flower panicles, sticky honeydew	Imidacloprid 0.3 ml/L (before flowers open)	*Beauveria bassiana* 5 g/L; neem oil 5 ml/L; clean orchard floor
Fruit fly	Maggots in ripening fruit	Follow label; avoid late sprays	Methyl eugenol traps 8-10 per acre; collect and bury fallen fruit
Malformation	Bunchy, deformed flower panicles	No pesticide cure	Remove affected panicles; NAA 200 ppm spray in October as advised locally
Bacterial canker	Raised angular spots on leaves and fruit	Streptocycline 0.1 g/L plus Copper oxychloride 3 g/L	Prune infected parts; avoid injury
Stem borer	Frass at holes in trunk, branch drying	Inject label-approved insecticide into holes	Clean holes; plug with neem-soaked cotton
Mealybug	White masses on panicles and shoots	Follow label	Sticky bands on trunk; *Lecanicillium* 5 g/L; clean orchard

62. Banana

Hindi / Telugu name	kelā / araṭi
Best season	Planting year-round with irrigation
Best soil	Deep fertile loam, well-drained, rich in organic matter
Soil pH	6.0-7.5
Main growing states	Tamil Nadu, Maharashtra, Gujarat, AP, Karnataka
Fertilizer (N : P2O5 : K2O)	About 200 g N, 60 g P2O5, 300 g K2O per plant in splits
Nutrient tip	High potash need; zinc and boron deficiency
Crop duration	11-14 months
Seed rate / spacing	1.8 x 1.8 m (about 3,000 plants/ha)
Water need	High
Typical yield	40-60 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Panama wilt (Fusarium)	Yellowing of older leaves, split pseudostem, brown inside	No reliable cure. Corm dip Carbendazim 2 g/L at planting	*Trichoderma* and *P. fluorescens* in pit; tissue-culture plants; resistant varieties
Sigatoka leaf spot	Yellow streaks turning brown on leaves	Propiconazole 1 ml/L; Mancozeb 2.5 g/L	Remove diseased leaves; *P. fluorescens* 10 g/L
Rhizome weevil	Tunnels in corm, plant falls	Follow label for soil treatment	*Beauveria* 5 g/L; clean planting material; split pseudostem traps

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Bunchy top virus	Narrow bunched leaves, dwarf plants	No cure. Control aphids: Imidacloprid 0.3 ml/L	Uproot and destroy infected plants; virus-free suckers
Bacterial soft rot (*Erwinia*)	Rotting pseudostem and corm, bad smell	Drench Copper oxychloride 3 g/L	Healthy suckers; good drainage
Nematodes	Root lesions, toppling plants	Follow label	Neem cake 250 g per plant; *Paecilomyces*; clean planting material
Leaf-eating caterpillar and thrips	Chewed leaves, scarred fruit	Follow label	Neem oil 5 ml/L; bunch covers (sleeves)

63. Grapes

Hindi / Telugu name	aṃgūr / drākṣa
Best season	Pruning twice a year; harvest Feb-Apr
Best soil	Light well-drained sandy loam to red loam; not heavy clay
Soil pH	6.5-7.5
Main growing states	Maharashtra, Karnataka, Tamil Nadu, AP
Fertilizer (N : P2O5 : K2O)	As per state grape-board schedule and soil test
Nutrient tip	Magnesium and zinc deficiency; avoid excess nitrogen
Crop duration	First crop in 2nd year
Seed rate / spacing	3 x 1.5 m on trellis
Water need	Moderate; drip
Typical yield	20-30 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Downy mildew	Oily yellow patches on leaves, white growth beneath	Metalaxyl plus Mancozeb 2.5 g/L; Cymoxanil plus Mancozeb 3 g/L; Bordeaux mixture 1%	Good canopy management; *Trichoderma* / *Bacillus subtilis* as support
Powdery mildew	White powder on leaves and berries	Wettable sulphur 2 g/L; Hexaconazole 1 ml/L	*Bacillus subtilis* 2-5 g/L; *Ampelomyces* product as per label
Anthracnose	Small dark spots on leaves, shoots and berries	Carbendazim 1 g/L; Mancozeb 2.5 g/L	Pruning and removal of infected parts
Mealybug	White cottony masses on bunches and shoots	Buprofezin 25 SC: 1-1.5 ml/L or as per label	*Cryptolaemus montrouzieri* beetles 10 per vine; *Lecanicillium* 5 g/L; control ants
Thrips and flea beetle	Scabby berries, notched buds	Follow label	Neem oil 5 ml/L; *Beauveria* 5 g/L
Bacterial leaf spot	Small angular dark spots	Copper oxychloride 3 g/L	Canopy management

64. Pomegranate

Hindi / Telugu name	anār / dānimma
Best season	Planting Jul-Aug or Feb-Mar; flowering seasons (Ambe, Mrig, Hasta)
Best soil	Light well-drained loam; tolerates moderate salinity and drought
Soil pH	6.5-7.5
Main growing states	Maharashtra, Karnataka, Gujarat, Rajasthan, AP
Fertilizer (N : P2O5 : K2O)	About 600 g N, 250 g P2O5, 250 g K2O per mature tree per year, split (state advice)
Nutrient tip	Boron and zinc for fruit set; calcium to reduce cracking
Crop duration	First fruit 2-3 years
Seed rate / spacing	4.5 x 3 m
Water need	Low to moderate; drip
Typical yield	12-20 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Bacterial blight (oily spot)	Oily dark spots on leaves and fruit, cracking	Streptocycline 0.5 g/L (or Bronopol 0.5 g/L) plus copper oxychloride 2.5 g/L	Disease-free planting material; orchard sanitation; prune infected twigs
Fruit spots (Cercospora, Alternaria)	Brown to black spots on fruit and leaves	Mancozeb 2.5 g/L; Carbendazim 1 g/L	*Bacillus subtilis*; avoid overhead water; remove infected fruit

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Wilt	Yellowing, drying branches, brown inside stem	Drench Carbendazim 2 g/L around root zone	*Trichoderma* and *P. fluorescens* in planting pit; good drainage
Fruit borer and thrips	Caterpillar inside fruit, scabby fruit surface	Spinosad 0.3 ml/L; Emamectin 0.4 g/L as per label	Bagging young fruits; neem oil 5 ml/L; collect fallen fruit
Aphids and whitefly	Honeydew, sooty mould	Imidacloprid 0.3 ml/L	Neem oil 5 ml/L; *Lecanicillium* 5 g/L
Nematodes	Root galls, decline	Follow label	Neem cake in basin; marigold

65. Citrus (Lime, Sweet orange, Mandarin)

Hindi / Telugu name	nīṃbū vargīya (nīṃbū, saṃtarā, mausamī) / nimmajāti (nimma, nāriṃja, battāyi)
Best season	Planting Jun-Aug; flowering Feb-Mar (Ambia) or Jul (Mrig)
Best soil	Deep well-drained loam; light to medium, no waterlogging
Soil pH	5.5-7.5
Main growing states	Maharashtra, AP, Punjab, MP, Karnataka, Assam
Fertilizer (N : P2O5 : K2O)	Mature tree: about 600-800 g N, 300 g P2O5, 300 g K2O per year plus FYM (state advice)
Nutrient tip	Zinc, iron and manganese deficiency common (yellow leaves); micronutrient spray
Crop duration	First fruit 3-4 years
Seed rate / spacing	5-6 m square
Water need	Moderate
Typical yield	10-20 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Citrus canker	Raised corky spots with yellow halo on leaves and fruit	Streptocycline 0.2-0.5 g/L plus copper oxychloride 3 g/L (follow local recommendation)	Prune and burn infected twigs; control leaf miner; clean planting material
Leaf miner	Silvery winding tunnels in young leaves	Imidacloprid 0.3 ml/L; Spinosad 0.3 ml/L	Neem oil 5 ml/L on new flush; *Beauveria* 5 g/L
Citrus psylla and greening	Curled leaves, yellow blotchy shoots, lopsided fruit	No cure for greening. Control psylla: Imidacloprid 0.3 ml/L	Disease-free plants; remove infected trees; neem oil 5 ml/L
Gummosis / Phytophthora	Gum oozing from trunk, bark cracking, decline	Bordeaux paste on trunk; drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma* in basin; raise graft union above soil; good drainage

66. Papaya

Hindi / Telugu name	papītā / boppāyi
Best season	Plant Jun-Jul, Sep-Oct or Feb-Mar
Best soil	Well-drained sandy loam; very sensitive to waterlogging
Soil pH	6.0-7.0
Main growing states	AP, Gujarat, Karnataka, Maharashtra
Fertilizer (N : P2O5 : K2O)	About 250 g N, 250 g P2O5, 500 g K2O per plant per year in splits
Nutrient tip	Boron deficiency causes deformed fruit; avoid waterlogging
Crop duration	8-10 months to fruit
Seed rate / spacing	1.8 x 1.8 m
Water need	Moderate
Typical yield	60-100 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Papaya ring spot virus	Mosaic leaves, ring spots on fruit, stunted trees	No cure. Control aphids with neem oil 5 ml/L or Imidacloprid 0.3 ml/L	Remove infected plants; tolerant varieties; keep papaya away from cucurbits
Papaya mealybug	White cottony masses on leaves and fruit	Follow label and local advice	Release parasitoid *Acerophagus papayae* via KVK; neem oil 5 ml/L
Foot / stem rot (Pythium)	Water-soaked rot at base, plant collapse	Drench Metalaxyl plus Mancozeb 2.5 g/L	Good drainage; raised beds; *Trichoderma* in pit
Anthracnose	Sunken spots on ripening fruit	Mancozeb 2.5 g/L; Carbendazim 1 g/L	Careful handling; avoid injury during harvest

67. Guava

Hindi / Telugu name	amarūd / jāma
Best season	Plant Jul-Aug; crops in rainy and winter seasons
Best soil	Wide range of soils; light loam to heavy; tolerates mild salinity
Soil pH	6.0-7.5
Main growing states	UP, MP, Bihar, Maharashtra, AP
Fertilizer (N : P2O5 : K2O)	About 600 g N, 300 g P2O5, 300 g K2O per mature tree per year
Nutrient tip	Zinc deficiency (small leaves); foliar zinc sulphate 5 g/L
Crop duration	First fruit 2-3 years
Seed rate / spacing	6 x 6 m (high density 3 x 3 m)
Water need	Low to moderate
Typical yield	15-25 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Wilt	Sudden yellowing and drying of branches	Drench Carbendazim 1-2 g/L around root zone	*Trichoderma* with FYM in basin; avoid root injury; good drainage
Fruit fly	Maggots in ripe fruit, premature drop	Follow label	Methyl eugenol traps 8-10 per acre; bagging of fruit; bury fallen fruit
Anthracnose / canker	Dark sunken spots on fruit and twigs	Mancozeb 2.5 g/L; Carbendazim 1 g/L	Pruning of infected twigs; copper oxychloride 3 g/L as preventive
Mealybug and scales	White cottony or crusty patches	Follow label and local advice	*Lecanicillium* 5 g/L; neem oil 5 ml/L; control ants

68. Apple

Hindi / Telugu name	seb / āpil
Best season	Dormant planting Dec-Feb; harvest Aug-Oct
Best soil	Well-drained loam in hills at 1500-2500 m with winter chill
Soil pH	5.5-6.8
Main growing states	Himachal, J&K;, Uttarakhand, Arunachal
Fertilizer (N : P2O5 : K2O)	As per state horticulture schedule and soil test
Nutrient tip	Calcium for bitter pit; boron and zinc
Crop duration	First fruit 4-5 years
Seed rate / spacing	5 x 5 m (high density 3 x 1 m)
Water need	Moderate
Typical yield	10-25 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Scab	Olive-green velvety spots on leaves and fruit, cracked fruit	Mancozeb 2.5 g/L; Captan 2 g/L; Hexaconazole 1 ml/L (sprays from green tip stage)	Remove and bury fallen leaves; resistant varieties; good air flow
Powdery mildew	White powder on shoots and flowers	Wettable sulphur 2 g/L; Hexaconazole 1 ml/L	Prune infected shoots; *Bacillus subtilis*
Woolly aphid	White cottony masses on branches and roots	Imidacloprid 0.3 ml/L as per label	Release *Aphelinus mali* parasitoid; protect natural enemies
Collar rot	Bark rot at base, tree decline	Drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma* in basin; avoid waterlogging and trunk injury
Codling moth	Larvae inside fruit	Follow label	Pheromone traps; fruit bagging; collect fallen fruit

69. Pineapple

Hindi / Telugu name	anānās / anāsa
Best season	Planting Apr-Jun or Sep-Oct
Best soil	Light sandy loam and laterite; well-drained acidic soils
Soil pH	4.5-6.5
Main growing states	Assam, West Bengal, Kerala, Tripura
Fertilizer (N : P2O5 : K2O)	About 8-12 g N and 4-6 g K2O per plant in splits; local advice
Nutrient tip	Iron deficiency in higher pH; foliar urea and iron
Crop duration	15-18 months
Seed rate / spacing	22,000-44,000 suckers/ha; double row 60 x 30 x 30 cm
Water need	Low to moderate
Typical yield	40-60 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Heart rot / root rot	Rot at the growing centre, wilting, red-brown leaves	Drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma* at planting; well-drained beds
Mealybug wilt	Reddish wilting leaves, mealybug on roots and base	Follow label and local advice	Control ants; *Lecanicillium* 5 g/L; neem oil 5 ml/L; healthy suckers
Fruit rot	Soft rot after injury or in humid weather	Carbendazim 1 g/L pre-harvest as per label	Careful handling; remove damaged fruit
Black rot	Water-soaked rot, black on cut fruit and slips	Slip dip: Carbendazim 1 g/L	Avoid injury; clean planting material

70. Watermelon and Muskmelon

Hindi / Telugu name	tarabūj / kharabūjā / puccakāya / kharbūjā
Best season	Zaid (Jan-Mar sowing); riverbed cultivation
Best soil	Light sandy loam and riverbeds; well-drained
Soil pH	6.0-7.0
Main growing states	UP, Punjab, Rajasthan, AP, Karnataka
Fertilizer (N : P2O5 : K2O)	80-100 : 50 : 50 kg/ha
Nutrient tip	Boron and calcium for fruit quality
Crop duration	80-100 days
Seed rate / spacing	Seed 3-4 kg/ha; 2.5 x 0.6-1 m
Water need	Moderate
Typical yield	25-40 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Anthracnose	Sunken dark spots on leaves and fruit	Mancozeb 2.5 g/L	*P. fluorescens*; crop rotation; clean seed
Fusarium wilt	Sudden vine wilt, brown vascular tissue	Drench Carbendazim 1 g/L	*Trichoderma* 4 g/kg seed and soil use; grafted seedlings; rotation
Powdery and downy mildew	White powder or yellow patches on leaves	Hexaconazole 1 ml/L; Metalaxyl plus Mancozeb 2.5 g/L	Wider spacing; avoid overhead water
Fruit fly	Maggots in fruit	Follow label	Cue-lure traps 8-10 per acre; collect and bury infested fruit
Mosaic virus (aphid spread)	Mottled, small, crinkled leaves	No cure. Control aphids with Imidacloprid 0.3 ml/L	Silver mulch; neem oil 5 ml/L; remove infected plants

71. Sapota (Chikoo)

Hindi / Telugu name	cīkū / sapoṭā
Best season	Planting Jun-Aug; crops twice a year
Best soil	Deep well-drained loam, red and black; tolerates coastal soils
Soil pH	6.0-8.0
Main growing states	Karnataka, Gujarat, Maharashtra, Tamil Nadu, AP
Fertilizer (N : P2O5 : K2O)	About 0.5-1 kg N per mature tree per year plus FYM
Nutrient tip	Iron chlorosis in alkaline soils
Crop duration	First fruit 4-5 years
Seed rate / spacing	10 x 10 m (8 x 8 m)
Water need	Low
Typical yield	10-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot	Purplish-brown spots with grey centres	Mancozeb 2.5 g/L	Pruning; clean orchard
Sooty mould	Black coating on leaves from honeydew	Control the sucking pest first	Starch spray 10 g/L; neem oil 5 ml/L
Bud borer and hairy caterpillar	Damaged buds, defoliation	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L; remove egg masses
Scale insect and mealybug	White or crusty patches on twigs	Follow label	*Lecanicillium* 5 g/L; neem oil 5 ml/L; ant control

72. Custard apple (Sitaphal)

Hindi / Telugu name	śarīphā / sītāphal / sītāphalaṃ
Best season	Plant Jun-Aug; fruit Sep-Nov
Best soil	Light shallow gravelly soils; drought tolerant; avoids waterlogging
Soil pH	6.0-8.0
Main growing states	Maharashtra, Telangana, AP, Gujarat
Fertilizer (N : P2O5 : K2O)	FYM 10-15 kg per tree plus modest NPK
Nutrient tip	Hardy; waterlogging is the main problem
Crop duration	First fruit 3-4 years
Seed rate / spacing	5 x 5 m
Water need	Low
Typical yield	5-8 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Mealybug	White cottony masses on fruit and shoots	Follow label and local advice	*Lecanicillium* 5 g/L; neem oil 5 ml/L; *Cryptolaemus* beetles; ant control
Anthracnose and fruit rot	Dark sunken spots on fruit and leaves	Carbendazim 1 g/L; Mancozeb 2.5 g/L	Prune infected twigs; avoid injury
Fruit borer	Larvae boring into fruit	Emamectin benzoate 0.4 g/L as per label	Collect infested fruit; bagging

73. Litchi

Hindi / Telugu name	līcī / licī
Best season	Plant Jul-Aug; fruit May-Jun
Best soil	Deep loam rich in organic matter with a high water table
Soil pH	5.5-7.5
Main growing states	Bihar, West Bengal, Uttarakhand, Assam
Fertilizer (N : P2O5 : K2O)	Mature tree: about 0.5-1.0 kg N, 0.25 kg P2O5, 0.5 kg K2O plus FYM
Nutrient tip	Boron and zinc (fruit cracking, small fruit)
Crop duration	First fruit 5-7 years
Seed rate / spacing	10 x 10 m
Water need	Moderate to high
Typical yield	8-12 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fruit and shoot borer	Larvae boring into young fruit and shoots	Spinosad 0.3 ml/L as per label	Pheromone traps; neem oil 5 ml/L; collect fallen fruit
Fruit cracking	Cracked fruit due to moisture swing	Borax 4 g/L spray at fruit set (as per local advice)	Regular light irrigation; mulching
Litchi erinose mite	Velvety brown patches on leaves, leaf curl	Wettable sulphur 2 g/L	Prune and burn infested shoots; neem oil 5 ml/L
Anthracnose	Brown spots on leaves and fruit	Carbendazim 1 g/L; Mancozeb 2.5 g/L	Orchard sanitation

74. Jackfruit

Hindi / Telugu name	kaṭahal / panasa
Best season	Plant Jun-Aug
Best soil	Well-drained deep loam; avoids waterlogging
Soil pH	6.0-7.5
Main growing states	Kerala, Karnataka, Odisha, Assam, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	FYM 20-30 kg per tree plus about 0.5 kg N
Nutrient tip	Avoid water stagnation
Crop duration	First fruit 5-7 years
Seed rate / spacing	10 x 10 m
Water need	Low
Typical yield	30-50 fruits per tree

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fruit rot (*Rhizopus*)	Soft rotting of young fruit	Mancozeb 2.5 g/L	Remove and bury infected fruit; avoid injury
Shoot and fruit borer	Bored shoots and fruit with frass	Follow label	neem oil 5 ml/L; cut and destroy affected parts
Bud rot and gummosis	Gum oozing, bark rot	Bordeaux paste on trunk; Metalaxyl plus Mancozeb 2.5 g/L drench	*Trichoderma* in basin; good drainage
Mealybug	White masses on tender parts	Follow label	neem oil 5 ml/L; ant control

75. Strawberry

Hindi / Telugu name	sṭra□berī / sṭrāberrī
Best season	Planting Sep-Oct; harvest winter-spring
Best soil	Light sandy loam rich in organic matter, well-drained
Soil pH	5.5-6.5
Main growing states	Maharashtra (Mahabaleshwar), Himachal, J&K;, Uttarakhand
Fertilizer (N : P2O5 : K2O)	About 100 : 60 : 100 kg/ha
Nutrient tip	Iron deficiency in higher pH; use mulch and drip
Crop duration	90-120 days to first fruit
Seed rate / spacing	30 x 30 cm raised beds (about 44,000 plants/ha)
Water need	Moderate; drip
Typical yield	15-25 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Grey mould (*Botrytis*)	Grey fluffy mould on ripe fruit	Carbendazim 1 g/L; Iprodione 2 g/L	Remove infected fruit; mulch; avoid overhead water
Powdery mildew	White powder, upward curling leaves	wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L
Red stele and crown rot	Wilting, red-core roots	Drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma*; disease-free runners; raised beds
Spider mites	Bronzed, webbed leaves	Spiromesifen 1 ml/L	neem oil 5 ml/L; predatory mite *Neoseiulus*

76. Peach and Plum

Hindi / Telugu name	āṛū / ālūbukhārā / pīc / plam
Best season	Dormant planting in winter
Best soil	Well-drained sandy loam in mid-hills
Soil pH	6.0-7.0
Main growing states	Himachal, Uttarakhand, J&K;, Punjab (low-chill)
Fertilizer (N : P2O5 : K2O)	As per state schedule
Nutrient tip	Zinc and boron; avoid waterlogging
Crop duration	First fruit 3-4 years
Seed rate / spacing	5 x 5 m
Water need	Moderate
Typical yield	8-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf curl (peach)	Red, thick, curled leaves in spring	Bordeaux mixture 1% or Mancozeb 2.5 g/L at bud swell	Remove and burn infected leaves; winter spray
Brown rot	Brown rotting fruit with grey spore powder	Carbendazim 1 g/L	Remove mummified fruit; pruning for air flow
Aphids	Curled leaves, honeydew	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L
Fruit fly	Maggots in fruit	Follow label	Methyl eugenol traps; collect fallen fruit

77. Ber (Indian jujube)

Hindi / Telugu name	ber / regu
Best season	Plant Jul-Aug; fruit Dec-Mar
Best soil	Light sandy loam; very drought tolerant; tolerates salinity
Soil pH	6.5-8.5
Main growing states	Rajasthan, UP, Haryana, Gujarat, Maharashtra
Fertilizer (N : P2O5 : K2O)	About 500 g N per mature tree plus FYM
Nutrient tip	Hardy; prune hard each year
Crop duration	First fruit 2-3 years
Seed rate / spacing	6 x 6 m
Water need	Very low
Typical yield	8-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves, flowers and fruit	Wettable sulphur 2 g/L; Hexaconazole 1 ml/L	Spray at first sign; prune dense growth
Fruit fly	Maggots in ripening fruit	Follow label	Methyl eugenol traps 8-10 per acre; collect fallen fruit
Leaf-eating caterpillars and lac insect	Defoliation, encrusted twigs	Emamectin benzoate 0.4 g/L as per label	neem oil 5 ml/L; *Bt* 2 g/L
Leaf spot (*Alternaria*)	Brown spots on leaves	Mancozeb 2.5 g/L	Orchard sanitation

78. Amla (Indian gooseberry)

Hindi / Telugu name	āṃvalā / usiri
Best season	Plant Jul-Aug; fruit Oct-Jan
Best soil	Wide soil range; tolerates saline and alkaline soils
Soil pH	6.0-8.0
Main growing states	UP, MP, Rajasthan, Gujarat, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	About 500-1000 g N per mature tree plus FYM
Nutrient tip	Boron for fruit quality; zinc
Crop duration	First fruit 4-5 years
Seed rate / spacing	8 x 8 m
Water need	Low
Typical yield	10-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Rust	Orange-brown pustules on leaves and fruit	Mancozeb 2.5 g/L; Hexaconazole 1 ml/L	Remove infected parts
Fruit rot	Soft rot of fruit before and after harvest	Carbendazim 1 g/L	Careful handling; avoid injury
Bark-eating caterpillar	Webbing and frass on branches	Inject label-approved insecticide in holes	Clean tunnels; plug with cotton soaked in neem oil
Shoot gall maker	Galls on shoot tips	Follow label	Prune and burn galls; neem oil 5 ml/L

79. Dragon fruit

Hindi / Telugu name	ḍraigana phrūṭ / kamalam / ḍrāgan phrūṭ
Best season	Plant year-round with irrigation; fruit May-Nov
Best soil	Well-drained sandy loam; avoid waterlogging
Soil pH	6.0-7.0
Main growing states	Gujarat, Maharashtra, Karnataka, Telangana, AP
Fertilizer (N : P2O5 : K2O)	FYM 10 kg per post plus balanced NPK in splits
Nutrient tip	Needs support posts; avoid root rot
Crop duration	12-18 months
Seed rate / spacing	3 x 3 m posts, 4 plants per post
Water need	Low to moderate
Typical yield	10-20 t/ha at full bearing

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Stem canker and rot	Brown to orange spots and soft rot on stems	Carbendazim 1 g/L; Mancozeb 2.5 g/L	Remove infected cladodes; improve air flow
Anthracnose	Sunken spots on fruit and stems	Carbendazim 1 g/L; Mancozeb 2.5 g/L	*P. fluorescens*; careful handling
Ants and mealybug	Cottony masses, ants on stems	Follow label	neem oil 5 ml/L; *Lecanicillium* 5 g/L
Fruit fly	Maggots in ripening fruit	Follow label	Traps; bagging; collect fallen fruit

80. Fig

Hindi / Telugu name	aṃjīr / aṃjūraṃ
Best season	Plant Jul-Aug or winter
Best soil	Well-drained medium loam; moderate climate
Soil pH	6.0-8.0
Main growing states	Maharashtra, Karnataka, Gujarat, UP
Fertilizer (N : P2O5 : K2O)	FYM 10-20 kg plus about 200-300 g N per tree
Nutrient tip	Boron deficiency causes fruit drop
Crop duration	First fruit 2-3 years
Seed rate / spacing	4.5 x 4.5 m
Water need	Moderate
Typical yield	10-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Rust	Yellow-orange pustules on leaves	Mancozeb 2.5 g/L	Remove fallen leaves; good air flow
Fruit and branch rot	Soft rot and dieback	Carbendazim 1 g/L	Prune infected parts; avoid injury
Mealybug and scale	Cottony or crusty patches	Follow label	neem oil 5 ml/L; ant control
Fig mosaic (mite spread)	Mottled, distorted leaves	No cure	Disease-free cuttings; remove infected plants

81. Pear

Hindi / Telugu name	nāśapātī / piyar
Best season	Dormant planting in winter
Best soil	Well-drained loam in hills; needs winter chill
Soil pH	5.5-7.0
Main growing states	Himachal, J&K;, Punjab, Uttarakhand
Fertilizer (N : P2O5 : K2O)	As per state schedule
Nutrient tip	Boron deficiency causes cork spots
Crop duration	First fruit 4-5 years
Seed rate / spacing	5 x 5 m
Water need	Moderate
Typical yield	10-20 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Scab	Olive-green spots on leaves and fruit	Mancozeb 2.5 g/L; Captan 2 g/L	Remove fallen leaves; resistant varieties
Pear psylla	Honeydew, sooty mould, leaf drop	Imidacloprid 0.3 ml/L as per label	neem oil 5 ml/L; protect predators
Fire blight	Blackened, burnt-looking shoots and blossoms	Copper sprays at bloom as per local advice	Prune 30 cm below infection; disinfect tools
Codling moth and fruit fly	Larvae or maggots in fruit	Follow label	Pheromone and methyl eugenol traps; bagging

82. Date palm

Hindi / Telugu name	khajūr / kharjūraṃ
Best season	Plant Feb-Mar or Jul-Aug
Best soil	Sandy loam with good drainage; tolerates high salinity; needs hot dry summer
Soil pH	7.0-8.5
Main growing states	Gujarat (Kutch), Rajasthan
Fertilizer (N : P2O5 : K2O)	FYM 25-50 kg per palm plus NPK as per soil test
Nutrient tip	Tolerant of salinity
Crop duration	First fruit 4-5 years
Seed rate / spacing	8 x 8 m
Water need	Moderate
Typical yield	80-100 kg per tree

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Graphiola leaf spot (false smut)	Black pustules on leaves	Mancozeb 2.5 g/L	Remove infected fronds; sanitation
Inflorescence rot (Khamedj)	Rotting of flower spathe before opening	Bordeaux mixture 1% or Carbendazim 1 g/L before spathe opening	Remove and burn infected spathes
Dust mite (*Oligonychus*)	Webbing and dust-like mites on fruit	Wettable sulphur dusting or spray 2 g/L	Water spray to wash mites; early treatment
Red palm weevil / rhinoceros beetle	Tunnels in crown, falling fronds	Follow label and local advice	Pheromone traps; clean crowns; *Metarhizium*

83. Lemon (Kagzi lime)

Hindi / Telugu name	nīṃbū (kāgajī) / nimma
Best season	Plant Jun-Aug; fruit year-round
Best soil	Light to medium well-drained loam
Soil pH	5.5-7.5
Main growing states	AP, Gujarat, Karnataka, Maharashtra, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	About 600 g N, 200 g P2O5, 200 g K2O per mature tree per year
Nutrient tip	Zinc and iron deficiency; micronutrient spray
Crop duration	First fruit 3-4 years
Seed rate / spacing	5 x 5 m
Water need	Moderate
Typical yield	10-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Citrus canker	Raised corky spots with yellow halo	Streptocycline 0.2-0.5 g/L plus copper oxychloride 3 g/L (local advice)	Prune and burn infected twigs; control leaf miner
Leaf miner	Silvery tunnels in new flush	Imidacloprid 0.3 ml/L; Spinosad 0.3 ml/L	neem oil 5 ml/L on new flush
Gummosis	Gum ooze, bark cracking	Bordeaux paste; Metalaxyl plus Mancozeb 2.5 g/L drench	*Trichoderma* in basin; raise graft union
Psylla and greening	Curled leaves, yellow blotches	No cure for greening. Control psylla with Imidacloprid 0.3 ml/L	Disease-free plants; remove infected trees

84. Sweet orange (Mosambi)

Hindi / Telugu name	mausamī / battāyi
Best season	Plant Jun-Aug; fruit Oct-Feb
Best soil	Deep well-drained loam; light textured; no waterlogging
Soil pH	6.0-7.5
Main growing states	AP, Telangana, Maharashtra, Punjab
Fertilizer (N : P2O5 : K2O)	About 600-800 g N, 300 g P2O5, 300 g K2O per mature tree per year
Nutrient tip	Zinc, iron, manganese deficiency; micronutrient spray
Crop duration	First fruit 3-4 years
Seed rate / spacing	6 x 6 m
Water need	Moderate
Typical yield	10-20 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Gummosis / Phytophthora	Gum ooze, bark rot, decline	Bordeaux paste on trunk; drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma* in basin; budding union above soil
Citrus psylla and leaf miner	Curled shoots, silvery tunnels	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L
Fruit drop and fruit sucking moth	Premature drop, pierced fruit	Follow label and local advice	Orchard sanitation; bagging; light traps
Citrus greening	Yellow blotchy leaves, lopsided fruit	No cure. Control psylla	Disease-free plants; remove infected trees

85. Jamun

Hindi / Telugu name	jāmun / nereḍu
Best season	Plant Jun-Aug; fruit May-Jul
Best soil	Deep loam; tolerates waterlogging and varied soils
Soil pH	6.0-8.0
Main growing states	UP, Maharashtra, Gujarat, Karnataka, AP
Fertilizer (N : P2O5 : K2O)	FYM 20-30 kg plus modest NPK per tree
Nutrient tip	Hardy; minimal inputs
Crop duration	First fruit 6-8 years (seedling), 3-4 (grafted)
Seed rate / spacing	8 x 8 m
Water need	Low
Typical yield	8-10 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot and anthracnose	Brown spots on leaves and fruit	Mancozeb 2.5 g/L; Carbendazim 1 g/L	Prune and clean orchard
Fruit fly	Maggots in ripe fruit	Follow label	Methyl eugenol traps; collect fallen fruit
Leaf-eating caterpillars	Defoliation	Emamectin benzoate 0.4 g/L as per label	*Bt* 2 g/L; neem oil 5 ml/L

86. Tamarind

Hindi / Telugu name	imalī / ciṃta
Best season	Plant Jun-Aug; harvest Feb-Apr
Best soil	Wide soil range; deep well-drained loam; drought tolerant
Soil pH	6.0-8.0
Main growing states	Karnataka, Tamil Nadu, AP, Chhattisgarh
Fertilizer (N : P2O5 : K2O)	FYM 20-30 kg per tree
Nutrient tip	Very hardy; minimal inputs
Crop duration	First fruit 5-7 years (grafted)
Seed rate / spacing	10 x 10 m
Water need	Very low
Typical yield	8-10 t/ha pods

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on young leaves and flowers	Wettable sulphur 2 g/L	Spray at first sign
Bark-eating caterpillar and borers	Tunnels and frass on trunk	Inject label-approved insecticide	Clean tunnels; plug with neem-oil soaked cotton
Fruit borer in storage	Holes in dried pods	Follow label	Proper drying; airtight storage

87. Avocado

Hindi / Telugu name	evokāḍo / avakāḍo
Best season	Plant Jun-Aug in hills
Best soil	Well-drained light loam; very sensitive to waterlogging; mid altitude
Soil pH	5.5-6.5
Main growing states	Karnataka (Coorg), Kerala, Tamil Nadu, Nagaland, Sikkim
Fertilizer (N : P2O5 : K2O)	FYM 20 kg plus about 300-500 g N per mature tree
Nutrient tip	Zinc and iron deficiency; avoid waterlogging
Crop duration	First fruit 3-5 years (grafted)
Seed rate / spacing	8 x 8 m
Water need	Moderate; no waterlogging
Typical yield	8-10 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Root rot (*Phytophthora*)	Wilting, leaf drop, rotted roots	Drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma*; raised mounds; excellent drainage
Anthracnose	Dark sunken spots on fruit and leaves	Carbendazim 1 g/L; copper oxychloride 3 g/L	Prune; careful harvesting
Thrips and mites	Scarred fruit, bronzed leaves	Follow label	neem oil 5 ml/L; *Beauveria* 5 g/L

88. Kiwi

Hindi / Telugu name	kīvī / kivi
Best season	Dormant planting in winter
Best soil	Deep well-drained loam in cool hills; sensitive to waterlogging
Soil pH	5.5-6.5
Main growing states	Arunachal, Himachal, J&K;, Sikkim
Fertilizer (N : P2O5 : K2O)	As per state schedule
Nutrient tip	Male and female vines needed for pollination
Crop duration	First fruit 3-4 years
Seed rate / spacing	4 x 4 m on T-bar trellis
Water need	Moderate
Typical yield	8-12 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Root and crown rot	Wilting, decline	Drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma*; raised beds; drainage
Leaf spot	Brown spots on leaves	Mancozeb 2.5 g/L	Prune for air flow
Scales and mealybug	Crusty or cottony patches	Follow label	neem oil 5 ml/L

89. Bael (Wood apple)

Hindi / Telugu name	bel / māreḍu
Best season	Plant Jun-Aug; fruit Apr-Jun
Best soil	Poor, rocky, saline and alkaline soils; very hardy
Soil pH	5.0-8.0
Main growing states	UP, Bihar, MP, Rajasthan, Gujarat
Fertilizer (N : P2O5 : K2O)	FYM 20-25 kg per tree plus modest NPK
Nutrient tip	Hardy; minimal inputs
Crop duration	First fruit 5-7 years
Seed rate / spacing	8 x 8 m
Water need	Very low
Typical yield	8-10 t/ha at full bearing

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot and canker	Brown spots on leaves and fruit	Mancozeb 2.5 g/L; Carbendazim 1 g/L	Prune and clean orchard
Fruit cracking and drop	Cracked fruit, premature drop	Borax 4 g/L spray at fruit set as per local advice	Regular light irrigation; mulching
Lemon butterfly caterpillar	Chewed young leaves	Follow label	*Bt* 2 g/L; hand-pick

90. Apricot

Hindi / Telugu name	khubānī / āprikāṭ
Best season	Dormant planting in winter
Best soil	Well-drained loam in cold-dry hills
Soil pH	6.0-7.5
Main growing states	J&K; (Ladakh), Himachal, Uttarakhand
Fertilizer (N : P2O5 : K2O)	As per state schedule
Nutrient tip	Late frost at flowering is the main risk
Crop duration	First fruit 4-5 years
Seed rate / spacing	6 x 6 m
Water need	Moderate
Typical yield	10-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Gummosis and canker	Gum oozing from bark	Bordeaux paste on wounds	*Trichoderma*; prune in dry weather
Brown rot	Rotting blossoms and fruit	Carbendazim 1 g/L	Remove mummified fruit
Aphids and fruit fly	Curled leaves, maggots in fruit	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; traps

91. Walnut

Hindi / Telugu name	akharoṭ / vālnaṭ
Best season	Dormant planting in winter
Best soil	Deep well-drained loam in hills
Soil pH	6.0-7.5
Main growing states	J&K;, Himachal, Uttarakhand
Fertilizer (N : P2O5 : K2O)	As per state schedule
Nutrient tip	Grafted trees bear much earlier than seedlings
Crop duration	First nuts 4-5 years (grafted), 7-10 (seedling)
Seed rate / spacing	10 x 10 m
Water need	Moderate
Typical yield	2-3 t/ha nuts

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Anthracnose (*Marssonina*)	Dark spots on leaves and husks, early leaf fall	Mancozeb 2.5 g/L; Carbendazim 1 g/L	Remove fallen leaves
Walnut blight (bacterial)	Black spots on nuts and shoots	copper oxychloride 3 g/L	Prune infected shoots
Codling moth and aphids	Larvae in nuts, curled leaves	Follow label	neem oil 5 ml/L; pheromone traps

92. Cherry

Hindi / Telugu name	cerī / cerrī
Best season	Dormant planting in winter
Best soil	Well-drained loam in hills with winter chill
Soil pH	6.0-7.0
Main growing states	J&K;, Himachal
Fertilizer (N : P2O5 : K2O)	As per state schedule
Nutrient tip	Rain at ripening causes fruit cracking
Crop duration	First fruit 4-5 years
Seed rate / spacing	6 x 6 m
Water need	Moderate
Typical yield	5-8 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Brown rot	Rotting fruit with grey powder	Carbendazim 1 g/L	Remove infected fruit; pruning
Cherry fruit fly	Maggots in ripe fruit	Follow label	Traps; clean harvest
Powdery mildew	White powder on shoots and leaves	wettable sulphur 2.5 g/L	*Bacillus subtilis*

93. Passion fruit

Hindi / Telugu name	paiśana phrūṭ / pāṣan phrūṭ
Best season	Plant Jun-Aug
Best soil	Well-drained loam; avoids waterlogging
Soil pH	5.5-7.0
Main growing states	Meghalaya, Mizoram, Nagaland, Kerala, Karnataka
Fertilizer (N : P2O5 : K2O)	FYM 10-15 kg per vine plus modest NPK in splits
Nutrient tip	Needs a strong trellis; prune after harvest
Crop duration	First fruit in 9-12 months
Seed rate / spacing	3 x 3 m trellis (about 1,100 vines/ha)
Water need	Moderate
Typical yield	10-15 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fusarium wilt and collar rot	Wilting, rotted base	Drench Carbendazim 1 g/L	*Trichoderma*; disease-free vines; good drainage
Woodiness virus (aphid spread)	Distorted, hard, small fruit	No cure	Remove infected vines; control aphids with neem oil 5 ml/L
Fruit fly	Maggots in fruit	Follow label	Traps; collect fallen fruit
Anthracnose	Brown sunken spots on fruit and leaves	Mancozeb 2.5 g/L; Carbendazim 1 g/L	Prune for air flow

PULSES (SEED CROPS)

94. Chickpea (Gram)

Hindi / Telugu name	canā / śanaga
Best season	Rabi (Oct-Nov)
Best soil	Well-drained loam and black soils; avoids waterlogging and salinity
Soil pH	6.0-8.0
Main growing states	MP, Maharashtra, Rajasthan, UP, Karnataka, AP
Fertilizer (N : P2O5 : K2O)	20 : 40-60 : 20 (sulphur 20 kg/ha)
Nutrient tip	Rhizobium plus PSB culture; zinc and sulphur
Crop duration	95-130 days
Seed rate / spacing	Seed 60-80 kg/ha; 30 x 10 cm
Water need	Low (1-2 irrigations)
Typical yield	1.5-2.5 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fusarium wilt	Drooping and yellowing, brown inside stem	Seed treatment: Carbendazim 2 g/kg (with Thiram as advised)	*Trichoderma* 4-10 g/kg seed; resistant varieties; sow at correct time
Ascochyta blight	Brown spots on leaves, stems and pods in cool humid weather	Mancozeb 2.5 g/L; Chlorothalonil 2 g/L	Clean seed; crop rotation; destroy residue
Pod borer (*Helicoverpa*)	Round holes in pods, caterpillar inside	Emamectin benzoate 0.4 g/L; Chlorantraniliprole 0.3 ml/L	HaNPV 250 LE/ha; *Bt* 2 g/L; pheromone traps; bird perches
Dry root rot	Sudden drying of plants, black rotted roots	Seed treatment as for wilt	*Trichoderma* seed treatment; avoid moisture stress at flowering
Cutworm	Seedlings cut at ground level	Follow label (soil application)	Deep summer ploughing; light traps
Botrytis grey mould	Grey growth on flowers and pods in humid weather	Carbendazim 1 g/L	Wide spacing; avoid late irrigation

95. Pigeon pea (Tur / Red gram)

Hindi / Telugu name	arahar / tuar / kaṃdulu
Best season	Kharif (Jun-Jul)
Best soil	Well-drained loam, red and black; avoids waterlogging
Soil pH	6.5-7.5
Main growing states	Maharashtra, Karnataka, Telangana, MP, UP
Fertilizer (N : P2O5 : K2O)	20-25 : 50 : 20 kg/ha
Nutrient tip	Rhizobium culture; zinc and sulphur
Crop duration	150-250 days
Seed rate / spacing	Seed 12-15 kg/ha; 90 x 20 cm
Water need	Low
Typical yield	1.2-2 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Wilt	Yellowing and drying, purple streak on stem	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4-10 g/kg seed; resistant varieties; rotation
Sterility mosaic	Bushy pale plants with no flowers	No cure. Control mite vector: wettable sulphur 2.5 g/L or as per label	Rogue out infected plants early; resistant varieties
Pod borer and pod fly	Holes in pods, damaged seeds	Emamectin benzoate 0.4 g/L; Spinosad 0.3 ml/L	HaNPV 250 LE/ha; *Bt* 2 g/L; neem oil 5 ml/L; pheromone traps
Phytophthora blight	Water-soaked stem lesions, patchy plant death in wet fields	Seed treatment: Metalaxyl 3 g/kg	Ridge sowing; good drainage; resistant varieties

96. Green gram and Black gram

Hindi / Telugu name	mūṃg / uṛad / pesalu / minumulu
Best season	Kharif, Rabi and Zaid
Best soil	Well-drained loam; avoids saline soils
Soil pH	6.5-7.5
Main growing states	Rajasthan, MP, Maharashtra, AP, UP, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	20 : 40 : 20 kg/ha
Nutrient tip	Rhizobium culture; sulphur and zinc
Crop duration	60-90 days
Seed rate / spacing	Seed 15-20 kg/ha; 30 x 10 cm
Water need	Low
Typical yield	0.8-1.2 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Yellow mosaic virus	Bright yellow patches on leaves, poor pods	No cure. Control whitefly: Imidacloprid 0.3 ml/L; Thiamethoxam 0.2 g/L	Resistant varieties; remove infected plants; yellow sticky traps; timely sowing
Powdery mildew	White powder on leaves	Wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L
Cercospora leaf spot	Brown spots with grey centre	Mancozeb 2.5 g/L; Carbendazim 1 g/L	Seed treatment *Trichoderma* 4 g/kg; crop rotation
Pod borer	Bored pods and seeds	Emamectin benzoate 0.4 g/L	HaNPV; *Bt* 2 g/L; neem oil 5 ml/L

97. Lentil (Masur)

Hindi / Telugu name	masūr / masūr
Best season	Rabi (Oct-Nov)
Best soil	Well-drained loam; light to medium
Soil pH	6.0-7.5
Main growing states	UP, MP, Bihar, West Bengal
Fertilizer (N : P2O5 : K2O)	20 : 40 : 20 kg/ha
Nutrient tip	Rhizobium culture; sulphur
Crop duration	100-130 days
Seed rate / spacing	Seed 30-40 kg/ha; 25 x 5 cm
Water need	Low
Typical yield	1-1.5 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Wilt and root rot	Drooping and drying of plants	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4-10 g/kg seed; resistant varieties; rotation
Rust	Reddish-brown pustules	Mancozeb 2.5 g/L	Resistant varieties; timely sowing
Pod borer	Bored pods	Emamectin benzoate 0.4 g/L	HaNPV; *Bt* 2 g/L; pheromone traps
Aphids	Clusters on tender shoots	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

98. Field pea

Hindi / Telugu name	maṭara (dāl) / baṭhānī
Best season	Rabi (Oct-Nov)
Best soil	Well-drained loam
Soil pH	6.0-7.5
Main growing states	UP, MP, Bihar, Rajasthan
Fertilizer (N : P2O5 : K2O)	20 : 40-50 : 20 kg/ha
Nutrient tip	Rhizobium culture; molybdenum
Crop duration	100-130 days
Seed rate / spacing	Seed 75-100 kg/ha; 30 x 5 cm
Water need	Low
Typical yield	1.5-2 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves	wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L; early sowing
Rust	Orange-brown pustules	Mancozeb 2.5 g/L	Resistant varieties
Leaf miner and pod borer	Mined leaves, bored pods	Thiamethoxam 0.2 g/L; Emamectin benzoate 0.4 g/L	neem oil 5 ml/L; *Bt* 2 g/L
Wilt	Yellowing, wilting	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4-10 g/kg seed

99. Horse gram

Hindi / Telugu name	kulathī / ulavalu
Best season	Kharif and late Kharif
Best soil	Poor gravelly red soils; drought tolerant
Soil pH	6.0-7.5
Main growing states	Karnataka, AP, Telangana, Tamil Nadu, Odisha
Fertilizer (N : P2O5 : K2O)	20 : 40 : 0-20 kg/ha
Nutrient tip	Hardy; Rhizobium
Crop duration	90-120 days
Seed rate / spacing	Seed 20-25 kg/ha; 30 x 10 cm
Water need	Very low
Typical yield	0.6-1 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew and rust	White powder, rusty pustules	wettable sulphur 2.5 g/L; Hexaconazole 1 ml/L	Resistant varieties; timely sowing
Yellow mosaic virus	Yellow mottle, poor pods	No cure. Control whitefly with Imidacloprid 0.3 ml/L	yellow sticky traps; remove infected plants
Pod borer	Bored pods	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L

100. Cowpea

Hindi / Telugu name	lobiyā / bobbarlu / alasaṃdalu
Best season	Kharif and Zaid
Best soil	Sandy loam to loam; tolerates poor soils
Soil pH	6.0-7.5
Main growing states	Rajasthan, Karnataka, Kerala, AP, UP
Fertilizer (N : P2O5 : K2O)	20 : 40-60 : 20 kg/ha
Nutrient tip	Rhizobium culture; fits as green manure or fodder too
Crop duration	70-100 days
Seed rate / spacing	Seed 20-25 kg/ha; 30 x 10 cm
Water need	Low
Typical yield	1-1.5 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Mosaic and yellow mosaic	Mottled, curled leaves	No cure. Control aphids and whitefly with Imidacloprid 0.3 ml/L	yellow sticky traps; neem oil 5 ml/L; resistant varieties
Anthracnose and leaf spot	Dark sunken spots on pods and leaves	Carbendazim 1 g/L; Mancozeb 2.5 g/L	*Trichoderma* seed treatment 4 g/kg
Pod borer (*Maruca*)	Bored flowers and pods	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L
Aphids	Dense colonies on shoots	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

101. Rajmash (Kidney bean)

Hindi / Telugu name	rājamā / rājmā
Best season	Rabi (Oct-Nov) in plains; summer in hills
Best soil	Well-drained loam; sensitive to waterlogging and salinity
Soil pH	6.0-7.5
Main growing states	Maharashtra, Himachal, J&K;, Uttarakhand
Fertilizer (N : P2O5 : K2O)	40-60 : 60 : 40 kg/ha
Nutrient tip	Zinc and molybdenum
Crop duration	100-120 days
Seed rate / spacing	Seed 80-100 kg/ha; 30 x 10 cm
Water need	Moderate
Typical yield	1.5-2 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Rust	Reddish-brown pustules	Hexaconazole 1 ml/L; Mancozeb 2.5 g/L	Resistant varieties; crop rotation
Anthracnose	Dark sunken spots on pods and stems	Carbendazim 1 g/L; Mancozeb 2.5 g/L	Clean seed; *Trichoderma* seed treatment
Bean common mosaic	Mottled, puckered leaves	No cure. Control aphids with Imidacloprid 0.3 ml/L	Virus-free seed; remove infected plants
Root rot	Brown rotted roots, wilting	Drench Carbendazim 1 g/L	*Trichoderma* seed treatment 4-10 g/kg; good drainage

102. Moth bean

Hindi / Telugu name	moṭh / mot
Best season	Kharif (Jun-Jul)
Best soil	Light sandy desert soils; very drought tolerant
Soil pH	7.0-8.5
Main growing states	Rajasthan, Gujarat, Haryana
Fertilizer (N : P2O5 : K2O)	15-20 : 30-40 : 0 kg/ha
Nutrient tip	Hardy crop for dry areas
Crop duration	60-90 days
Seed rate / spacing	Seed 12-15 kg/ha; 30 x 10 cm
Water need	Very low
Typical yield	0.4-0.8 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Yellow mosaic virus	Yellow patches on leaves, poor pods	No cure. Control whitefly with Imidacloprid 0.3 ml/L	yellow sticky traps; resistant varieties; rogue
Cercospora leaf spot	Brown spots with grey centre	Mancozeb 2.5 g/L; Carbendazim 1 g/L	*Trichoderma* seed treatment 4 g/kg
Root rot	Drying of plants	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4-10 g/kg seed
Pod borer	Bored pods	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L

OILSEEDS

103. Groundnut

Hindi / Telugu name	mūṃgaphalī / veruśanaga
Best season	Kharif (Jun-Jul); Rabi and summer in south
Best soil	Light well-drained sandy loam and red soils with calcium; avoid heavy clay
Soil pH	6.0-7.5
Main growing states	Gujarat, AP, Karnataka, Tamil Nadu, Rajasthan
Fertilizer (N : P2O5 : K2O)	20-25 : 40-50 : 40-50 (gypsum 400-500 kg/ha at flowering)
Nutrient tip	Calcium (pops, empty pods), sulphur, iron chlorosis; apply gypsum
Crop duration	100-130 days
Seed rate / spacing	Kernels 100-120 kg/ha; 30 x 10 cm
Water need	Moderate
Typical yield	1.5-2.5 t/ha pods

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Tikka (early and late leaf spot)	Brown to black spots with yellow halo	Chlorothalonil 2 g/L; Hexaconazole 1 ml/L	*P. fluorescens* 10 g/L; crop rotation
Collar rot / stem rot	Seedling collapse, white growth at collar	Seed treatment: Thiram 3 g/kg	*Trichoderma viride* 4-10 g/kg seed
Rust	Orange pustules under leaves	Chlorothalonil 2 g/L; Hexaconazole 1 ml/L	Resistant varieties
Spodoptera / leaf miner	Chewed leaves, mined blotches	Chlorantraniliprole 0.3 ml/L; Emamectin 0.4 g/L	SlNPV 250 LE/ha; NSKE 5%; light and pheromone traps
Bud necrosis virus (thrips spread)	Ring spots, bud death, stunting	No cure. Control thrips with Imidacloprid seed treatment as per label	Border crop of sorghum or bajra; blue sticky traps
Root rot and stem rot	Wilting, white growth at base	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4-10 g/kg seed; rotation
White grub	Yellowing, plants pulled easily	Follow label for soil use	*Metarhizium*; summer ploughing

104. Soybean

Hindi / Telugu name	soyābīn / soyācikkuḍu
Best season	Kharif (Jun-Jul)
Best soil	Medium to deep black soils with good drainage
Soil pH	6.0-7.5
Main growing states	MP, Maharashtra, Rajasthan, Karnataka
Fertilizer (N : P2O5 : K2O)	20-30 : 60-80 : 40 (sulphur 20 kg/ha)
Nutrient tip	Rhizobium culture; sulphur, zinc and molybdenum
Crop duration	90-110 days
Seed rate / spacing	Seed 60-80 kg/ha; 45 x 5 cm
Water need	Moderate
Typical yield	2-2.5 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Rust	Small rusty pustules under leaves	Hexaconazole 1 ml/L; Propiconazole 1 ml/L	Resistant varieties; timely sowing
Charcoal rot / root rot	Wilting, grey-black dry rot at base	Seed treatment: Thiram 3 g/kg plus Carbendazim as advised	*Trichoderma* 4-10 g/kg seed; avoid drought stress
Defoliators (Spodoptera, semilooper)	Chewed leaves, skeleton leaves	Chlorantraniliprole 0.3 ml/L; Emamectin 0.4 g/L	SlNPV 250 LE/ha; *Bt* 2 g/L; neem oil 5 ml/L
Girdle beetle and stem fly	Girdled branches, wilting tops	Chlorantraniliprole 0.3 ml/L as per label	Early sowing; remove affected plant parts
Yellow mosaic virus	Yellow mottled leaves, poor pods	No cure. Control whitefly: Imidacloprid 0.3 ml/L	Resistant varieties; yellow sticky traps

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Stem fly	Wilting of young plants, tunnels in stems	Seed treatment with Thiamethoxam as per label	Early sowing; resistant varieties
Bacterial pustule and frogeye leaf spot	Small pustules, round spots on leaves	Copper oxychloride 3 g/L; Carbendazim 1 g/L	Rotation; clean seed

105. Mustard

Hindi / Telugu name	sarasoṃ / āvālu
Best season	Rabi (Oct-Nov)
Best soil	Light to medium loam; sandy loam; avoids waterlogging
Soil pH	6.5-8.0
Main growing states	Rajasthan, UP, Haryana, MP, West Bengal
Fertilizer (N : P2O5 : K2O)	60-80 : 40 : 40 (sulphur 30-40 kg/ha)
Nutrient tip	Sulphur is important for oil; boron
Crop duration	110-140 days
Seed rate / spacing	Seed 4-5 kg/ha; 30 x 10 cm
Water need	Low (2-3 irrigations)
Typical yield	1.5-2.5 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Alternaria blight	Dark concentric spots on leaves and pods	Mancozeb 2.5 g/L; Iprodione 2 g/L	Clean seed; early sowing; crop rotation
White rust	White blisters on leaf underside, staghead on flowers	Metalaxyl plus Mancozeb 2.5 g/L	Resistant varieties; remove infected plants
Mustard aphid	Dense colonies on flower stalk and pods	Imidacloprid 0.3 ml/L as per label	Neem oil 5 ml/L; *Beauveria* 5 g/L; early sowing; ladybirds and syrphid flies help
Powdery mildew	White powder on leaves	Wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L
Painted bug and sawfly	Chewed seedlings, sap sucking	Follow label	Neem oil 5 ml/L; irrigate to reduce
Stem rot (Sclerotinia)	White mouldy stem rot, bleached stems	Carbendazim 1 g/L at flowering	Crop rotation; avoid dense sowing

106. Sunflower

Hindi / Telugu name	sūrajamukhī / poddutiruguḍu
Best season	Kharif, Rabi and summer
Best soil	Well-drained medium black and red soils
Soil pH	6.5-8.0
Main growing states	Karnataka, AP, Telangana, Maharashtra
Fertilizer (N : P2O5 : K2O)	60-90 : 60 : 40 kg/ha
Nutrient tip	Boron deficiency causes hollow stem and poor seed set
Crop duration	90-110 days
Seed rate / spacing	Hybrid seed 5-6 kg/ha; 60 x 30 cm
Water need	Moderate
Typical yield	1.5-2.5 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Downy mildew	Pale patches, stunting, white fuzz under leaves	Seed treatment: Metalaxyl 6 g/kg	Resistant hybrids; field sanitation; rotation
Alternaria leaf spot and rust	Brown spots on leaves, orange pustules	Mancozeb 2.5 g/L	*P. fluorescens* 10 g/L; resistant hybrids
Head borer (*Helicoverpa*)	Larvae feeding on developing head	Emamectin benzoate 0.4 g/L; Spinosad 0.3 ml/L	HaNPV 250 LE/ha; *Bt* 2 g/L; pheromone traps
Thrips and necrosis virus	Leaf curling, necrotic spots, stunted plants	Imidacloprid 0.3 ml/L; Fipronil 1.5 ml/L	Blue sticky traps; early sowing; border crop of sorghum or maize

107. Sesame (Til)

Hindi / Telugu name	til / nuvvulu
Best season	Kharif and summer
Best soil	Light well-drained sandy loam; avoids waterlogging
Soil pH	5.5-8.0
Main growing states	Rajasthan, Gujarat, UP, MP, West Bengal
Fertilizer (N : P2O5 : K2O)	40-50 : 20-30 : 20 kg/ha
Nutrient tip	Sulphur improves oil
Crop duration	80-100 days
Seed rate / spacing	Seed 4-5 kg/ha; 30 x 10 cm
Water need	Low
Typical yield	0.6-1 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Phyllody	Flowers turn into leafy structures, no seed	No cure. Control leafhopper with Imidacloprid 0.3 ml/L	Rogue out infected plants early; early sowing
Alternaria leaf spot	Brown spots with yellow halo	Mancozeb 2.5 g/L	*P. fluorescens*; clean seed
Stem and root rot (*Macrophomina*)	Wilting, black dry rot at base	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed; avoid drought stress
Leaf webber and capsule borer	Webbed leaves, bored capsules	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L

108. Safflower

Hindi / Telugu name	kusum / kusuma
Best season	Rabi (Sep-Oct)
Best soil	Deep black soils; drought tolerant
Soil pH	6.0-8.0
Main growing states	Maharashtra, Karnataka
Fertilizer (N : P2O5 : K2O)	40-50 : 25-30 : 0-20 kg/ha
Nutrient tip	Hardy under residual moisture
Crop duration	120-140 days
Seed rate / spacing	Seed 15-20 kg/ha; 45 x 20 cm
Water need	Low
Typical yield	1-1.5 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Alternaria leaf spot	Brown spots with rings on leaves	Mancozeb 2.5 g/L	*P. fluorescens*; clean seed
Wilt and root rot	Yellowing, wilting	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed; crop rotation
Aphids	Dense colonies on shoots	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L; ladybirds help
Capsule borer	Bored capsules	Emamectin benzoate 0.4 g/L	HaNPV; *Bt* 2 g/L

109. Castor

Hindi / Telugu name	araṃḍī / āmudaṃ
Best season	Kharif (Jun-Aug)
Best soil	Well-drained sandy loam and red soils; avoids waterlogging
Soil pH	5.0-6.5
Main growing states	Gujarat, Rajasthan, Telangana, AP
Fertilizer (N : P2O5 : K2O)	60-80 : 30-40 : 30 kg/ha
Nutrient tip	Responds to nitrogen and sulphur
Crop duration	150-210 days
Seed rate / spacing	Seed 5-6 kg/ha; 90 x 60 cm

Water need	Low
Typical yield	1.5-2.5 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Wilt	Yellowing, drooping and plant death	Drench Carbendazim 1 g/L	*Trichoderma* seed treatment 4 g/kg; resistant hybrids; rotation
Alternaria leaf blight	Concentric brown spots on leaves	Mancozeb 2.5 g/L	*P. fluorescens* 10 g/L
Grey mould (*Botrytis*)	Grey mould on spikes in humid weather	Carbendazim 1 g/L	Resistant varieties; good air flow
Semilooper, *Spodoptera* and capsule borer	Defoliation, bored capsules	Emamectin benzoate 0.4 g/L	SlNPV 250 LE/ha; *Bt* 2 g/L
Jassids	Yellowing, curling	Imidacloprid 0.3 ml/L	neem oil 5 ml/L

110. Linseed

Hindi / Telugu name	alasī / avise
Best season	Rabi (Oct-Nov)
Best soil	Medium to heavy loam and black soils
Soil pH	6.0-7.5
Main growing states	MP, UP, Chhattisgarh, Maharashtra
Fertilizer (N : P2O5 : K2O)	50-60 : 30 : 20 kg/ha
Nutrient tip	Sulphur and boron for yield
Crop duration	110-130 days
Seed rate / spacing	Seed 25-30 kg/ha; 25 x 5-7 cm
Water need	Low
Typical yield	0.8-1.2 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Rust	Orange pustules on leaves and stems	Mancozeb 2.5 g/L; Hexaconazole 1 ml/L	Resistant varieties; early sowing
Powdery mildew	White powder on leaves	wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L
Wilt	Yellowing, wilting	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed
Bud fly	Damaged buds, no flowers	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; early sowing

111. Niger (Ramtil)

Hindi / Telugu name	rāmatil / verrinuvvulu
Best season	Kharif (Jul-Aug)
Best soil	Poor shallow red and lateritic soils; tolerates poor drainage better than most
Soil pH	5.5-7.0
Main growing states	Odisha, Chhattisgarh, Karnataka, Maharashtra, MP
Fertilizer (N : P2O5 : K2O)	20 : 20 : 10 kg/ha
Nutrient tip	Low-input tribal-area crop
Crop duration	90-110 days
Seed rate / spacing	Seed 5-7 kg/ha; 30 x 10 cm
Water need	Low
Typical yield	0.4-0.7 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spots (*Cercospora*, *Alternaria*)	Brown spots on leaves	Mancozeb 2.5 g/L	Clean seed; rotation
Powdery mildew	White powder on leaves	wettable sulphur 2.5 g/L	*Bacillus subtilis*
Caterpillars and aphids	Defoliation, curling	Emamectin benzoate 0.4 g/L	neem oil 5 ml/L; *Bt* 2 g/L

112. Toria and Taramira (Rapeseed)

Hindi / Telugu name	toriyā / tārāmīrā / toriyā
Best season	Toria: Sep sowing; Taramira: Oct (Rabi, dry areas)
Best soil	Light to medium loam; sandy loam; taramira on dry sandy soils
Soil pH	6.5-8.0
Main growing states	Rajasthan, Haryana, UP, Punjab, MP
Fertilizer (N : P2O5 : K2O)	40-60 : 30 : 0 (sulphur 20 kg/ha)
Nutrient tip	Sulphur improves oil; taramira suits low rainfall
Crop duration	Toria 90-100 days; taramira 130-150 days
Seed rate / spacing	Seed 4-5 kg/ha; 30 x 10 cm
Water need	Low
Typical yield	Toria 1-1.5 t/ha; taramira 0.8-1.2 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Alternaria blight	Dark spots with rings on leaves and pods	Mancozeb 2.5 g/L	Clean seed; early sowing
White rust	White blisters under leaves	Metalaxyl plus Mancozeb 2.5 g/L	Resistant varieties; crop rotation
Aphids	Dense colonies on flower stalk and pods	Imidacloprid 0.3 ml/L as per label	neem oil 5 ml/L; *Beauveria* 5 g/L
Painted bug	Sap-sucking on seedlings	Follow label	Irrigate early; neem oil 5 ml/L

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113. Turmeric

Hindi / Telugu name	haldī / pasupu
Best season	Plant Apr-Jun; harvest Jan-Mar
Best soil	Well-drained sandy loam and clay loam rich in organic matter
Soil pH	5.5-7.5
Main growing states	Telangana, AP, Tamil Nadu, Karnataka, Odisha
Fertilizer (N : P2O5 : K2O)	120 : 60 : 120 (plus FYM 25 t/ha)
Nutrient tip	Zinc and iron in alkaline soils; avoid waterlogging
Crop duration	7-9 months
Seed rate / spacing	Seed rhizomes 2-2.5 t/ha; 45 x 25 cm
Water need	Moderate
Typical yield	20-30 t/ha fresh

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Rhizome rot	Yellowing, soft rotting rhizomes	Seed rhizome dip: Mancozeb 2.5 g/L for 30 minutes; drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma* in soil with FYM; healthy seed rhizomes; drainage
Leaf blotch and leaf spot	Brown blotches and spots on leaves	Mancozeb 2.5 g/L; Propiconazole 1 ml/L	*P. fluorescens*; remove infected leaves
Shoot borer	Dead hearts, frass at holes	Emamectin benzoate 0.4 g/L	neem oil 5 ml/L; remove affected shoots
Rhizome scale	White crusty scales on stored rhizomes	Dip seed rhizomes in label-approved solution	neem oil 5 ml/L dip; discard infested rhizomes

114. Ginger

Hindi / Telugu name	adarak / allaṃ
Best season	Plant Apr-May; harvest Dec-Jan
Best soil	Well-drained loam rich in organic matter; avoids waterlogging
Soil pH	5.5-6.5
Main growing states	Kerala, Karnataka, Assam, Odisha, Meghalaya
Fertilizer (N : P2O5 : K2O)	75 : 50 : 50 (plus FYM 25-30 t/ha)
Nutrient tip	Zinc; needs excellent drainage
Crop duration	8-9 months
Seed rate / spacing	Seed rhizomes 1.5-2.5 t/ha; 30 x 20 cm
Water need	Moderate
Typical yield	15-25 t/ha fresh

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Soft rot (*Pythium*)	Yellowing, water-soaked, foul-smelling rhizomes	Drench Metalaxyl plus Mancozeb 3 g/L; seed dip Mancozeb 3 g/L for 30 min	*Trichoderma* in soil with neem cake; healthy seed; raised beds
Bacterial wilt	Sudden green wilting, milky ooze from stem	No effective spray; remove infected clumps	Healthy seed; *P. fluorescens*; rotation
Leaf spot	Oval brown spots on leaves	Mancozeb 2.5 g/L	Remove infected leaves; mulching
Shoot borer	Dead hearts in shoots	Emamectin benzoate 0.4 g/L	neem oil 5 ml/L; remove affected shoots

115. Black pepper

Hindi / Telugu name	kālī mirc / miriyālu
Best season	Plant with monsoon (Jun-Jul)
Best soil	Well-drained loam, laterite and forest soils with shade; heavy rainfall regions
Soil pH	5.0-6.5
Main growing states	Kerala, Karnataka, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	About 50 g N, 50 g P2O5, 150 g K2O per vine per year plus FYM
Nutrient tip	Avoid waterlogging; lime for acidic soils
Crop duration	Bears from 3-4 years
Seed rate / spacing	2.5 x 2.5 m on live standards
Water need	Moderate rainfall 2000+ mm
Typical yield	1-2 kg dry pepper per vine

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Quick wilt / foot rot (*Phytophthora*)	Sudden leaf drop and vine collapse in monsoon	Bordeaux mixture 1% on leaves; drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma* and *P. fluorescens* in basin; good drainage; healthy cuttings
Anthracnose (*Colletotrichum*)	Brown spots on leaves and berries	Carbendazim 1 g/L; Bordeaux mixture 1%	Prune and shade management
Pollu beetle	Hollow, light berries	Follow label and local advice	neem oil 5 ml/L; *Beauveria* 5 g/L
Slow decline (nematodes)	Gradual yellowing and decline	Follow label	Neem cake in basin; *Pasteuria* / *Paecilomyces* as per KVK

116. Cardamom

Hindi / Telugu name	ilāyacī / yālakulu
Best season	Plant Jun-Jul; harvest Oct-Jan
Best soil	Forest loam, humus rich, under shade at 600-1500 m
Soil pH	5.0-6.5
Main growing states	Kerala, Karnataka, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	About 75 : 75 : 150 per ha
Nutrient tip	Needs shade, moisture and mulch
Crop duration	Bears from about 3 years
Seed rate / spacing	2 x 2 m to 2.5 x 2.5 m under shade
Water need	Moderate with shade
Typical yield	150-300 kg/ha dry

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Katte (mosaic virus)	Pale streaks on leaves, stunted clumps	No cure. Control aphid vector with neem oil 5 ml/L	Rogue and destroy infected clumps; virus-free suckers
Capsule rot (Azhukal)	Rotting capsules in monsoon	Bordeaux mixture 1% before monsoon	Prune and thin shade; remove infected capsules
Thrips	Corky, scabby capsules	Follow label	neem oil 5 ml/L; *Beauveria* 5 g/L; trash removal
Rhizome rot	Yellowing clumps, rotted rhizomes	Drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma* in soil; drainage

117. Coriander

Hindi / Telugu name	dhaniyā / kottimīra / dhaniyālu
Best season	Rabi (Oct-Nov); leaves year-round
Best soil	Well-drained loam and black soils
Soil pH	6.5-7.5
Main growing states	Rajasthan, MP, Gujarat, AP, Karnataka
Fertilizer (N : P2O5 : K2O)	60 : 40 : 20 kg/ha
Nutrient tip	Sulphur for oil content
Crop duration	90-110 days (seed); 30-45 days (leaf)
Seed rate / spacing	Seed 10-15 kg/ha; 30 x 10 cm
Water need	Low to moderate
Typical yield	1-1.5 t/ha seed; 6-8 t/ha leaf

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves and umbels	wettable sulphur 2.5 g/L; Hexaconazole 1 ml/L	*Bacillus subtilis* 2-5 g/L
Stem gall	Swellings and galls on stems, umbels	Hexaconazole 1 ml/L	*Trichoderma* seed treatment 4 g/kg; clean seed
Wilt	Yellowing and wilting	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed; rotation
Aphids	Clusters on umbels	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

118. Cumin

Hindi / Telugu name	jīrā / jīlakarra
Best season	Rabi (Nov)
Best soil	Light well-drained sandy loam and loam; cool dry weather
Soil pH	7.0-8.5
Main growing states	Gujarat, Rajasthan
Fertilizer (N : P2O5 : K2O)	30 : 20 : 0 kg/ha
Nutrient tip	Sensitive to humidity and disease; avoid cloudy weather
Crop duration	100-120 days
Seed rate / spacing	Seed 12-15 kg/ha; 30 x 10 cm
Water need	Low (3-4 irrigations)
Typical yield	0.6-1 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Wilt (Fusarium)	Drooping and drying of seedlings and plants	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed; crop rotation; summer ploughing
Alternaria blight	Brown spots, drying of whole plant in humid weather	Mancozeb 2.5 g/L	Early sowing; avoid late irrigation
Powdery mildew	White powder on leaves	wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L
Aphids	Clusters on tender parts	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

119. Fennel

Hindi / Telugu name	saumph / soṃpu
Best season	Rabi (Oct-Nov)
Best soil	Well-drained loam; black and alluvial
Soil pH	6.5-8.0
Main growing states	Gujarat, Rajasthan, UP
Fertilizer (N : P2O5 : K2O)	90 : 40 : 30 kg/ha
Nutrient tip	Sulphur improves oil content
Crop duration	150-180 days
Seed rate / spacing	Seed 8-10 kg/ha; 45 x 20 cm
Water need	Low
Typical yield	1.5-2 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Ramularia blight	Brown spots on leaves and umbels	Mancozeb 2.5 g/L	Clean seed; crop rotation
Powdery mildew	White powder on leaves and umbels	wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L
Aphids	Clusters on umbels	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

120. Fenugreek (seed crop)

Hindi / Telugu name	methī / meṃtulu
Best season	Rabi (Oct-Nov)
Best soil	Well-drained loam; tolerates a range of soils
Soil pH	6.0-7.5
Main growing states	Rajasthan, Gujarat, MP, UP
Fertilizer (N : P2O5 : K2O)	20 : 40 : 20 kg/ha
Nutrient tip	Rhizobium culture
Crop duration	100-120 days
Seed rate / spacing	Seed 20-25 kg/ha; 30 x 10 cm
Water need	Low
Typical yield	1-1.5 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves	wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L
Root rot	Yellowing, rotted roots	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed
Aphids	Clusters on tender shoots	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

121. Ajwain (Carom)

Hindi / Telugu name	ajavāin / vāma
Best season	Rabi (Oct-Nov)
Best soil	Well-drained loam and black soils
Soil pH	6.5-8.0
Main growing states	Rajasthan, Gujarat, MP, AP
Fertilizer (N : P2O5 : K2O)	40-60 : 30 : 0 kg/ha
Nutrient tip	Sensitive to waterlogging
Crop duration	120-150 days
Seed rate / spacing	Seed 2.5-3 kg/ha; 45 x 30 cm
Water need	Low
Typical yield	0.8-1.2 t/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves and umbels	wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L
Root rot	Yellowing, rotted roots	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed
Aphids	Clusters on umbels	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

122. Clove

Hindi / Telugu name	lauṃg / lavaṃgālu
Best season	Plant Jun-Jul
Best soil	Deep loam, laterite and humus-rich soils with shade in humid tropics
Soil pH	5.5-6.5
Main growing states	Tamil Nadu (Kanyakumari), Kerala, Karnataka
Fertilizer (N : P2O5 : K2O)	As per Spices Board advice with FYM
Nutrient tip	Needs shade for young plants; avoid waterlogging
Crop duration	Flowering from 6-8 years
Seed rate / spacing	6 x 6 m to 7 x 7 m
Water need	High rainfall regions
Typical yield	3-4 kg dry buds per tree at full bearing

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Die-back and leaf rot	Drying of shoots, leaf rot	Bordeaux mixture 1% or Carbendazim 1 g/L	*Trichoderma*; prune infected shoots
Seedling wilt	Wilting and death of young plants	Drench Carbendazim 1 g/L	*Trichoderma*; shade and drainage
Stem borer	Frass at holes in stem	Follow label and local advice	Clean holes; plug with neem-soaked cotton

123. Nutmeg

Hindi / Telugu name	jāyaphal / jājikāya
Best season	Plant Jun-Jul
Best soil	Deep loam rich in humus under shade; well-drained
Soil pH	5.5-7.0
Main growing states	Kerala, Tamil Nadu, Karnataka, Andaman
Fertilizer (N : P2O5 : K2O)	As per Spices Board advice with FYM 25-50 kg per tree
Nutrient tip	Needs partial shade and mulching
Crop duration	Bears from 6-8 years
Seed rate / spacing	8 x 8 m
Water need	Moderate; rainfall 1500+ mm
Typical yield	Varies widely; ask Spices Board for local yield

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fruit rot and shoot die-back	Rotting fruit, drying shoots	Bordeaux mixture 1%	*Trichoderma*; prune infected shoots; drainage
Thread blight	Thread-like fungus webbing on shoots	Carbendazim 1 g/L	Prune infected parts; improve air flow
Scale insects	Crusty patches on stems	Follow label	neem oil 5 ml/L; *Lecanicillium* 5 g/L

124. Cinnamon

Hindi / Telugu name	dālacīnī / dālcina cekka
Best season	Plant Jun-Jul
Best soil	Sandy loam and laterite; well-drained in humid tropics
Soil pH	5.0-6.5
Main growing states	Kerala, Karnataka, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	As per Spices Board advice with FYM
Nutrient tip	Coppice every 2 years for bark
Crop duration	First bark harvest at 2-3 years
Seed rate / spacing	2 x 2 m (bush)
Water need	Moderate; humid climate
Typical yield	200-300 kg/ha dry bark

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot and die-back	Brown spots, drying shoots	Mancozeb 2.5 g/L; Bordeaux mixture 1%	Prune infected parts
Stripe canker	Longitudinal cankers on stems	Bordeaux paste on cankers	Remove infected stems
Leaf miner and caterpillars	Mined and chewed leaves	Follow label	neem oil 5 ml/L; *Bt* 2 g/L

125. Vanilla

Hindi / Telugu name	venilā / venillā
Best season	Plant Jun-Sep
Best soil	Loose humus-rich mulched soil under 50% shade; good drainage
Soil pH	6.0-7.0
Main growing states	Kerala, Karnataka, Tamil Nadu, Northeast
Fertilizer (N : P2O5 : K2O)	FYM and mulch plus balanced NPK as per Spices Board
Nutrient tip	Hand pollination is required for beans
Crop duration	Flowers from 3 years
Seed rate / spacing	Cuttings 1.5-2 m long at 2 x 1.2 m on live supports
Water need	Moderate; humid
Typical yield	300-500 kg green beans/ha at full bearing

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Root and stem rot (*Fusarium* / *Phytophthora*)	Rotting roots and stem, vine drops	Bordeaux mixture 1%; drench Carbendazim 1 g/L	*Trichoderma* in mulch; drainage; healthy cuttings
Anthracnose and leaf blight	Brown spots on leaves and beans	Mancozeb 2.5 g/L	Prune for air flow; remove infected parts
Slugs and snails	Chewed buds and young shoots	Follow label	Hand-pick; clean surroundings

126. Saffron

Hindi / Telugu name	kesar / kuṃkumapuvvu
Best season	Plant corms Aug-Sep; flowers Oct-Nov
Best soil	Well-drained loamy (karewa) soils in cool, dry climate
Soil pH	6.0-8.0
Main growing states	J&K; (Pampore, Kishtwar)
Fertilizer (N : P2O5 : K2O)	FYM 20 t/ha plus modest NPK
Nutrient tip	Needs cool climate and sharp drainage; very limited areas
Crop duration	Flowers Oct-Nov; corm cycle of several years
Seed rate / spacing	Corms 3-4 t/ha; planting depth 10-15 cm
Water need	Low; avoid waterlogging
Typical yield	2-3 kg dry saffron/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Corm rot (*Fusarium*)	Rotted corms, poor flowers	Corm dip: Carbendazim 1 g/L for 30 minutes	*Trichoderma*; healthy corms; rotation
Rodents	Corms eaten in field	Follow local advice	Clean surroundings; traps
Leaf rust	Orange spots on leaves	Mancozeb 2.5 g/L	Remove infected leaves

127. Tea

Hindi / Telugu name	cāy / teyāku
Best season	Year-round; flush from Mar-Nov
Best soil	Deep acidic well-drained loam at hill and plain estates
Soil pH	4.5-5.5
Main growing states	Assam, West Bengal, Tamil Nadu, Kerala
Fertilizer (N : P2O5 : K2O)	Per Tea Board advice: about 100-200 N, 30-60 P2O5, 60-100 K2O
Nutrient tip	Needs acidic soil; zinc and magnesium
Crop duration	First plucking 3-4 years
Seed rate / spacing	About 11,000 bushes/ha; 1.2 x 0.75 m
Water need	High rainfall
Typical yield	1,500-2,500 kg made tea/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Blister blight	Translucent spots turning to blisters on young leaves	Follow Tea Board / UPASI approved fungicide schedule	Shade and plucking rounds; resistant clones
Red spider mite	Bronzed leaves in dry weather	Follow label and local advice (Tea Board schedule)	neem oil 5 ml/L; wash with water spray
Tea mosquito bug (*Helopeltis*)	Dark sunken spots on shoots	Follow label and local advice (Tea Board schedule)	neem oil 5 ml/L; *Beauveria* 5 g/L
Looper caterpillar	Chewed leaves	Follow label and local advice	*Bt* 2 g/L; NPV; hand collection

128. Coffee

Hindi / Telugu name	ka□phī / kāphī
Best season	Plant Jun-Aug; harvest Nov-Feb
Best soil	Well-drained loam rich in organic matter under shade
Soil pH	6.0-6.5
Main growing states	Karnataka, Kerala, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	Per Coffee Board schedule and soil test
Nutrient tip	Needs shade and mulch; lime for acid soils
Crop duration	Bears from 3-4 years
Seed rate / spacing	2 x 2 m (arabica); 2.5 x 2.5 m (robusta)
Water need	1500-2500 mm rainfall
Typical yield	0.8-1.5 t/ha clean coffee

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf rust	Orange powder under leaves, early leaf fall	Bordeaux mixture 1%; Hexaconazole 1 ml/L as per Coffee Board schedule	Resistant varieties; shade management; timely sprays
Coffee berry borer	Bored berries with powder frass	Follow Coffee Board schedule	*Beauveria* 5 g/L; strip picking; clean harvest; traps
White stem borer	Frass at stem, yellowing	Follow label and local advice	Shade regulation; scraping bark; destroy infested stems
Black rot	Black rotting of leaves and berries in wet weather	Bordeaux mixture 1%	Pruning for air flow
Mealybug	White cottony masses	Follow label	*Lecanicillium* 5 g/L; ant control

129. Rubber

Hindi / Telugu name	rabar / rabbaru
Best season	Plant with monsoon (Jun-Jul)
Best soil	Deep well-drained laterite and red soils in humid tropics
Soil pH	4.5-6.5
Main growing states	Kerala, Tamil Nadu, Tripura
Fertilizer (N : P2O5 : K2O)	Per Rubber Board schedule
Nutrient tip	Needs mulch; zinc and magnesium
Crop duration	Tapping from 6-7 years
Seed rate / spacing	About 450 trees/ha; 4.9 x 4.9 m
Water need	High rainfall
Typical yield	1.5-2 t/ha dry rubber

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Abnormal leaf fall (*Phytophthora*)	Leaf fall, rotting fruit in monsoon	Copper oxychloride oil-based spray as per Rubber Board schedule	Timely spraying before monsoon; clean planting
Powdery mildew	White powder on new leaves	Sulphur dusting as per Rubber Board	Follow board schedule
Pink disease	Pink crust and cankers on branches	Bordeaux paste or approved fungicide on stem	Prune infected branches; improve drainage
Tapping panel dryness	Drying of tapping cut	Follow Rubber Board advice	Reduce tapping intensity; balanced nutrition

130. Coconut

Hindi / Telugu name	nāriyal / kobbari
Best season	Plant with monsoon or Sep-Oct
Best soil	Sandy loam, laterite and coastal alluvium with good drainage
Soil pH	5.2-8.0
Main growing states	Kerala, Karnataka, Tamil Nadu, AP
Fertilizer (N : P2O5 : K2O)	About 0.5 kg N, 0.3 kg P2O5, 1.0 kg K2O per palm per year (CPCRI)
Nutrient tip	Boron deficiency; apply borax 50 g per palm where needed
Crop duration	Bears in 3-4 years (dwarf and hybrid), 5-7 years (tall)
Seed rate / spacing	7.5 x 7.5 m (about 175 palms/ha)
Water need	Moderate
Typical yield	80-150 nuts per palm

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Bud rot	Rotting of central spindle in monsoon	Remove rotten tissue; Bordeaux mixture 1% to crown	*Trichoderma* in crown; good drainage
Basal stem rot (*Ganoderma*)	Bleeding patches on trunk, yellowing fronds	Root feeding: Hexaconazole 5 EC 2 ml in 100 ml water (CPCRI advice)	*Trichoderma* with neem cake in basin; avoid injury
Red palm weevil	Tunnels in crown, falling fronds, frass	Follow label and local advice	Pheromone traps; clean crown; do not injure palm
Rhinoceros beetle	V-cut fronds from feeding	Follow label	Clean breeding sites; *Metarhizium* in manure pits; pheromone traps
Eriophyid mite	Triangular yellow patches, small nuts	Follow CPCRI advice (root feeding or spray)	neem oil 5 ml/L with soap spray on bunches; balanced nutrition

131. Arecanut

Hindi / Telugu name	supārī / poka / vakka
Best season	Plant with monsoon
Best soil	Well-drained loam and laterite with shade
Soil pH	5.0-6.5
Main growing states	Karnataka, Kerala, Assam, West Bengal
Fertilizer (N : P2O5 : K2O)	About 100 g N, 40 g P2O5, 140 g K2O per palm per year plus FYM
Nutrient tip	Needs irrigation and good drainage
Crop duration	Bears from 5-7 years
Seed rate / spacing	2.7 x 2.7 m (about 1,370 palms/ha)
Water need	Moderate
Typical yield	2-3 t/ha processed nut

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Koleroga (fruit rot)	Dark water-soaked rotting nuts, nut drop in monsoon	Bordeaux mixture 1% before and during monsoon	Remove infected bunches; pruning; drainage
Yellow leaf disease	Yellowing of fronds, poor nuts	No cure	Remove severely affected palms; healthy seedlings
Foot rot (*Ganoderma*)	Rotting at base of trunk	Drench Hexaconazole 1 ml/L	*Trichoderma* in basin
Spindle bug and mites	Damaged leaves and nuts	Follow label and local advice	neem oil 5 ml/L

132. Cashew

Hindi / Telugu name	kājū / jīḍimāmiḍi
Best season	Plant with monsoon
Best soil	Sandy loam and laterite; well-drained; coastal
Soil pH	5.0-6.5
Main growing states	Maharashtra, Kerala, Karnataka, AP, Odisha
Fertilizer (N : P2O5 : K2O)	About 500 g N, 200 g P2O5, 200 g K2O per tree per year
Nutrient tip	Hardy; responds to nitrogen and zinc
Crop duration	Bears from 3-4 years
Seed rate / spacing	7 x 7 m (high density 5 x 4 m)
Water need	Low
Typical yield	1-1.5 t/ha raw nuts

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Tea mosquito bug	Dark lesions on shoots, flowers and nuts	Follow DCR / local KVK spray schedule	neem oil 5 ml/L; timely sprays during flush
Anthracnose and die-back	Dark lesions on shoots and leaves	Mancozeb 2.5 g/L; Carbendazim 1 g/L	Prune dead twigs
Stem and root borer	Gum, frass at base, yellowing	Follow label and local advice	Clean tunnels; remove severely infested trees
Powdery mildew	White powder on flowers	wettable sulphur 2.5 g/L	Spray at first sign

133. Oil palm

Hindi / Telugu name	yala pām / āyil pām
Best season	Plant with monsoon
Best soil	Deep loam with good drainage and heavy irrigation
Soil pH	4.0-7.5
Main growing states	AP, Telangana, Karnataka, Tamil Nadu, Kerala
Fertilizer (N : P2O5 : K2O)	Per state advice: about 1-2 kg N, 0.5 kg P2O5, 2 kg K2O per palm
Nutrient tip	Boron deficiency (hook leaf)
Crop duration	Bears from 3-4 years
Seed rate / spacing	9 m triangular (about 143 palms/ha)
Water need	High (1500-2000 mm)
Typical yield	15-25 t/ha fruit bunches

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Basal stem rot (*Ganoderma*)	Fronds hang, bracket fungus at trunk base	Follow state horticulture advice	*Trichoderma* at planting; remove infected stumps
Bud rot / spear rot	Rotting of spear leaf	Remove rotten tissue; Bordeaux paste	*Trichoderma*; drainage
Rhinoceros beetle	Damage to spear leaves	Follow label	Pheromone traps; *Metarhizium* in breeding sites
Bagworm and leaf eating caterpillars	Defoliation	Follow label and local advice	*Bt* 2 g/L; NPV

134. Cocoa

Hindi / Telugu name	koko / koko
Best season	Plant Jun-Aug under shade (coconut or arecanut garden)
Best soil	Deep well-drained loam rich in humus; humid tropics
Soil pH	6.0-7.0
Main growing states	Kerala, Karnataka, Tamil Nadu, AP
Fertilizer (N : P2O5 : K2O)	About 100 g N, 40 g P2O5, 140 g K2O per mature tree per year
Nutrient tip	Grown as intercrop under shade; mulch
Crop duration	Pods from 3-4 years
Seed rate / spacing	3 x 3 m as intercrop under coconut or arecanut
Water need	Moderate
Typical yield	1-2 kg dry beans per tree

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Black pod (*Phytophthora*)	Dark rotting pods in wet weather	Bordeaux mixture 1%; copper oxychloride 3 g/L	Remove infected pods; prune for air flow
Vascular streak dieback	Yellow leaves, dying branches	Prune infected branches; follow local advice	Remove and burn infected parts
Mealybug and stem borer	White masses, frass on stems	Follow label	neem oil 5 ml/L; *Lecanicillium* 5 g/L

FLOWERS

135. Rose

Hindi / Telugu name	gulāb / gulābī
Best season	Plant Oct-Nov; year-round with care
Best soil	Well-drained loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	Karnataka, Maharashtra, Tamil Nadu, West Bengal
Fertilizer (N : P2O5 : K2O)	About 40 g N per plant in splits plus FYM
Nutrient tip	Iron deficiency in alkaline soils
Crop duration	Flowers from 3-4 months; productive 5-8 years
Seed rate / spacing	75 x 60 cm
Water need	Moderate
Typical yield	Flowers in 2-3 flushes a year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Black spot	Black spots with yellow halo, leaf drop	Mancozeb 2.5 g/L; Hexaconazole 1 ml/L	Remove fallen leaves; avoid overhead water
Powdery mildew	White powder on shoots and buds	Wettable sulphur 2 g/L; Hexaconazole 1 ml/L	*Bacillus subtilis* 2-5 g/L
Dieback	Drying of shoots from tip	Bordeaux paste on cut ends after pruning	Prune 2-3 cm below dead portion; burn clippings
Thrips and mites	Browned petals, bronzed leaves	Spiromesifen 1 ml/L; Fipronil 1.5 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L; blue sticky traps

136. Marigold

Hindi / Telugu name	geṃdā / baṃti
Best season	Kharif, Rabi and summer
Best soil	Well-drained sandy loam
Soil pH	6.5-7.5
Main growing states	Karnataka, West Bengal, Tamil Nadu, Maharashtra
Fertilizer (N : P2O5 : K2O)	100 : 60 : 60 kg/ha
Nutrient tip	Easy crop; avoid waterlogging
Crop duration	90-120 days
Seed rate / spacing	Seed 1.5-2 kg/ha nursery; 45 x 30 cm
Water need	Moderate
Typical yield	20-25 t/ha flowers

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Damping off	Seedling collapse in nursery	Drench Metalaxyl plus Mancozeb 2 g/L	*Trichoderma* 4 g/kg seed; raised beds
Leaf spot and blight	Brown spots, blighting of leaves and flowers	Mancozeb 2.5 g/L	*P. fluorescens*; avoid overhead water
Red spider mite	Webbing and bronze leaves	Propargite 2 ml/L	neem oil 5 ml/L; water spray
Helicoverpa bud worm	Bored flower buds	Emamectin benzoate 0.4 g/L	HaNPV; *Bt* 2 g/L; pheromone traps

137. Jasmine

Hindi / Telugu name	camelī / mogarā / malle
Best season	Plant Jun-Jul
Best soil	Well-drained red loam and sandy loam
Soil pH	6.0-7.0
Main growing states	Tamil Nadu, Karnataka, AP
Fertilizer (N : P2O5 : K2O)	About 120 : 120 : 240 per ha per year in splits
Nutrient tip	Prune after each flush
Crop duration	Flowers from 1-2 years
Seed rate / spacing	2 x 2 m
Water need	Moderate
Typical yield	3-8 t/ha flowers

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Rust	Orange pustules under leaves	Mancozeb 2.5 g/L	Remove infected leaves; pruning
Leaf blight	Brown blight on leaves and shoots	Mancozeb 2.5 g/L; Carbendazim 1 g/L	*P. fluorescens*
Bud worm	Larvae feeding on buds	Spinosad 0.3 ml/L as per label	*Bt* 2 g/L; neem oil 5 ml/L
Mites and thrips	Curled, bronzed leaves	Spiromesifen 1 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

138. Chrysanthemum

Hindi / Telugu name	guladāudī / cemaṃti
Best season	Planting Jun-Aug (short-day flowering)
Best soil	Well-drained loam rich in organic matter
Soil pH	6.0-7.0
Main growing states	Karnataka, Tamil Nadu, Maharashtra, West Bengal
Fertilizer (N : P2O5 : K2O)	120 : 80 : 120 kg/ha
Nutrient tip	Responds to pinching and balanced nutrition
Crop duration	90-120 days to flower
Seed rate / spacing	30 x 30 cm
Water need	Moderate
Typical yield	10-15 t/ha loose flowers

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot (*Septoria*)	Brown to black spots on lower leaves	Mancozeb 2.5 g/L	Remove lower infected leaves; avoid overhead water
Rust	Pustules under leaves	Hexaconazole 1 ml/L	Resistant varieties
Wilt	Yellowing and wilting	Drench Carbendazim 1 g/L	*Trichoderma*; disease-free cuttings
Aphids	Clusters on buds and tips	Imidacloprid 0.3 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

139. Tuberose

Hindi / Telugu name	rajanīgaṃdhā / rajanīgaṃdha
Best season	Plant Feb-Apr
Best soil	Well-drained sandy loam
Soil pH	6.5-7.5
Main growing states	West Bengal, Tamil Nadu, Karnataka, AP
Fertilizer (N : P2O5 : K2O)	200 : 100 : 100 kg/ha
Nutrient tip	Needs sunny location
Crop duration	90-120 days to flower
Seed rate / spacing	Bulbs 60,000-100,000/ha; 30 x 30 cm
Water need	Moderate
Typical yield	10-15 t/ha loose flowers

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot and stem rot	Brown spots on leaves, rot at base	Mancozeb 2.5 g/L; Carbendazim 1 g/L	Healthy bulbs; drainage
Flower bud borer	Bored buds	Emamectin benzoate 0.4 g/L	*Bt* 2 g/L; neem oil 5 ml/L
Thrips	Silvery streaks on flowers	Follow label and local advice	Blue sticky traps; neem oil 5 ml/L
Root knot nematode	Galls on roots, stunted growth	Follow label	Neem cake in soil; marigold intercrop

140. Gerbera (polyhouse)

Hindi / Telugu name	jaraberā / jerberā
Best season	Year-round in polyhouse; plant Sep-Oct
Best soil	Light well-drained soil or cocopeat beds; protected cultivation
Soil pH	5.5-6.5
Main growing states	Karnataka, Maharashtra, Tamil Nadu, West Bengal
Fertilizer (N : P2O5 : K2O)	Fertigation: about 20 : 10 : 40 g per sq m per month (grower advice)
Nutrient tip	Iron deficiency at high pH; keep EC low
Crop duration	First flower in 3-4 months; productive 2-3 years
Seed rate / spacing	About 8 plants per sq m beds
Water need	Fertigation; moderate
Typical yield	200-250 stems per sq m per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves and buds	Hexaconazole 1 ml/L; wettable sulphur 2 g/L	Good ventilation; *Bacillus subtilis*
Botrytis grey mould	Grey mould on petals and buds	Carbendazim 1 g/L	Lower humidity; remove infected flowers
Whitefly and thrips	Silvery, distorted flowers	Spiromesifen 1 ml/L as per label	yellow sticky traps; neem oil 5 ml/L; *Beauveria* 5 g/L
Crown and root rot	Wilting, rotted crown	Drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma*; shallow planting; good drainage

141. Carnation (polyhouse)

Hindi / Telugu name	kārneśan / kārneṣan
Best season	Year-round in polyhouse in cool hills; plant Sep-Oct
Best soil	Light well-drained soil or coco beds; protected cultivation
Soil pH	6.0-7.0
Main growing states	Karnataka, Maharashtra, Himachal, Tamil Nadu (Nilgiris)
Fertilizer (N : P2O5 : K2O)	Fertigation as per grower advice
Nutrient tip	Calcium and boron for stem strength
Crop duration	First flower in 4-5 months; productive 1.5-2 years
Seed rate / spacing	About 30-40 plants per sq m
Water need	Fertigation; moderate
Typical yield	200-250 stems per sq m per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fusarium wilt	Yellowing, wilting, brown vascular tissue	Drench Carbendazim 1 g/L	*Trichoderma*; disease-free cuttings; steam or solarize beds
Rust	Brown pustules on leaves	Hexaconazole 1 ml/L	Remove infected leaves; reduce humidity
Red spider mite and thrips	Webbing, silvery petals	Spiromesifen 1 ml/L	neem oil 5 ml/L; *Beauveria* 5 g/L

142. Gladiolus

Hindi / Telugu name	glaiḍiyolas / glāḍiyolas
Best season	Plant Oct-Nov (plains); spring in hills
Best soil	Light sandy loam rich in organic matter; well-drained
Soil pH	6.0-7.0
Main growing states	West Bengal, UP, Maharashtra, Karnataka, Himachal
Fertilizer (N : P2O5 : K2O)	100-200 : 100 : 100 kg/ha
Nutrient tip	Needs full sun and good drainage
Crop duration	70-100 days to spike
Seed rate / spacing	Corms 1.5-2 lakh/ha; 30 x 20 cm
Water need	Moderate
Typical yield	1.5-2 lakh spikes/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fusarium corm rot	Yellowing, rotted corms	Corm dip: Carbendazim 1 g/L for 30 minutes	*Trichoderma*; healthy corms; rotation
Leaf spot and blight	Brown spots on leaves and spikes	Mancozeb 2.5 g/L	Remove infected leaves
Thrips	Silvery streaks on flowers	Fipronil 1.5 ml/L	Blue sticky traps; neem oil 5 ml/L
Botrytis	Grey mould on petals in humid weather	Carbendazim 1 g/L	Lower humidity; remove infected spikes

143. Anthurium (polyhouse)

Hindi / Telugu name	eṃthuriyam / āṃthūriyaṃ
Best season	Year-round in polyhouse
Best soil	Cocopeat or brick-charcoal media, well-drained; protected cultivation
Soil pH	5.5-6.5
Main growing states	Kerala, Karnataka, Tamil Nadu, Maharashtra
Fertilizer (N : P2O5 : K2O)	Fertigation with balanced nutrients as per grower advice
Nutrient tip	Needs 70-80% humidity and 75% shade
Crop duration	First flower in 10-12 months
Seed rate / spacing	About 40 x 40 cm beds or pots
Water need	Humid; moderate irrigation
Typical yield	6-8 flowers per plant per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Bacterial blight (*Xanthomonas*)	Water-soaked spots, yellowing, wilting	Streptocycline 0.1 g/L plus copper oxychloride 3 g/L	Disease-free plants; avoid overhead water
Root rot	Yellowing, rotted roots	Drench Metalaxyl plus Mancozeb 2.5 g/L	*Trichoderma*; well-drained media
Mealybug, thrips and mites	Cottony masses, scarred flowers	Spiromesifen 1 ml/L as per label	neem oil 5 ml/L; *Beauveria* 5 g/L

144. Orchids (Dendrobium, Cymbidium)

Hindi / Telugu name	□rkiḍ / ārkiḍ
Best season	Year-round in polyhouse
Best soil	Not soil grown: bark, coconut husk or charcoal media with good aeration
Soil pH	5.5-6.5
Main growing states	Sikkim, Kerala, Karnataka, Maharashtra, Arunachal
Fertilizer (N : P2O5 : K2O)	Dilute balanced fertilizer about 0.5 g/L weekly (grower advice) kg/ha
Nutrient tip	Needs 60-70% humidity and 50-75% shade
Crop duration	2-3 years to flower from seedlings
Seed rate / spacing	Pots or benches
Water need	Humid; moderate
Typical yield	2-3 flower spikes per plant per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Black rot (*Phytophthora*)	Black soft rot at base, leaf collapse	Drench Metalaxyl plus Mancozeb 2 g/L	Good air flow; avoid overwatering
Botrytis petal blight	Brown spots on petals	Carbendazim 1 g/L	Lower humidity; remove infected flowers
Scale, mealybug and mites	Crusty or cottony patches	Follow label	neem oil 5 ml/L; clean plants
Snails and slugs	Chewed buds and roots	Follow label	Hand-pick; clean benches

145. Crossandra (Kanakambaram)

Hindi / Telugu name	kanakāmbaram / abolī / kanakāṃbaraṃ
Best season	Plant Jun-Jul or Oct
Best soil	Red sandy loam, well-drained
Soil pH	6.0-7.5
Main growing states	Tamil Nadu, Karnataka, AP, Telangana
Fertilizer (N : P2O5 : K2O)	FYM 10 t/ha plus about 100 : 40 : 40 per year
Nutrient tip	Needs warm weather; avoid waterlogging
Crop duration	3-4 months to first flower
Seed rate / spacing	30 x 30 cm
Water need	Moderate
Typical yield	3-5 t/ha flowers per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Fusarium wilt	Yellowing, wilting	Drench Carbendazim 1 g/L	*Trichoderma*; healthy cuttings; drainage
Leaf spot	Brown spots on leaves	Mancozeb 2.5 g/L	Remove infected leaves
Mealybug, whitefly and mites	Cottony masses, curling	Follow label	neem oil 5 ml/L; *Beauveria* 5 g/L
Bud worm	Bored buds and spikes	Emamectin benzoate 0.4 g/L as per label	*Bt* 2 g/L; neem oil 5 ml/L

MEDICINAL AND AROMATIC CROPS

146. Ashwagandha

Hindi / Telugu name	aśvagaṃdhā / aśvagaṃdha
Best season	Kharif (Jul-Sep sowing)
Best soil	Light sandy loam and red soils; avoids waterlogging
Soil pH	7.5-8.0
Main growing states	MP, Rajasthan, Gujarat
Fertilizer (N : P2O5 : K2O)	20 : 20 : 0 (low input, mostly FYM)
Nutrient tip	Hardy dryland crop
Crop duration	150-180 days
Seed rate / spacing	Seed 5-10 kg/ha; 30 x 5-10 cm
Water need	Low
Typical yield	300-500 kg/ha dry roots

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Seedling rot and blight	Seedlings collapse at base	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed; raised beds; drainage
Leaf spot	Brown leaf spots	Mancozeb 2.5 g/L	*P. fluorescens*
Aphids and mites	Curling, yellowing	Follow label and local advice	neem oil 5 ml/L; *Beauveria* 5 g/L

147. Aloe vera

Hindi / Telugu name	eloverā / ghṛtakumārī / kalabaṃda
Best season	Plant Jul-Aug or Feb-Mar
Best soil	Light sandy loam with good drainage
Soil pH	7.0-8.5
Main growing states	Rajasthan, Gujarat, Tamil Nadu, AP
Fertilizer (N : P2O5 : K2O)	FYM 10-15 t/ha plus modest N
Nutrient tip	Avoid waterlogging
Crop duration	First leaf cut at 8-12 months
Seed rate / spacing	About 10,000 suckers/ha; 60 x 40 cm
Water need	Low
Typical yield	40-50 t/ha fresh leaf per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot and rot	Dark sunken spots, soft rot of leaves	Mancozeb 2.5 g/L; Carbendazim 1 g/L	Avoid waterlogging; remove infected leaves
Root rot	Yellowing, rotted roots	Drench Carbendazim 1 g/L	*Trichoderma*; raised beds with good drainage
Mealybug and scale	White masses at leaf base	Follow label	neem oil 5 ml/L; *Lecanicillium* 5 g/L

148. Stevia

Hindi / Telugu name	sṭīviyā / sṭīviyā
Best season	Plant Feb-Mar or Jul
Best soil	Well-drained sandy loam rich in organic matter
Soil pH	6.5-7.5
Main growing states	Karnataka, Maharashtra, MP, Haryana
Fertilizer (N : P2O5 : K2O)	FYM 10 t/ha plus 100 : 50 : 50
Nutrient tip	Prefer organic inputs (leaves are the product)
Crop duration	About 4 months per cut (3-4 cuts a year)
Seed rate / spacing	40,000-60,000 plants/ha
Water need	Moderate
Typical yield	2-4 t/ha dry leaf per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot (*Septoria*)	Brown spots on leaves	Prefer bio products; chemicals only as per label	*P. fluorescens* 10 g/L; remove infected leaves
Root and stem rot	Yellowing, wilting, rotted base	Avoid chemicals where possible	*Trichoderma*; raised beds; good drainage
Aphids and whitefly	Curling, honeydew	Avoid chemicals (leaves are the product)	neem oil 5 ml/L; yellow sticky traps

149. Betel vine (Paan)

Hindi / Telugu name	pān / tamalapāku
Best season	Plant Feb-Mar
Best soil	Well-drained loam in shaded humid garden (baraja)
Soil pH	6.0-7.5
Main growing states	West Bengal, Karnataka, AP, Tamil Nadu, Bihar
Fertilizer (N : P2O5 : K2O)	FYM plus oil cakes with modest NPK
Nutrient tip	Needs shade and humidity; avoid waterlogging
Crop duration	Perennial; leaves from 3-4 months
Seed rate / spacing	Cuttings on supports in shaded baraja
Water need	Frequent light irrigation
Typical yield	Leaves picked every 2-3 weeks

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Foot rot (*Phytophthora*)	Wilting vines, rotted base	Bordeaux mixture 0.5-1% as per local advice	*Trichoderma* and *P. fluorescens* in soil; drainage
Leaf spot	Brown spots on leaves	Bordeaux mixture 0.5%	Remove infected leaves
Scale insect	Crusty patches on stems	Follow label	neem oil 5 ml/L; remove infested vines
Note	Leaves are chewed fresh	Use only recommended products and keep waiting periods	Prefer bio products and clean cultivation

150. Mushroom (Button and Oyster)

Hindi / Telugu name	maśarūm / khumb / puṭṭagoḍugulu
Best season	Button: winter; Oyster: most of the year
Best soil	Not soil grown: compost or straw substrate indoors
Soil pH	Compost pH 7.0-7.5
Main growing states	Punjab, Haryana, UP, Tamil Nadu, Karnataka
Fertilizer (N : P2O5 : K2O)	Not applicable
Nutrient tip	Hygiene and temperature control matter more than nutrition
Crop duration	Button 40-60 days; oyster 25-35 days per cycle
Seed rate / spacing	On compost or straw substrate indoors
Water need	Humidity control
Typical yield	Button about 15-20 kg per 100 kg compost; oyster 60-80% biological efficiency

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Green mould (*Trichoderma* contaminant)	Green patches on compost or bag	Hygiene and compost pasteurization; follow mushroom research advice	Clean room, sterile tools, fresh spawn; remove infected bags
Bacterial blotch	Brown sticky blotches on caps	Spray water with bleaching powder (about 150 ppm chlorine)	Avoid wet caps; good ventilation
Flies and mites	Maggots, tunnelling in stalks	Follow advice	Screening of doors; neem oil 5 ml/L; sticky traps

151. Mulberry (for silkworm rearing)

Hindi / Telugu name	śahatūt / malbarī / tūta
Best season	Plant Jun-Aug; leaf harvest 5-6 times a year
Best soil	Fertile well-drained loam, red and alluvial
Soil pH	6.2-6.8
Main growing states	Karnataka, Tamil Nadu, AP, West Bengal
Fertilizer (N : P2O5 : K2O)	Irrigated: about 300 : 120 : 120 per ha per year in splits
Nutrient tip	Nitrogen drives leaf yield; boron and zinc
Crop duration	Leaf harvest every 3-4 months
Seed rate / spacing	90 x 90 cm or 60 x 60 cm
Water need	Moderate
Typical yield	30-50 t leaf/ha per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot and powdery mildew	Brown spots and white powder on leaves	Use only products allowed by the sericulture department	Remove infected leaves; observe waiting period before feeding silkworms
Root rot	Wilting and rotted roots	Drench Carbendazim 1 g/L as advised	*Trichoderma* in soil; drainage
Mealybug	White cottony masses on tips (tukra)	Follow sericulture advice	Release *Acerophagus* / *Cryptolaemus* via KVK; prune affected tips
Thrips and mites	Silvering, curling leaves	Use only approved products	neem oil 5 ml/L (observe waiting period)

152. Lemongrass

Hindi / Telugu name	nīṃbū ghās / lemana grās / nimmagaḍḍi
Best season	Plant Jul-Aug or Feb-Mar; harvest 3-4 cuts a year
Best soil	Light sandy loam to red soils; drought tolerant
Soil pH	5.0-8.5
Main growing states	Kerala, Karnataka, Uttar Pradesh, Maharashtra
Fertilizer (N : P2O5 : K2O)	150 : 60 : 60 per year in splits
Nutrient tip	Responds to nitrogen after each cut
Crop duration	First cut at 90-120 days; 3-4 cuts a year
Seed rate / spacing	Slips 60 x 30 cm (about 40,000/ha)
Water need	Low
Typical yield	15-25 t/ha herb; oil 80-120 kg/ha per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf blight and rust	Brown spots and pustules	Mancozeb 2.5 g/L	Remove infected leaves
Root rot	Yellowing, rotted roots	Drench Carbendazim 1 g/L	*Trichoderma*; drainage
Termites	Damage to clumps	Follow label	Neem cake in soil; *Beauveria* 5 g/L

153. Mentha (Mint)

Hindi / Telugu name	pudīnā / meṃthā / pudīnā / miṃṭ
Best season	Plant Jan-Feb for oil crop; leaf mint year-round
Best soil	Well-drained fertile sandy loam and loam
Soil pH	6.0-7.5
Main growing states	UP, Punjab, Haryana, Uttarakhand
Fertilizer (N : P2O5 : K2O)	120 : 60 : 60 kg/ha
Nutrient tip	Needs nitrogen and irrigation after each cut
Crop duration	90-110 days per cut (2-3 cuts a year)
Seed rate / spacing	Suckers 400-500 kg/ha; 45 x 20 cm
Water need	Moderate to high
Typical yield	30-40 t/ha herb; oil 100-150 kg/ha

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Powdery mildew	White powder on leaves	wettable sulphur 2.5 g/L	*Bacillus subtilis* 2-5 g/L
Rust and leaf spot	Orange pustules, brown spots	Mancozeb 2.5 g/L	Remove infected leaves
Hairy caterpillar and aphids	Defoliation, curling	Emamectin benzoate 0.4 g/L only for oil crop; avoid on culinary mint	neem oil 5 ml/L; *Bt* 2 g/L

154. Tulsi (Holy basil)

Hindi / Telugu name	tulasī / tulasi
Best season	Sow Jun-Jul or Feb-Mar
Best soil	Well-drained sandy loam; hardy
Soil pH	6.0-7.5
Main growing states	UP, MP, Gujarat, Karnataka, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	FYM 10 t/ha plus about 60 : 30 : 30
Nutrient tip	Use organic inputs where leaves are used as herb or tea
Crop duration	90-100 days per cut
Seed rate / spacing	Seed 300-400 g/ha nursery; 45 x 45 cm
Water need	Low to moderate
Typical yield	20-25 t/ha fresh herb

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Root rot and damping off	Seedling collapse, wilting	Seed treatment: Carbendazim 2 g/kg	*Trichoderma* 4 g/kg seed; drainage
Leaf spot	Brown spots on leaves	Prefer bio products	*P. fluorescens* 10 g/L; remove infected leaves
Aphids and caterpillars	Curling, chewed leaves	Avoid chemicals where possible	neem oil 5 ml/L; *Bt* 2 g/L

155. Palmarosa

Hindi / Telugu name	pāmārojā / pāmarosā
Best season	Plant Jun-Aug
Best soil	Light well-drained soils; tolerates poor and moderately saline soils
Soil pH	5.5-8.5
Main growing states	MP, Maharashtra, UP, AP, Karnataka
Fertilizer (N : P2O5 : K2O)	100-150 : 40 : 40 per year in splits
Nutrient tip	Hardy aromatic grass; 3-4 cuts a year
Crop duration	First cut at 4-5 months; 3-4 cuts a year
Seed rate / spacing	Seed 3-4 kg/ha or slips; 60 x 30 cm
Water need	Low
Typical yield	Oil 80-100 kg/ha per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf blight and rust	Brown spots and orange pustules	Mancozeb 2.5 g/L	Remove infected leaves
Root rot	Yellowing, rotted roots	Drench Carbendazim 1 g/L	*Trichoderma*; drainage
Termites	Damage to clumps	Follow label	Neem cake in soil; *Beauveria* 5 g/L

156. Patchouli

Hindi / Telugu name	pacaulī / pacauli
Best season	Plant Jun-Aug
Best soil	Well-drained loam rich in humus with partial shade
Soil pH	5.5-7.0
Main growing states	Karnataka, Kerala, Assam, Tamil Nadu
Fertilizer (N : P2O5 : K2O)	100 : 50 : 50 per year in splits
Nutrient tip	Needs mulching and partial shade; avoid waterlogging
Crop duration	First harvest at 5-6 months; 2-3 harvests a year
Seed rate / spacing	60 x 60 cm cuttings
Water need	Moderate
Typical yield	Oil 25-35 kg/ha per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Root rot (*Rhizoctonia*)	Wilting, rotted roots	Drench Carbendazim 1 g/L	*Trichoderma*; well-drained beds
Leaf spot	Brown spots on leaves	Mancozeb 2.5 g/L	Remove infected leaves
Root knot nematode	Galls on roots, stunting	Follow label	Neem cake 250 kg/ha; marigold intercrop

157. Citronella

Hindi / Telugu name	siṭronelā / siṭronellā
Best season	Plant Jun-Aug
Best soil	Light to red loam; well-drained; tolerates moderately poor soils
Soil pH	5.5-7.5
Main growing states	Assam, Karnataka, Kerala, MP
Fertilizer (N : P2O5 : K2O)	100 : 50 : 50 per year in splits
Nutrient tip	Hardy grass; 3-4 cuts a year
Crop duration	First cut at 4-6 months; 3-4 cuts a year
Seed rate / spacing	Slips 60 x 45 cm
Water need	Moderate
Typical yield	Oil 80-100 kg/ha per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf blight	Brown drying of leaf tips	Mancozeb 2.5 g/L	Remove infected leaves
Termites	Damage to clumps	Follow label	Neem cake in soil; *Beauveria* 5 g/L
Root rot	Yellowing, rotted roots	Drench Carbendazim 1 g/L	*Trichoderma*; drainage

158. Safed musli

Hindi / Telugu name	sapheda mūsalī / saphed muslī
Best season	Plant tubers Jun-Jul
Best soil	Light sandy loam and red soils; well-drained
Soil pH	6.0-7.5
Main growing states	MP, Chhattisgarh, Maharashtra, Gujarat
Fertilizer (N : P2O5 : K2O)	FYM 20 t/ha plus modest NPK
Nutrient tip	Harvest when leaves dry; avoid waterlogging
Crop duration	90-120 days (harvest at leaf drying)
Seed rate / spacing	Fingers 30 x 15 cm
Water need	Moderate
Typical yield	5-8 t/ha fresh roots

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Tuber rot (*Fusarium*)	Rotted tubers, wilting	Tuber dip: Carbendazim 1 g/L for 30 minutes	*Trichoderma*; healthy tubers; drainage
Leaf spot	Brown spots on leaves	Mancozeb 2.5 g/L	Remove infected leaves
Nematodes	Galls on roots	Follow label	Neem cake in soil; rotation

159. Giloy (Guduchi)

Hindi / Telugu name	giloy / tippatīga
Best season	Plant Jun-Aug (vine)
Best soil	Almost any soil; well-drained; hardy
Soil pH	6.0-7.5
Main growing states	All states
Fertilizer (N : P2O5 : K2O)	FYM plus modest NPK
Nutrient tip	Give support (trees or trellis)
Crop duration	Harvest stems from about 12 months
Seed rate / spacing	Stem cuttings; 2 x 2 m on support
Water need	Low
Typical yield	3-5 t/ha dry stem

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot	Brown spots on leaves	Prefer bio products	*P. fluorescens*; remove infected leaves
Caterpillars and aphids	Chewed or curled leaves	Avoid chemicals where possible	*Bt* 2 g/L; neem oil 5 ml/L

160. Brahmi (Bacopa)

Hindi / Telugu name	brāhmī / brahmi
Best season	Plant Jun-Sep
Best soil	Moist loam and marshy soils; tolerates waterlogging
Soil pH	6.0-7.5
Main growing states	Kerala, Tamil Nadu, UP, AP
Fertilizer (N : P2O5 : K2O)	FYM plus about 60 : 30 : 30
Nutrient tip	Needs constant moisture; mulch
Crop duration	3-4 months per cut
Seed rate / spacing	Cuttings 15 x 15 cm
Water need	High moisture
Typical yield	10-15 t/ha fresh herb per year

Disease / pest	Key signs	Chemical control (per litre)	Bio / organic option
Leaf spot and root rot	Brown spots, rotted roots	Prefer bio products	*Trichoderma*; *P. fluorescens*; avoid stagnant dirty water
Aphids and caterpillars	Curled or chewed shoots	Avoid chemicals where possible	neem oil 5 ml/L; *Bt* 2 g/L

Quick dose chart for a 15 L pump

If the dose per litre is	You need for a 15 L pump
0.3 ml/L	4.5 ml
0.5 ml/L	7.5 ml
1 ml/L	15 ml
2 ml/L	30 ml
5 ml/L	75 ml
0.4 g/L	6 g
1 g/L	15 g
2.5 g/L	37.5 g
3 g/L	45 g

Good practices that reduce disease and pesticide use

- Use certified, disease-free seed and treat it with *Trichoderma* or *Pseudomonas* before sowing.
- Rotate crops. Do not grow the same crop or its relatives (tomato, chilli, brinjal, potato) in the same field repeatedly.
- Keep fields clean: remove infected plants, crop residue and weeds that carry pests.
- Use balanced fertilizer based on a soil test. Excess nitrogen increases blast, sheath blight, planthoppers and aphids.
- Scout the field weekly and spray only when pests cross the threshold or disease first appears.
- Use traps and bio-control first and chemicals only when needed. Keep spray records.
- Use drip or avoid overhead irrigation in disease-prone crops to keep leaves dry.

Disclaimer: this guide is for general information. Recommendations differ by state, season, variety and registered products in India (CIB and RC). Always follow the product label and the advice of your local agricultural officer, KVK, or the relevant board (Tea, Coffee, Rubber, Spices, CPCRI).